

Custom LDO Voron guide — print edition

Unofficial. Hardware verification gates remain open. Not a physically validated configuration.

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LDO Voron 2.4R2 — integrated custom assembly guide

Unofficial derivative, prepared 2026-10-09. Readiness: documentation complete with hardware verification gates; not a physically validated machine configuration.

This is the chronological build path for the fourteen reported purchases. Start here and follow the chapters in order. The target is provisionally 350-class, Voron design **2.4R2**, LDO kit **Rev D-family, unverified**. The current West3D D+ listing is a research lead. It does not establish the size, batch, electronics, or options in the purchased kit.

The [original Voron assembly manual](#) remains authoritative for unchanged mechanical details. This guide replaces its affected steps before assembly. Source figures are labeled **STOCK REFERENCE**; they do not depict the CNC upgrades. Vendor instructions stay linked where their redistribution license is unknown. Local original schematics describe relationships, not unverified dimensions or manufacturing geometry.

Build in this order

1. [Inventory and evidence](#): identify the kit and allocate purchases.
2. [Printed parts and preparation](#): print retained parts and resolve interfaces.
3. [Frame, Z rails, deck and Z drives](#).
4. [Modified gantry, titanium backers, Z supports and belts](#).
5. [MRW bed, kinematics and Wings](#).
6. [Rapido 2 Fiber UHF, StealthBurner, CNC carriage and Cartographer](#).
7. [Electronics and cable routing](#).
8. [Software and configuration preparation](#).
9. [Pre-power, first start and calibration](#).
10. [Enclosure, filament path and first print](#).

Gates are part of the procedure

STOP at a gate before its affected step. Continue independent chapters when their prerequisites are satisfied. Closing a gate requires the named receipt detail, board marking, manufacturer instruction, measurement, or physical check; an assumption does not close it. No actual installed quantities or measurements have been supplied. All allocations are plans, not claims of completed assembly.

The most consequential decisions are the bed's magnet arrangement, UHF duct and mount geometry, clamp overlaps and rigid-joint alignment, actual PCB revisions, and hotend startup load versus the complete electrical path. A blocked heater gate prohibits heater commissioning even if other software checks pass.

Build records

- [All fourteen purchase dispositions](#) and [machine manifest](#).
- [Compatibility decisions](#).
- [Printed-parts matrix](#).
- [Hardware and missing requirements](#).
- [Every stock manual page mapped to this path](#).
- [Unresolved questions and closure evidence](#).
- [Pinned sources and illustration provenance](#), [source lockfile](#).
- [Non-deployable configuration examples](#).
- [Build commands and publication](#), [actual validation results](#).

Upstream design and manual: Voron Design contributors, GPL-3.0. Original custom Markdown, diagrams and tooling: GPL-3.0, modified 2026-10-09. See [attribution and reuse terms](#). This documentation carries no warranty. Static documentation checks do not validate physical fit, electrical safety, firmware, or the completed printer.

1. Inventory and evidence

Prerequisites: unopened or safely disconnected kit and all upgrade packages. **Tools:** ruler/calipers, camera for private inspection, labels and bins. No power. **Parts:** one LDO kit (01), one three-pack of backers (02), one each of purchases 03–07, Wings notation 2x (08), and one purchased set/item each of 09–14. Use the [manifest](#) for unit distinctions.

I1 — verify the build target before size-dependent preparation

No order documents, photos, packing lists or relevant printer configurations were found in the workspace filename/content search. This project did not access email or past conversations. The six Vitalii selections are purchase evidence provided by the user; the underlying private receipt is not redistributed.

1. Lay out the extrusions and measure them against the 350 BOM lead: ten A=470 mm, four B=530 mm, two C=450 mm, D=430 mm and E=340 mm. The [LDO 350 BOM directory](#) maps kit serial/batch to its BOM. Select the matching batch using the actual label; do not infer a serial from the order date.
2. Record the actual size, LDO revision and batch separately from design 2.4R2. If dimensions differ, stop printing 350 skirts and sizing belts. Retain the purchased MRW350 bed and resolve its fit before changing the target.
3. Identify rail labels, A/B motor model and shaft projection, X/Y belt width, Z belt width, PSU model, mains region, heater/fuse labels, and host options.
4. Read the mainboard and toolboard silkscreen on both sides while disconnected. Do not use a current product listing to pick firmware or pinouts. Photograph the boards privately and transcribe only technical model/revision fields.
5. Count each CNC package separately. Match the product-linked manual revision to the shipped part, and resolve conflicting package/fastener lists at the gate in [motion](#) before assembling those joints.
6. Identify backer lengths; bed standard versus embedded magnets; heater and magnetic-sheet options; Wings pieces versus sets; Rapido revision/UHF/PT1000 labels; CNC carriage 6/9 mm and UHF-extender contents; Cartographer firmware generation/interface. Leave unverified fields `null`.

Label bins before assembly

KEEP the kit frame, rails, four stock Z reduction drives, compatible motion belts, host/electronics and CW2/SB components after their revision gates. **REPLACE** printed A/B drives, front idlers, live XY joints, Z upper idlers, stock rigid-interface Z joint pieces, stock bed assembly, hotend and carriage. **OMIT** the stock Omron/Klicky/TAP probe path, Hall endstop magnets, stock nozzle Z endstop and their macros for the Cartographer homing path described later. Keep unused kit parts in a labeled surplus bin; do not discard them.

ADD only requirements documented in the [hardware matrix](#). An extra requirement is not a claimed purchase. Do not peel a bonded heater or magnetic sheet off the supplied LDO bed for reuse on the MRW plate.

Completion: all fourteen IDs accounted for, package discrepancies recorded, and no assumed PCB revision or private purchase data entered into Git. Unknown details remain tracked in [the questions register](#).

Sources: [West3D kit listing](#), [LDO batch BOM](#), [Rev D BOM lead](#).

2. Printed parts and preparation

Prerequisites: I1 size gate; parts inventory; disconnected hardware. **Tools:** calibrated ABS printer or PIF parts, deburring tools, insert soldering tool, hex drivers 1.5–4 mm (ball ends for access, straight ends for final fastening), calipers, square, straightedge, lubricant approved by the rail manufacturer, threadlocker compatible with metal joints. Keep solvent/threadlocker off plastics.

P1 — print only parts with a resolved interface

Parts and quantities: the [printed-parts matrix](#) enumerates filenames, planned quantities, pinned sources and dispositions. Quantity suffixes describe piece counts, not the number of download files. Core printed parts are retained; CNC-replaced files are explicitly omitted. Do not print every file under `STLs/` : that includes other sizes and alternate boards.

INTRODUCTION

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PART PRINTING GUIDELINES

The Voron Team has provided the following print guidelines for you to follow in order to have the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times community members that have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running.
Check Discord if you have any interest in having someone help you out.

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1. Use upstream p4: ABS, 0.2 mm layers, forced 0.4 mm extrusion width, 40% infill, four walls and five top/bottom layers. Use Grid, Gyroid, Honeycomb, Triangle or Cubic infill. Honor a component's own documented guidance where different.
2. Print the four Z drives, retained skirts/panel mounts and rail alignment tools first. Sort mirrored A/B parts into two of each; do not substitute an MGN9 X carriage for the R2 MGN12 design.

3. Audit LDO-provided parts against the actual carton. The Rev D supplement lists brackets, DIN clips, spacers and a clear PETG LED diffuser as provided, but its Nitehawk references target original SB. Verify V2 parts against its own repository before assuming the old cover/adaptor mount fits.
4. **OMIT** plastic A/B drives, front-idler stacks, XY-joint halves, stock Z upper tensioners, all four stock Z-joint assemblies (upper/lower print types) and all stock Z belt clips; CNC subassemblies supply the resolved replacement functions. **OMIT** stock inductive probe retention and nozzle Z-endstop printing for this path.
5. **STOP before printing the UHF hotend housing/duct.** The official standard Rapido v2 housing is HF. Follow T1/T2 in [toolhead](#) to verify the Fiber UHF adapter and airflow/nozzle geometry with the CNC mount.
6. **STOP before printing the replacement XY endstop support.** The Vitalii download contains STEP models rather than print-ready STL instructions. Resolve PCB revision, dimensions, export and fit at M6 in [motion](#).
7. Test inserts on the retained practice print. Install the inserts shown in the relevant stock figure only in retained plastic parts; threaded CNC holes do not accept stock heatset inserts. Clear the built-in skirt supports shown on upstream p211. Ensure bearing/rail interfaces are free of debris.

Completion: print list marked obtained/printed/blocked; no incompatible stock part prepared for installation; fasteners counted by type and length, not just diameter. The matrix's `included_by_ldo` designation is a BOM claim until counted.

Sources: original manual pp4–10,31; [LDO Rev D print supplement](#), [LDO supplement repository](#).

3. Frame, Z rails, deck and Z reduction drives

Prerequisites: I1 size gate closed; retained Z-drive prints prepared; rails identified. Keep the printer disconnected throughout this chapter. **Tools:** flat reference surface, square, ruler/calipers, 2.5/3/4 mm drivers, rail guides, rail lubricant, metal-compatible threadlocker and insert tool.

Frame — KEEP with a bed-support gate

Parts, provisional 350: ten A extrusions 470 mm (eight for cube, two bed supports), four B 530 mm, M5x16 BHCS (16 blind-joint screws for eight frame A pieces; four for the bed brackets), four bed corner brackets, four M5x10 BHCS and four M5 T-nuts at the bracket/frame interfaces. The kit's C/D/E extrusions belong to the gantry. These dimensions are the Rev D BOM lead, not shipped proof.



1. Sort and label extrusions as p13 shows. Confirm tapped ends and access holes. Test-fit roll-in nuts now: LDO warns that extrusion/nut tolerances may make preloading necessary. Remove burrs without damaging rail seating surfaces.
2. Start M5x16 BHCS in both ends of eight frame A pieces, p14. Leave the heads proud enough to enter the adjoining slots. Assemble blind joints by inserting heads into slots and tightening through the access holes, p10/p15.

3. On the flat surface, join two A pieces and a B vertical as p15 shows; add the remaining bottom-square A pieces and four B verticals, p16. Keep the access holes usable. Fit the upper four A pieces, p17. Check the faces and diagonals before final tightening. No torque is specified here.

FRAME

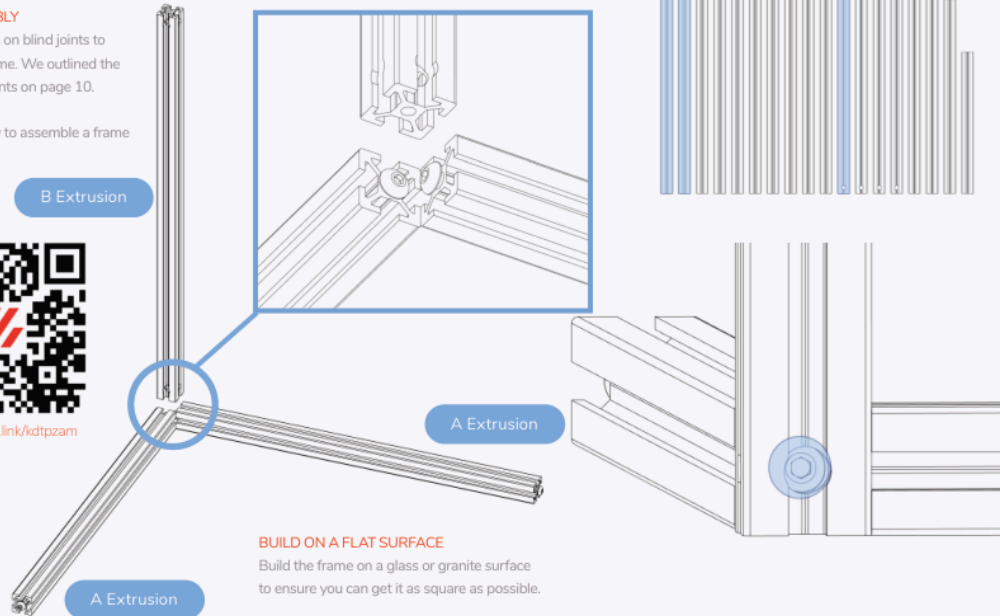
FRAME ASSEMBLY

This design relies on blind joints to assemble the frame. We outlined the basics of blind joints on page 10.

More tips on how to assemble a frame are linked below.



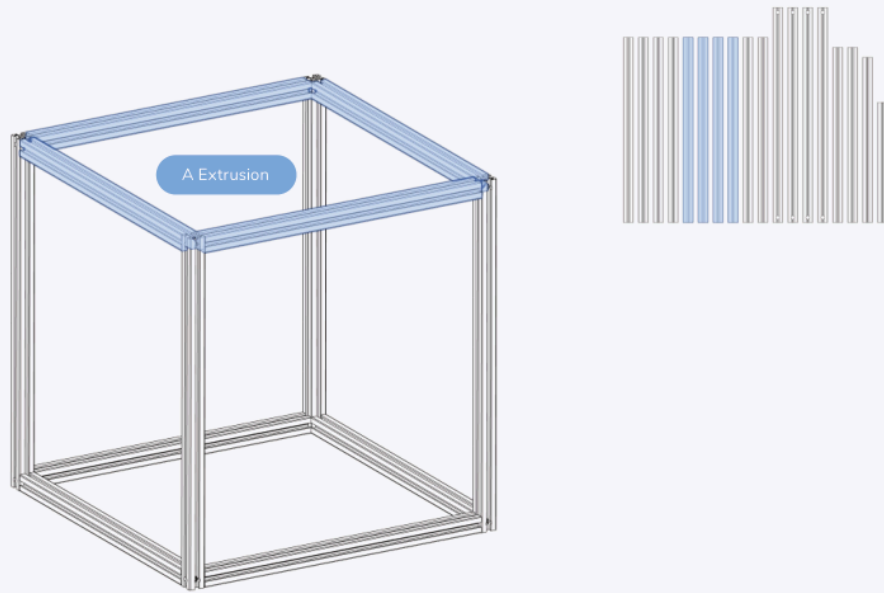
<https://voron.link/kdtpzam>



BUILD ON A FLAT SURFACE

Build the frame on a glass or granite surface to ensure you can get it as square as possible.

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F1 — settle the Wings support layout before fixing bed extrusions

REPLACE stock p20's 130 mm support-spacing instruction with the resolved MRW/Wings layout in [bed](#). The stock drawing uses two front-to-back supports and stock mounting holes; Wings use different plate holes. Do not lock those supports at the stock positions or drill the deck until B2/B3 is closed. After loosely fitting the supports in step 4 below, perform [bed identification and the B2/B3 cold support dry-fit](#) before final deck/DIN installation; heater installation remains later. The deck/DIN screws anchor into these supports, so verify that their fixed deck holes agree with the resolved support spacing. If they do not, obtain a documented replacement deck layout before fastening. Do not improvise slots or lock the supports against the Wings layout merely to reuse the stock holes.

4. Prepare the two bed A extrusions with corner brackets, four M5x16 BHCS in total, p18. Insert the four M5 T-nuts in the frame and loosely fit the four M5x10 bracket screws, p19. LDO substitutes its 1 mm brass precision spacer for the stock M5 shim **only at retained stock interfaces**. Keep these bed supports adjustable until the bed support geometry is verified.
5. Square all six frame faces and compare diagonals, p21. Correct twist while the frame is unloaded. Record measured diagonals locally. Recheck after the bed-support gate and final bed-support tightening.

Completion: cube square and untwisted, blind joints accessible, supports labeled provisional if their gate is open; no CNC assembly received stock spacer assumptions.

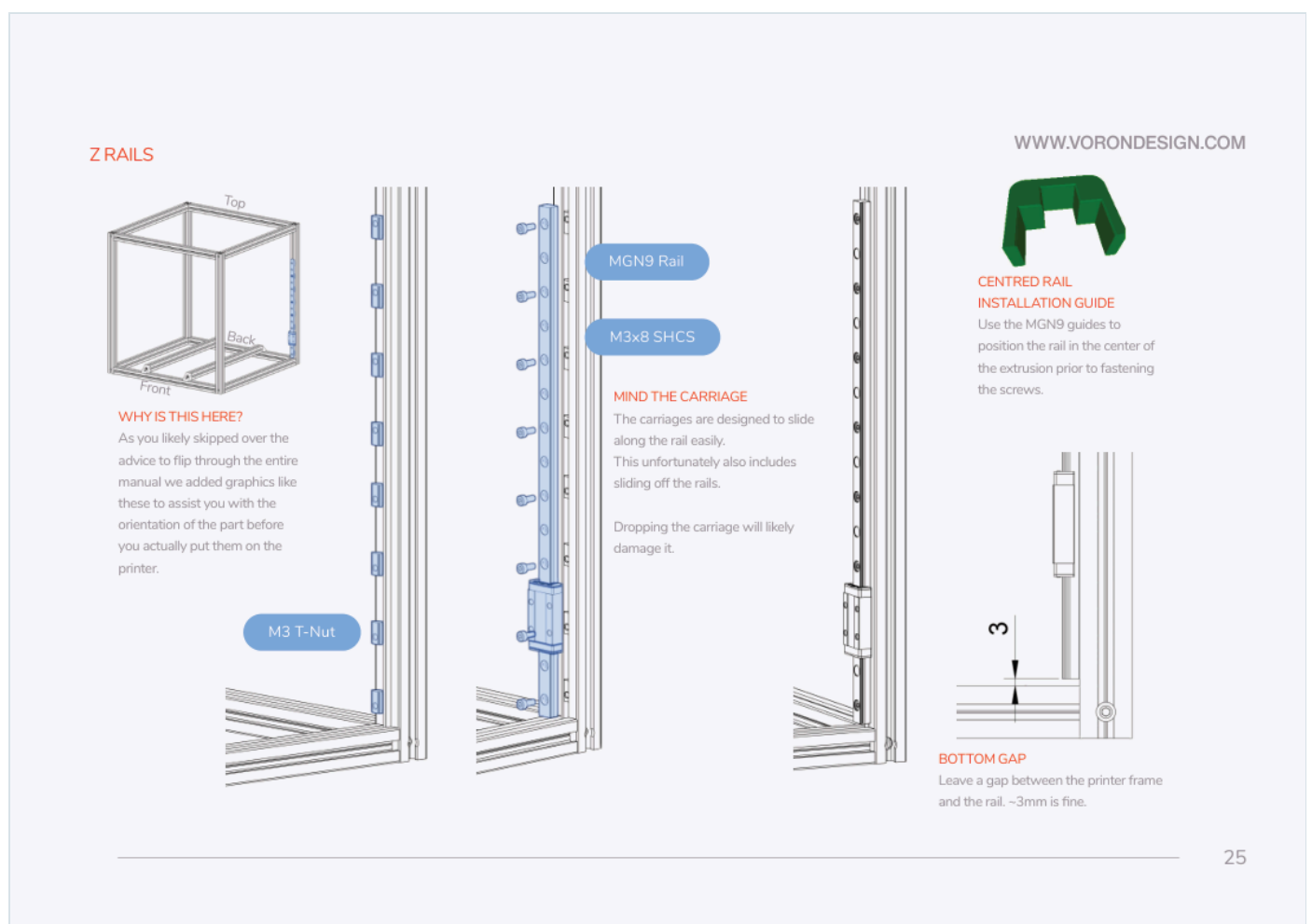
Z rails and deck — KEEP, measure the actual panel

Parts: four MGN9 Z rails/carriages (400 mm in the provisional BOM); M3x8 SHCS and M3 roll-in nuts at the alternating mounting holes shown on p25, one of each per occupied hole (count actual rail holes before taking hardware); two MGN9 guides; one deck panel, two DIN rails, four M5 T-nuts/four M5x10 BHCS; eight deck supports selected for the actual panel thickness.

F2 — panel thickness differs between the supplementary documents

The FAQ/print supplement says 4 mm deck; the retrieved Rev D BOM calls the deck 3 mm and the bottom panel 4 mm. Measure the shipped deck. Use `deck_support_4mm_x8.stl` only for 4 mm; use the 3 mm file for 3 mm. This is a source discrepancy, not permission to force the wrong clips onto a panel.

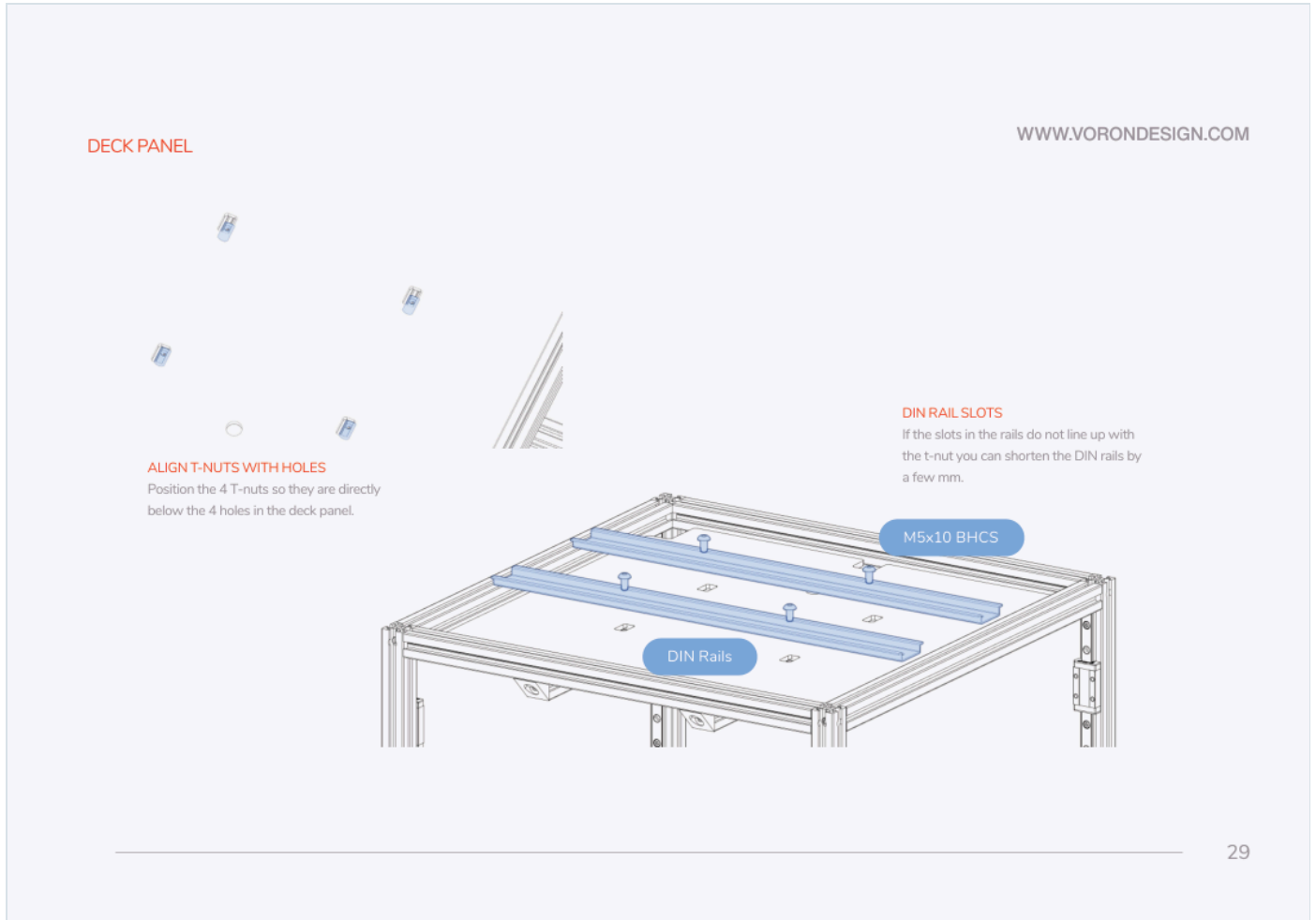
1. Follow rail preparation p24 and the rail maker's lubrication instructions. Secure carriage stops before changing orientation. Never slide the carriage off to clean the rail or let its ball bearings escape.
2. Center each Z rail on the inner vertical face with the MGN9 guides, p25. Keep approximately 3 mm clearance above the bottom frame as shown. Install screws center-outward in the selected alternating holes. Keep the rail flat; do not use a screw to draw a bent rail onto the extrusion.
3. Mirror all four rails inward as p27 shows. Test each carriage over its whole safe rail length by hand; it must not catch or change resistance abruptly. Leave temporary rail stops installed until rigid Z joints retain the carriages.



4. **STOP here if F1 is open:** keep the deck removable and unfastened; do not close the lower assembly around an unresolved support layout. Bench assembly of the lower Z drives may continue independently. Before the deck obstructs access, preload the verified bed-support nuts and install the chosen deck support clips. Fit the

deck with its notch toward the back, p28. Keep bed-wire pass-through accessible to the final bed cable route; do not apply stock p60's wire-hole coordinate to the Wings assembly.

5. With carriages positively secured, invert the frame to fit DIN rails beneath the deck. Align four M5 T-nuts to the four holes, attach rails with four M5x10 BHCS, p29. Keep the slot openings accessible for electronics service.



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Completion: four smooth captive carriages, panel supported without sag, deck notch rearward and bed/cable access retained. Do not install the stock bed.

Four lower Z reduction drives — KEEP

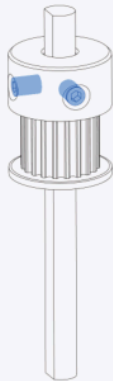
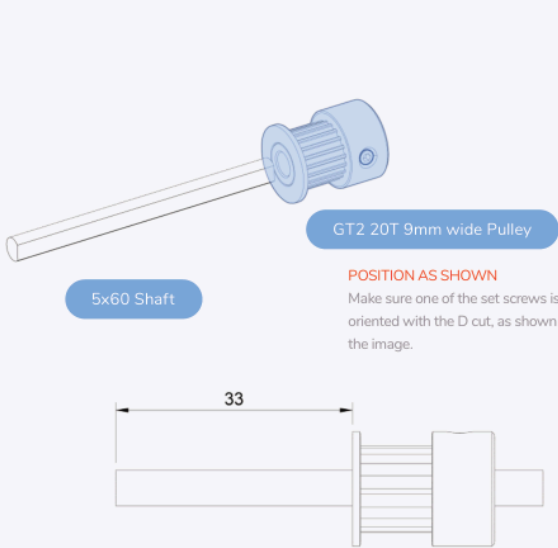
Parts, totals for four: two each mirrored drive main/retainer/motor mount/baseplate/tensioner print; four 5×60 mm shafts, four 20T **9 mm** shaft pulleys, four 80T **6 mm** reduction pulleys, twelve 625 bearings, eight M5 spacers, four 188 mm × 6 mm closed loops, four Z motors, four 16T **6 mm** motor pulleys, four M5 nuts, twenty-four M3x40 SHCS for main housings, twelve M3x8 SHCS for retainers, twelve M3x8 SHCS for motor mounts (three per motor in p39), eight M5x40 SHCS, eight M5x10 BHCS and sixteen M5 T-nuts for frame attachments, four rubber feet/four M5x16 BHCS. Check the actual retained print against the figure before issuing screws; do not transfer these counts to CNC A/B mounts.

1. Install inserts in the retained plastic Z-drive parts as p31 shows. Make two A and two mirrored B drives; do not reverse them to avoid cable exits.
2. Position the 20T/9 mm pulley on the 5×60 shaft as p32 shows: the drawing's 33 mm dimension is from the long exposed shaft end to the outer pulley flange face on that side, not the hub or tooth centre. One set screw meets the D-flat. Use the specified metal threadlocker on set screws, keeping it away from plastic and

bearings.

BELT DRIVE ASSEMBLY

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SET SCREWS
AKA THE ROOT OF ALL ISSUES
Insert both set screws and use thread locker on all set screws.

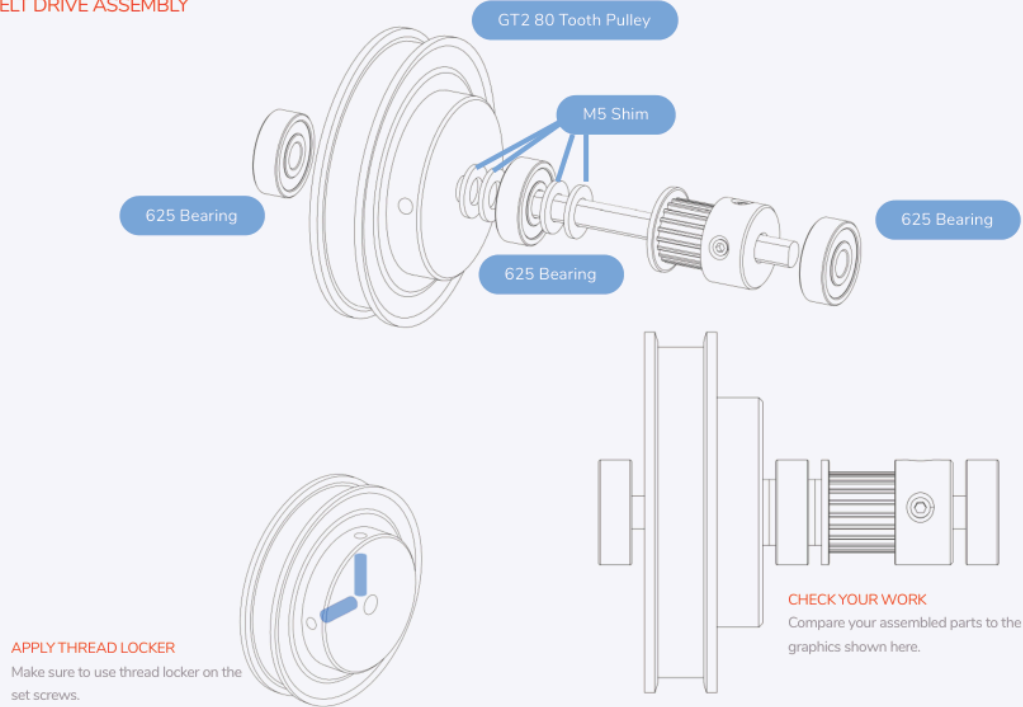
Use a high quality hex driver to prevent the hex profile from stripping. Ball-end drivers are not recommended.

Loose set screws account for the majority of issues that our users report. Save yourself hours of troubleshooting and apply thread locker to all set screws during the build.

See the product's application notes for instructions - keep away from printed parts.

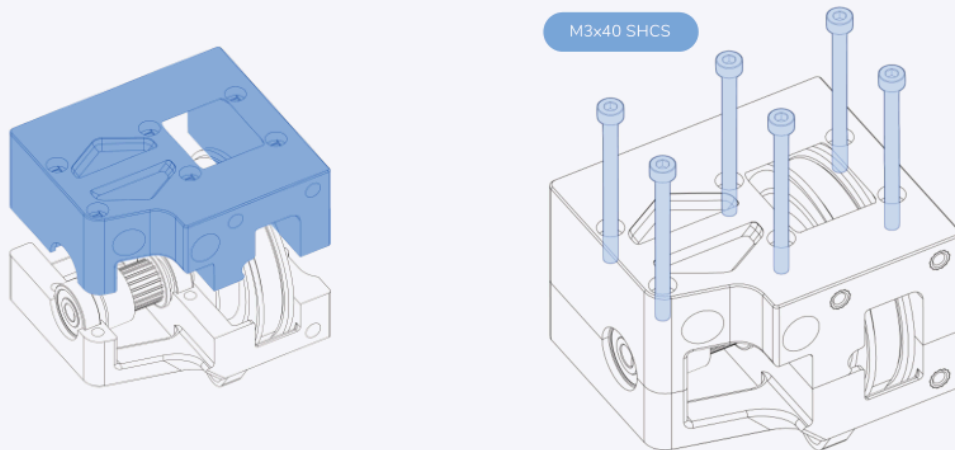
BELT DRIVE ASSEMBLY

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33

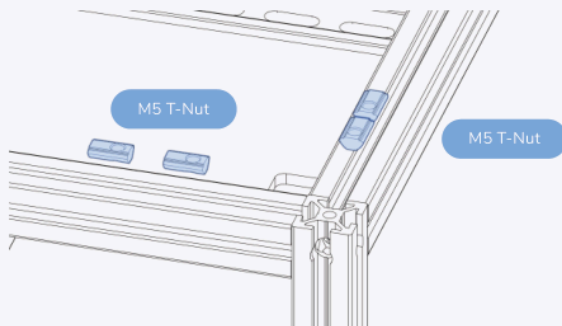
3. Build the 625/spacer/80T assembly in the exact p33 order. LDO's 1 mm brass precision spacer replaces the stock M5 shim at this retained interface. Use one spacer on each side of the middle 625 bearing (two per drive). Compare the side view; all three 625 bearings per drive must sit in their intended pockets. Set screws must not rub the housing.
4. Place the 188 mm closed loop in the body **before** closing it, p34–35. Align the shaft/bearing assembly to p35. Close the main housing with six M3x40 SHCS per drive, p36. Fit the retainer and M5 nut with three M3x8 SHCS, p37. Verify the closed loop remains captured and turns smoothly.



5. Fit each motor's 16T/6 mm pulley, p38. These four motors are the only place this build uses 16T motor pulleys. Align its tooth plane with the 80T pulley and closed loop; pulley hub orientation may differ with the actual motor. Set one screw on the flat and secure both. Mount the motor in the orientation on p39 so its cable exits clear the frame. Engage the closed loop before closing the stock lower-drive tensioning latch.
6. Support and invert the frame, p40–41. Preload four M5 nuts per drive as illustrated; fit two M5x40 SHCS. Fit the two M5x10 BHCS loosely at the base, p42–43, close the lower drive latch, p44, then secure frame screws and the foot with M5x16 BHCS, p45. Recheck drive location after latch closure, p46.
7. Repeat opposing and mirrored corners, p47. Turn each drive gently by hand and check closed-loop alignment and free shaft rotation. Label all four motor cables by their physical corners without assuming software Z order.

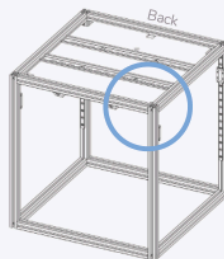
Z DRIVE

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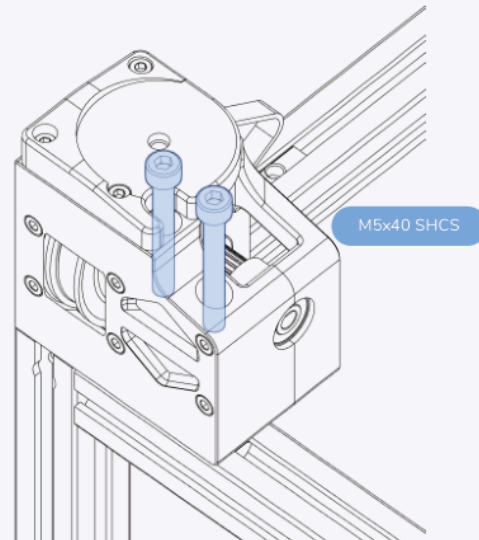
WHICH CORNER IS THIS?

We highlighted the corner with a circle.



UPSIDE DOWN ASSEMBLY

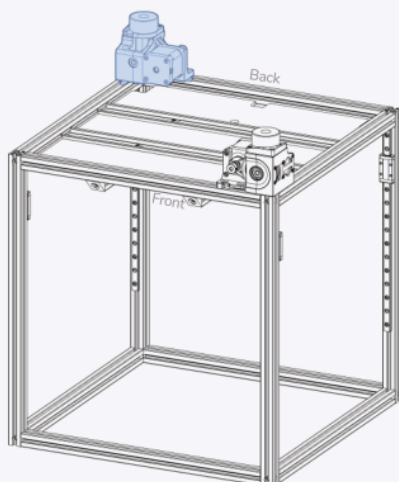
For ease of assembly we recommend flipping the printer on its head for the next steps.



41

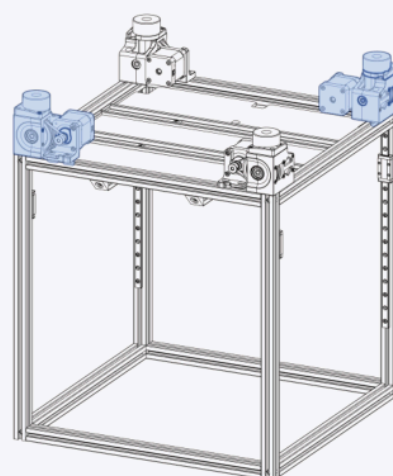
OTHER Z DRIVES

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REPEAT INSTRUCTIONS FOR OPPOSING CORNER

Build another Z drive, following the same instructions.



REPEAT INSTRUCTIONS FOR THE MIRRORRED DRIVES

Build two more Z drives following the instructions that came before. The printed parts are mirrored.

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REPLACE upper Z idler/tensioner assembly pp48–50 with purchase 12 at the start of [motion](#). Retain the lower reduction latch just assembled: it tensions the short 6 mm loop; the new upper unit tensions the long 9 mm Z belt.

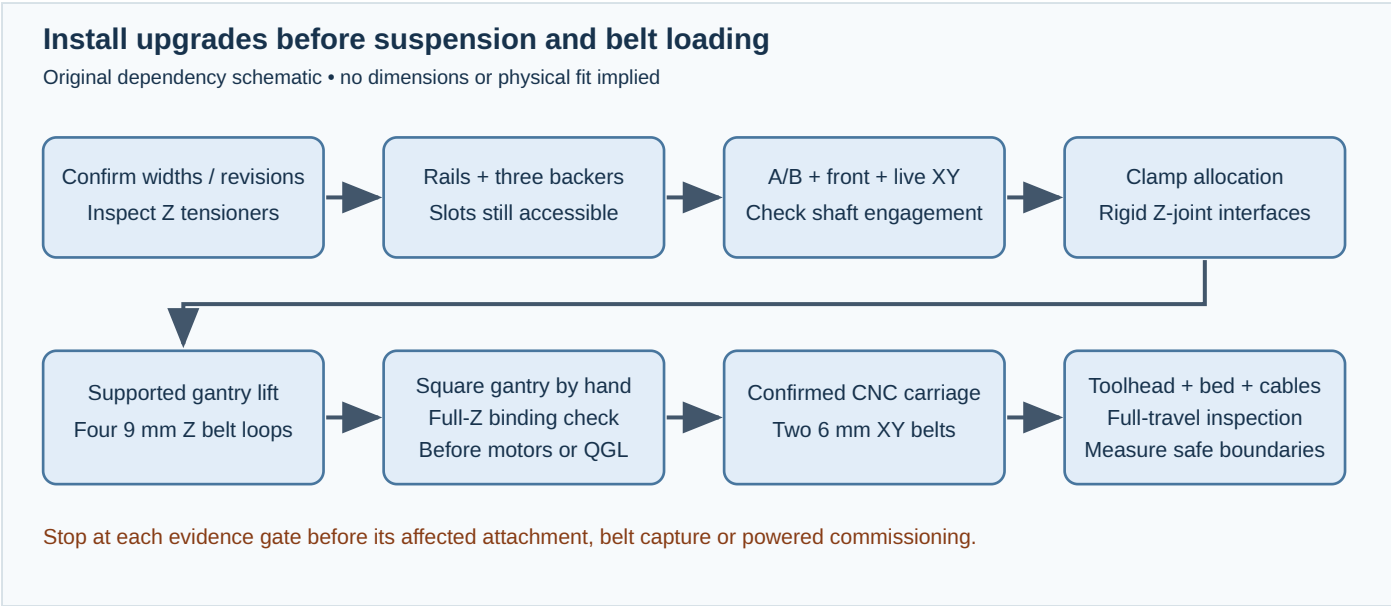
Completion: four correctly mirrored lower drives, short loops free and aligned, rail stops retained, no stock upper tensioner built, gantry not yet suspended.

Sources: original manual pp10,13–50 (all printed page numbers; PDF index = p–1), [LDO FAQ](#), [LDO Rev D BOM](#).

04 — Modified Z suspension, gantry and belts

This chapter installs purchases **02, 09, 10, 11, 12, 13 and 14** before the gantry is suspended. The planning target is a 350-class, single-front-MGN12-X / underside-MGN9-Y Voron 2.4R2 gantry. That geometry is documented by the pinned upstream manual; it still has to match the parts in the box. None of the combined CNC/backer/carriage arrangements below has been physically validated on this printer.

KEEP the frame, rail preparation and four lower Z reduction drives from the preceding chapter. **REPLACE** the stock upper Z idlers, A/B drive frames, front idlers, XY joints, Z joints and relevant belt clamps. **OMIT** their printed counterparts from the initial build. **ADD** the titanium backers while extrusion slots are accessible. Keep the frame open and leave the top panel off: the purchased Z tensioners adjust from above.



M1 — Verify the motion parts before assembling their stacks

Gate M1, blocks the affected subassembly only. Compare the machined parts with the product-linked manufacturer PDFs, not only the store photographs. Record actual revisions, bag contents and belt widths. The governing documents retrieved on 2026-10-09 are [A/B revision 1.0](#), [front-idler revision 1.0](#), [live-XY revision 1.1](#), and [Z tensioners, unversioned](#). Hashes, page mappings and conflicts are in [motion research](#).

The purchased **XY joints are the 6 mm version**. The A/B linked manual is also for 6 mm belts. The Z tensioner manufacturer independently specifies **9 mm Z belts**. The stock manual agrees: printed p.131 (PDF index 130) specifies 6 mm A/B belts, while p.111 (index 110) specifies 9 mm Z belts. Measure the kit belts and each pulley running width; do not infer Z width from XY width. A 9 mm Cartographer carriage variant would conflict with this 6 mm XY path unless the actual manufacturer documentation establishes a different arrangement.

The live manufacturer pages and assembly PDFs disagree in several places:

Subassembly	Evidence to reconcile before use
A/B pair	PDF package table says seven F695 bearings; its diagrams require seven per drive , including the shaft support bearing. The live product listing offers fourteen. Confirm fourteen are in the purchased bearing variant.

Subassembly	Evidence to reconcile before use
Front idlers	Live page lists four M3 slot nuts, M3x16 and M3x10 mounting bolts and four clamps; PDF lists two roll-in nuts and different internal fasteners, with clamps marked add-on. Count the shipped hardware and identify its revision.
Live XY	PDF package table says eight ultra-low M3x3 bearing-retention screws, but procedure says M3x4. Live product text also gives differing counts. Ask for a revision-matched retention specification; do not choose a screw length by guessing.
Rigid Z joints	Store gives hardware quantities but no linked assembly drawing. Obtain the screw-to-hole assignment and approved gantry/rail fit from the supplied instructions or manufacturer before fastening.

Tools for this chapter: quality straight hex drivers matching actual screws, two drivers for the Z tensioner axle, calipers, square, ruler/tape, needle-nose pliers or tweezers, supported work surface, rail stops/tape, four equal-height gantry supports and temporary retaining straps. Use medium threadlocker only on the metal screw joints where the relevant Vitalii PDF recommends it. Keep it out of bearings. No manufacturer torque values were found; none are invented here.

Install the upper Z tensioners before closing the top frame

Prerequisites: squared frame, Z rails and lower Z drives completed; M1 confirms nine-millimetre Z belts and matching tensioners. The top must remain accessible. **REPLACE** upstream printed pp.48–50 (PDF indices 47–49), including their M5x30 / printed-idler hardware, with purchase **12**.

Parts: one purchased set contains four preassembled tensioners and eight M5x25 button-head screws according to the live [product page](#). The PDF lists four tensioner blocks, two left and two right fixed blocks, four 9 mm GT2 idlers, four 20 mm threaded pins, eight 0.8 mm M5 shims, eight M3x6, four M5x16 and eight M5x25 screws. Treat those internal pieces as contents of the four assemblies, not an additional set. Confirm the M5x16 frame attachment and mating T-nuts from revision-matched instructions before installation; the PDF does not depict attachment to a full frame.

Gate M2, blocks mounting these tensioners. Confirm the left/right corner orientation, mating frame holes/T-nuts and frame-attachment screw length against the supplied product revision. A stock plastic-idler screw is not a substitute. Check that the idler plane aligns with the lower Z-drive pulley and that a straight driver reaches the top adjuster without colliding with frame braces or later panels.

1. Inspect the four preassembled units rather than unnecessarily dismantling them. Each idler should rotate freely. The PDF's assembly leaf 4 (printed "4", index 3) shows **one 0.8 mm shim on each side of the idler**, inside the tensioner fork.
2. If a unit was supplied loose, insert one M3x6 into the threaded axle (leaf 3/index 2), place the shim–idler–shim assembly in the fork (leaf 4/index 3), then secure the axle from the opposite side with the second M3x6 while holding the first with another driver (leaf 5 also printed "4", index 4).
3. Join each movable block to its matching fixed block with two M5x25 screws as shown on printed p.6 (index 5). Keep sufficient adjuster engagement; the stock instruction to back a plastic adjuster out and turn it four turns is **omitted** for these CNC units.
4. After M2 is cleared, attach the two left and two right assemblies in their confirmed top corners. Start with usable adjustment in both directions. Do not use the listing's promotional suggestion of overtightening as a belt-tension instruction.

Completion check: four idlers turn freely; all four belt planes align with their lower drive; no screw bottoms out; adjusters remain accessible from above. Leave the nine-millimetre Z belts uncut until the suspension and clamp gates are cleared.

Prepare the gantry rails and titanium backers while slots remain open

Prerequisites: M1 confirmed gantry rail types; unassembled Y and X extrusions; purchase **02** available. **KEEP** stock rail alignment/preparation on printed p.24 (index 23), p.88 (index 87) and p.101 (index 100). Rail carriages must remain on their rails and be restrained whenever an assembly is inverted. **ADD** three backers before the CNC ends and chain supports obstruct their slots.

Parts: one purchased three-pack, planned installation three pieces; actual ordered build-size variant is unknown. [West3D](#) describes approximately 3.2 mm thickness, twenty-two M3x8 flat-head screws and ten M3x6 flat-head screws, with approximately thirty M3 T-nuts required separately. Do not claim those T-nuts were purchased. The number actually installed equals the appropriate holes in the delivered backers, not the number of screws in the bag.

Gate M3, blocks backer attachment and gantry end closure. Confirm the ordered 350 build-size variant, actual strip lengths, thickness, countersinks, hole spacing and extrusion-slot clearance. The 350 variant name is a printer-size choice, not proof that the physical strip is exactly 350 mm long. Confirm T-nut type fits the actual extrusion and rail screws; account for each occupied hole. Select any cable-bridge clearance part only after confirming the actual chain/umbilical arrangement.

Backers belong opposite their rails

Original non-dimensional cross-section schematic • verify the actual rail arrangement

Top: titanium backer



Y extrusion

Underside: MGN9 rail

Documented Y screws: M3x8 FHCS

Rear: titanium backer



X extrusion

Front: MGN12 rail

Documented X screws: M3x6 FHCS

Ordered size, actual holes, thickness, nuts and all CNC/chain clearances still require measurement.

1. Clean/lubricate and mount the two Y MGN9 rails to the **undersides** of their extrusions using the stock rail hardware and alignment procedure. Work from the centre outward as the manual directs; restrain both carriages.
2. Install each Y backer on the **top** of its extrusion, opposite the underside rail. The [designer's pinned installation instructions](#) specify M3x8 flat-head screws for Y.
3. Install the single MGN12 X rail on the **front** face as stock p.101 shows. Attach its backer to the **rear** face, using the documented M3x6 flat-head screws. Do not put this backer on top: the source explicitly warns that this is the wrong compensating side for a single front MGN12 rail.

4. Seat each countersunk screw without bowing the extrusion, check engagement and slide each rail carriage through its length. Check access for CNC fasteners and endstop wiring before adding end units. Do not drill/tap a purchased prefinished titanium strip unless its manufacturer explicitly requires that operation.

Completion check: all three backers are opposite their respective rails; screw heads are seated; rail travel is smooth; no fastener reaches a rail race or obstructs the mating CNC unit. Candidate designer chain bridges are `XY_cable_chain_bridge-3hole-3mm_backer.stl` and `XY_cable_chain_bridge-Igus-3mm_backer.stl` at the pinned source; their 3 mm naming does not establish fit to an approximately 3.2 mm purchased backer. They remain unresolved until the actual routing is selected and clearance is checked.

Assemble the A/B double-shear drives and front idlers

REPLACE stock pp.64–80 (indices 63–79) with purchases **09** and **11**. Those pages remain stock belt-topology and motor-orientation references; their plastic-frame fasteners and stackups are not instructions for the machined parts.

A/B parts for the pair, after M1: four plates, four precision threaded pins, twelve 1 mm M5 shims, two 10 mm M5 sleeves, six 20 mm M3 sleeves, six cup shims, eight M3x8 flat-head screws, six M3x35 cap screws, four ultra-flat M3x4 bearing-retention screws and fourteen F695 bearings. The bearing count is a diagram-derived requirement for both complete double-shear drives, subject to the package-table conflict above. Retain two kit XY motors and two 20-tooth pulleys only after proving their fit.

Gate M4, blocks motor installation and belt loading. Measure each A/B motor's exposed shaft from its mounting face. With the motor plate, pulley and upper support bearing in their documented positions, prove that the shaft actually engages the support bearing's inner race and turns without bottoming, axial preload or radial misalignment. A customer review's "35 mm minimum" is a research lead, not a dimensional specification from the manufacturer. Obtain the actual required shaft/plate/pulley dimensions if the fit cannot be established. Do not substitute motors silently or claim double-shear operation with an unengaged shaft. Confirm motor-to-plate screws and thread engagement; the CNC PDF does not specify these installation screws or a numeric pulley height.

1. Match the A and B top/bottom plates to PDF leaf 2/index 1. Starting with the illustrated right-hand drive, attach its two threaded pins to the lower plate with M3x8 flat-head screws (leaf 3/index 2).
2. On the two-level smooth-idler pin, stack bottom to top: **1 mm shim** → **F695 flange down** → **F695 flange up** → **1 mm shim** → **1 mm shim** → **F695 flange down** → **F695 flange up** → **1 mm shim** (leaf 4/index 3). On the one-level pin: **10 mm M5 sleeve** → **1 mm shim** → **F695 flange down** → **F695 flange up** → **1 mm shim** (leaf 5/index 4).
3. Insert three 20 mm M3 sleeves; bring the upper plate down squarely and hold it with the three M3x35/cup-shim fasteners (leaf 6/index 5). Secure the pin ends with the illustrated M3x8 flat-head screws (leaf 7/index 6).
4. **Before the next step:** the manufacturer warns that only a little force is needed on the support-bearing retainers. Seat the seventh F695 in the support recess and capture it with the two ultra-flat M3x4 screws as leaf 8/index 7 shows. Do not clamp its inner race or distort the bearing.
5. Build the other drive using leaf 10/index 9: its single bearing pair occupies the **lower** belt level. It is not a second identical stack. Rotate all idlers freely, then perform M4's motor/pulley fit check. Motor leads should have bend room and clear the moving belts.

Front-idler parts for the pair, after M1: two Part A plates, two Part B plates, two tensioning forks, two side pieces, two pins, four F695 bearings, four 1 mm M5 shims, two 10 mm M5 sleeves, four M3x5 flat-head screws, four M3x45 cap screws and eight M3x6 button-head screws, as PDF leaf 2/index 1 shows. Confirm its revision-dependent extrusion screws/T-nuts and clamp supply separately.

1. On each pin stack: **10 mm M5 sleeve** → **1 mm shim** → **F695 flange down** → **F695 flange up** → **1 mm shim** (front PDF leaf 3/index 2). Fit into the tensioning fork and retain with two M3x5 flat-head screws (leaf 4/index 3).
2. Place the fork between its Part A/Part B plates and fit the two M3x45 adjustment screws (leaf 6/index 5). Add the side piece with four M3x6 screws (leaf 7/index 6), following the PDF's medium-threadlocker recommendation for the indicated metal joints.
3. Assemble both idlers the same way; install the **left unit upside down** relative to the right as leaf 9/index 8 shows. Check free rotation across the intended adjustment range.
4. **Gate M5, blocks final gantry attachment:** confirm CNC-to-extrusion mounting screw lengths and slot-nut arrangement for the actual revision. Preserve stock extrusion alignment and flush-end references on pp.85–95, but do not transfer plastic hardware dimensions where the CNC drawings do not establish them. Dry-fit all four corner units with backers present; verify their upper/lower belt planes, clamp access and rail clearances together.

Completion check: both drive assemblies have the correct opposite single-idler level; both shaft supports genuinely engage; front idlers have the documented opposite orientation; every idler spins freely; no screw, backer or extrusion lip touches a belt plane.

Assemble the revision 1.1 live-idler XY joints and endstop support

Prerequisites: M1's bearing-retention conflict cleared, X rail/backer installed and Y rails restrained. **REPLACE** stock pp.96–100 (indices 95–99) and their printed XY joints with purchase **14**. Mounting hardware on pp.103–106 also becomes revision-specific.

Parts from the revision 1.1 PDF for one purchased left/right set: four machined plates; two press-fit toothed idler shafts; two separate stainless pins; eight F695 bearings; four 1 mm M5 shims; two 20 mm M3 sleeves; four 8 mm M5 sleeves; four springs; four M3x8 flat-head screws; two M3x30 flat-head screws; six M5x10 flat-head mounting screws; eight M3x8 button-head mounting screws; eight ultra-low bearing-retention screws whose length is unresolved by the conflicting PDF text. Keep each spring and sleeve with its associated stack. Do not press the purchased idlers off their shafts.

1. Match the **right lower plate** to XY PDF leaf 3/index 2. Once the retention-length gate is cleared, fit its flanged shaft-support bearing without overtightening the two retainers. Fix the separate threaded pin with M3x8 as leaf 4/index 3 shows.
2. On that smooth-idler pin, bottom to top: **1 mm shim** → **F695 flange down** → **F695 flange up** → **1 mm shim** → **8 mm M5 sleeve** → **spring** (leaf 5/index 4). On the adjacent rotating shaft axis fit **spring** → **8 mm M5 sleeve** → **press-fit toothed-idler shaft** (leaf 6/index 5), seated in the lower support bearing.
3. Insert the 20 mm M3 sleeve, position the top plate and start M3x30 without fully tightening (leaf 7/index 6). Align the top, secure the pin with M3x8, tighten M3x30, and fit the upper shaft-support bearing retainers (leaf 8/index 7). Observe the manufacturer's low-force warning.
4. Build the left joint using the **reversed vertical levels** in leaf 10/index 9, not by making an identical right joint. Both smooth and toothed idlers must turn freely after retaining the bearings.
5. Set aside the completed left and right joints for the gantry-extrusion steps immediately below. Mount them to the X extrusion and Y carriages only after M5 establishes the revision-specific screws. Use the PDF's listed hardware where confirmed; do not install stock M5x40 plastic-idler bolts. Leave alignment fasteners adjustable until the gantry is manually squared.

Gate M6, blocks endstop installation and XY homing. Identify the actual LDO X/Y endstop PCB, connector orientation and mounting-hole pattern. Vitalii's linked [supporting-parts archive](#) contains **STEP CAD only**, not printable STLs: `Endstop_PCB_Mount_heat_inserts.step`, `2x_PCB_Spacer.step`, `Endstop_Bump.step`, `Endstop_Bump_Deep.step`. No print orientation, fastener/insert specification or reuse license was found. Obtain/select a matching printable export and approved hardware after checking the actual PCB and its trigger position. The deeper bump is an unresolved alternative, not an additional mandatory part. STEP filenames establish what was supplied; no geometric STEP fit validation was possible with the available tools.

KEEP separate X and Y endstop functions for this build; **REPLACE** the stock printed PCB/pod mount procedure on pp.162–164 (indices 161–163) with the confirmed CNC-compatible support. The stock right-XY-joint's probe-wire channel and stock Hall magnet pockets are not present by assumption. Do not tape a PCB into the belt path. Cartographer changes Z sensing, not automatically X/Y switches.

The selected **X authority is the physical switch on the CNC carriage**, after T1 verifies its bracket, cable and trigger target. The selected **Y authority is the confirmed Vitalii-compatible Y switch/PCB support**, after M6 verifies its rear trigger geometry. If a combined LDO XY PCB is retained for Y, leave its X channel unused and omit that duplicate X configuration. Confirm the carriage X-switch bump on the right live joint and Y trigger at the rear before homing; the supplied STEP models do not establish those interfaces by themselves.

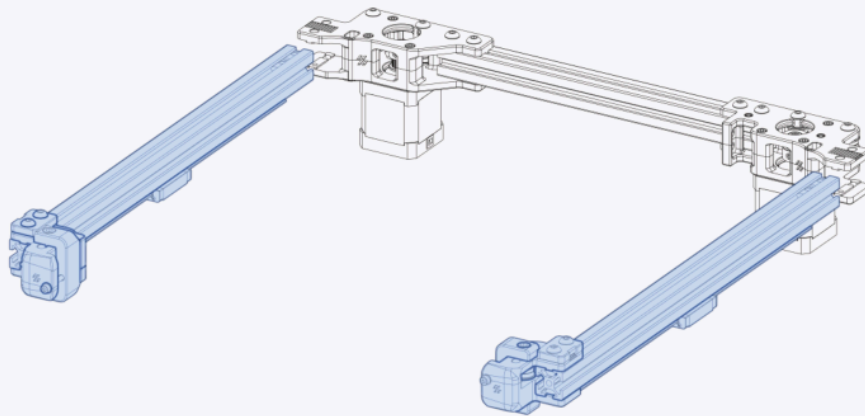
Completion check: both assembled joints turn freely; left/right belt levels differ as drawn; shaft bearings are retained without drag. Once mounted in the following gantry steps, the endstop support must clear the backer, belts and carriage through travel. Homing remains blocked until M6 is cleared and wiring/configuration agrees.

Join the gantry extrusions to the modified corners

Prerequisites: rails/backers prepared above, four CNC corners assembled, M4 and M5 cleared. **Parts:** two remaining **C** extrusions for the Y beams, one **E** rear crossmember, one **D** moving X beam, two MGN9 Y rail/carriages and one MGN12 X rail/carriage. These letter assignments follow the actual manual's p.13 reference, p.85 rear crossmember, p.88 Y beams and p.101 X beam. Match their lengths to the confirmed kit-size/batch cut list; the manual drawings do not provide a universal 350-kit cut list. Use the verified CNC mounting screws and mating slot nuts from M5, not a new copy of the stock plastic parts' hardware.

KEEP the outer printer frame's already installed M5x16 blind-joint screws from the preceding frame chapter. The gantry steps below use the corner plates and extrusion slot nuts, not a second set of frame blind joints. For comparison, the stock rear assembly on pp.85–87 uses eight M5 slot nuts/M5x10 button-head screws; stock pp.91–95 uses M5x16 in its gantry-corner and upper-clamp interfaces. Those stock dimensions are **not verified CNC replacement screw lengths**.

1. Lay the **E rear crossmember** between the two completed A/B drives, with **B at rear left and A at rear right when viewed from the front**, as p.83/index 82 identifies. Stage the appropriate CNC slot nuts before covering the extrusion ends. Seat the crossmember in the plate interfaces without twisting either corner, then start its M5-interface fasteners only after M5 confirms their actual specification. The views on pp.85–87 show this rear crossmember relationship.
2. Bring the two **C Y beams** forward from those rear corners, parallel to one another. Their MGN9 rails remain on the **undersides**, with backers on top. Stage the extrusion nuts needed for both CNC-corner attachment and the eventual Z belt captures before access closes; pp.89–90 show why both sides of the stock slots need early access. Determine the CNC count/position from M5/M8 rather than copying a stock plastic clamp's nut order.
3. Fit the front-idler units to the front ends of the C beams: front-left inverted relative to front-right, as the manufacturer front-idler PDF specifies. Seat all four C-beam ends at their confirmed CNC datum surfaces. Compare the resulting rear-E / two-C U-shaped assembly to the stock layout on p.94. Check equal diagonals and parallel Y rails; leave alignment fasteners adjustable. Install upper clamps only after M8 is cleared.



4. Prepare the moving **D X beam** with its MGN12 rail facing the printer front, its titanium backer behind and its carriage restrained. Fit the completed left and right live XY joints to the D-beam ends as p.103's arrangement shows. The revision 1.1 XY PDF lists six M5x10 flat-head and eight M3x8 button-head mounting screws; their assignment to the actual D beam and actual Y carriages must be established at M5. **REPLACE**, rather than copy, the stock p.104 M5x10/M5x16/M5x30 button-head stack and washers.
5. Tilt the completed D-beam/live-joint assembly into the U-shaped gantry and align each joint to its Y carriage, following the access concept on p.106/index 105. Support it while engaging the confirmed CNC-to-carriage screws. Do not use the stock "two bolts only" instruction or M3x16 length for the CNC part: the actual PCB support must be reconciled at M6. Move X slowly through Y travel, checking parallelism and that the joints do not lift or distort either carriage. Final squaring happens on supported suspension before belt tension is applied.

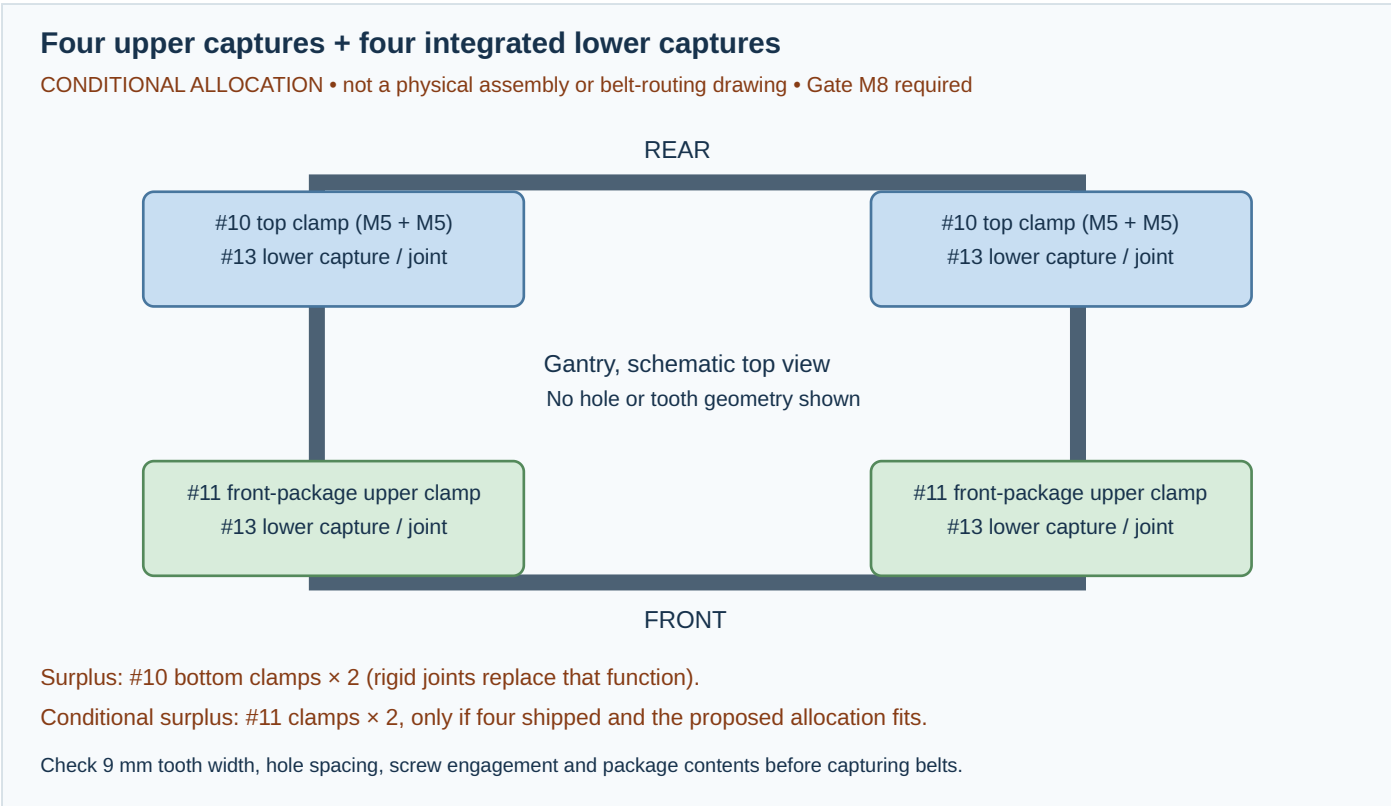
Completion check: one E rear beam, two C side beams and one moving D X beam are present; A is rear right, B rear left; both Y rails face downward and the X rail faces forward; corners meet their verified mounting datums; rail carriages move without a forced fit; the assembled gantry is square before tightening the remaining alignment fasteners.

Allocate the clamps and install rigid Z joints before suspension

REPLACE stock printed pp.110–116 (indices 109–115) with purchase **13**. **OMIT** stock upper/lower Z-joint plastic parts, spherical joint stack and separate lower belt clips. [Vitalii's rigid-joint page](#) explicitly says the complete stock joint and **bottom belt clamp** are replaced. A lower purchased CNC clamp must not be stacked onto that integrated lower function.

Parts: one purchased rigid-joint set is four joints, sixteen M3x8, four M3x14 and four M5x14 screws according to the listing. The actual head types and screw-to-hole assignments need revision-matched installation evidence; the store BOM alone is insufficient. Four equal-height supports and temporary straps are essential while the gantry is unpowered.

Gate M7, blocks rigid-joint fastening and cutting Z belts. Obtain the actual product installation drawing or supplied instructions confirming: rail carriage compatibility, orientation, which supplied screws mount each interface, integrated belt-capture geometry, allowed adjustment and fastener engagement. Confirm that the machined corners can be squared without forcing the four rail carriages into misalignment. No rigid-joint installation PDF/CAD was linked on the retrieved product page.

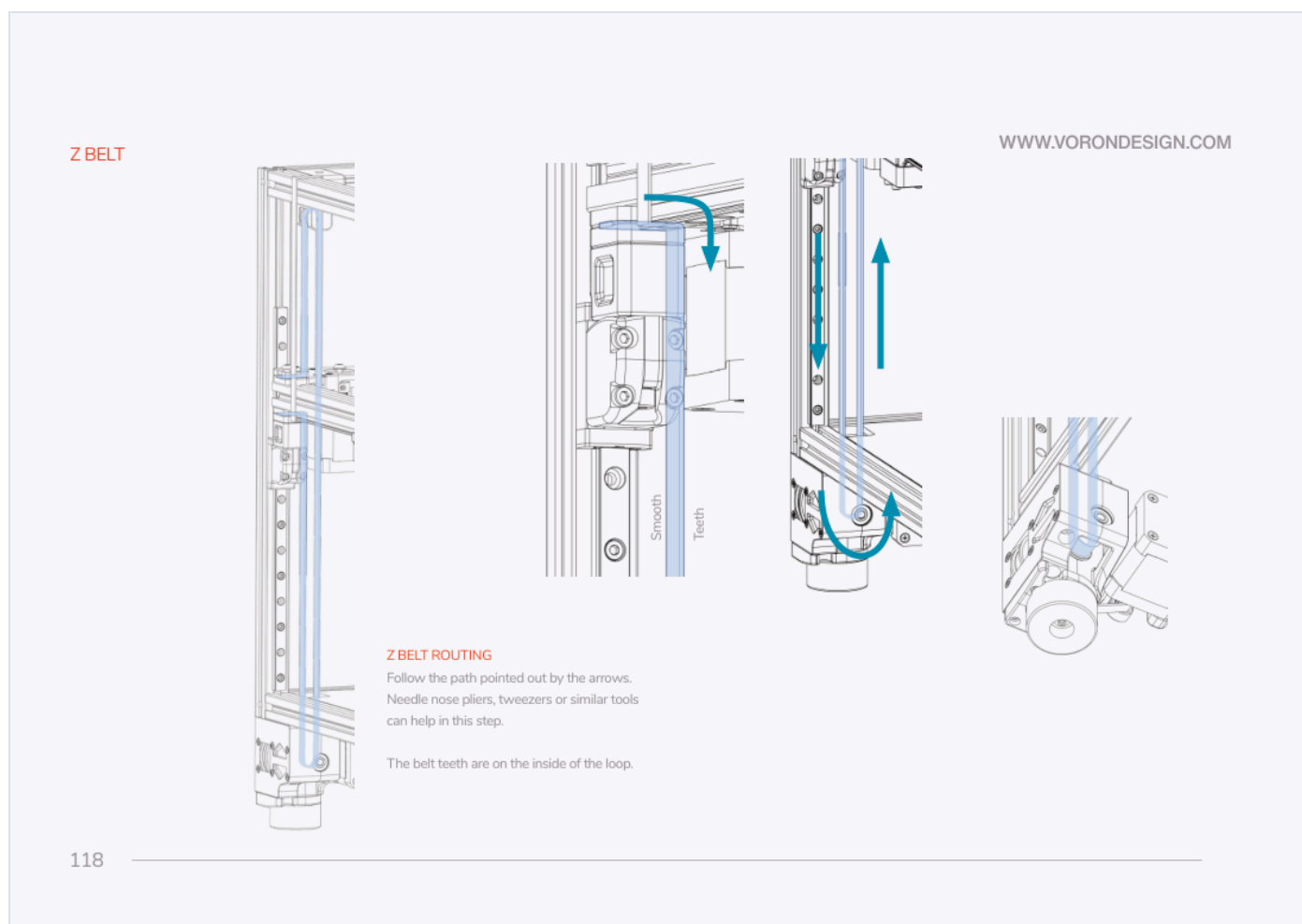


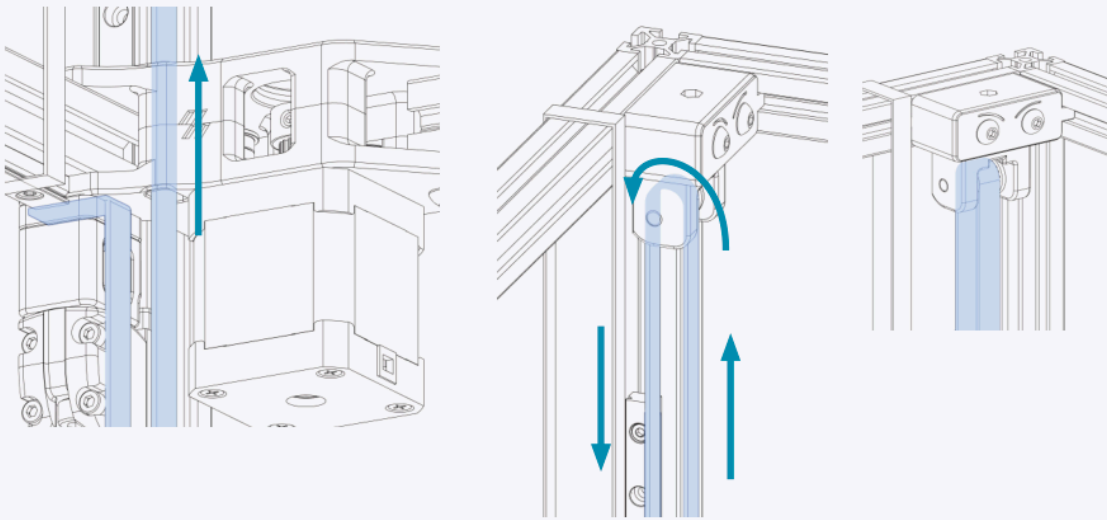
After M7/M8 are cleared, seat the four rigid joints against their matched Z carriages with their approved screws, align the gantry on equal-height supports, and capture one end of each nine-millimetre Z belt in its integrated lower interface. Keep belt teeth engaged in the confirmed serrations. Hoist the gantry past the uprights with help, using straps as stock p.114 (index 113) recommends. Finalize the joint/gantry interfaces only while the gantry is square and supported. Preserve access to the upper clamps.

Completion check: there are exactly four lower and four upper Z capture locations; no redundant bottom clamp; all four carriages sit naturally on their rails; all joints are secure without a forced twist. Surplus pieces are bagged and labelled by purchase ID.

Route the Z belts, square by hand, then route A/B belts

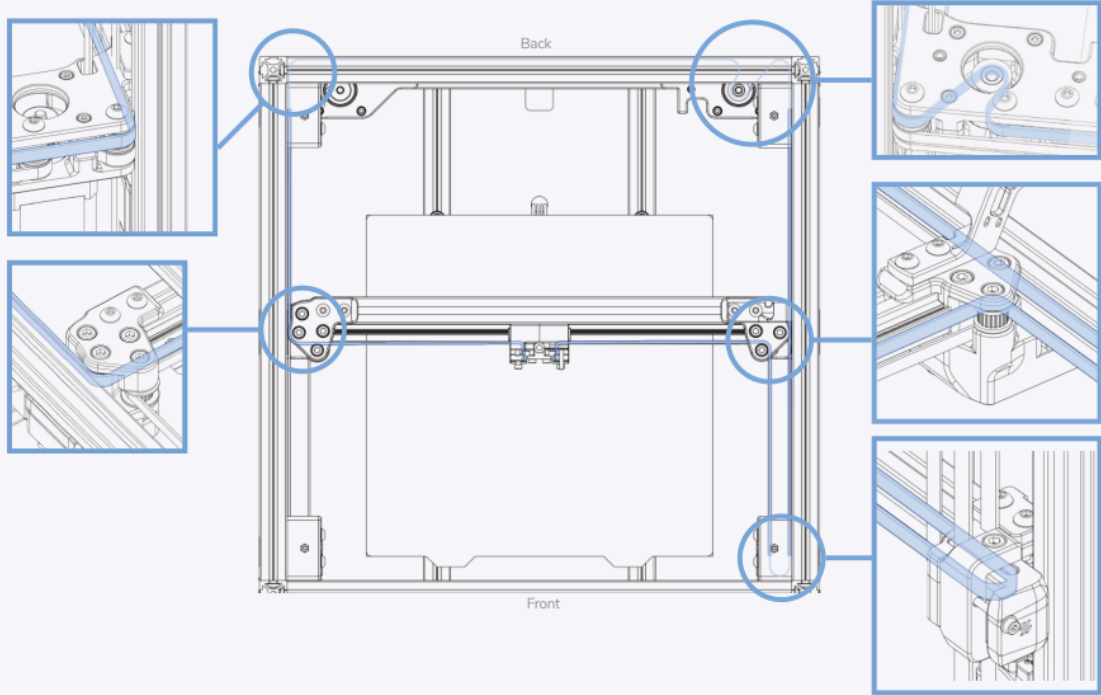
Prerequisites: M1–M5, M7 and M8 cleared for the affected assembly; gantry securely supported. M6 continues to block endstop mounting/homing but does not prevent independent bare-gantry preparation. Before the A/B belting step, complete the **bare CNC carriage identification, rail mounting and belt-capture preparation** in [the toolhead chapter](#), with its six-millimetre variant confirmed. The Rapido, extruder and ducts can be fitted later; do not build a stock carriage or require a complete hotend simply to thread the belts. **KEEP** the functional Z loops in stock pp.118–121 (indices 117–120) and stacked CoreXY topology pp.125–127 / pp.132–138 (indices 124–126 / 131–137). These are stock reference drawings, not pictures of the modified CNC assembly.



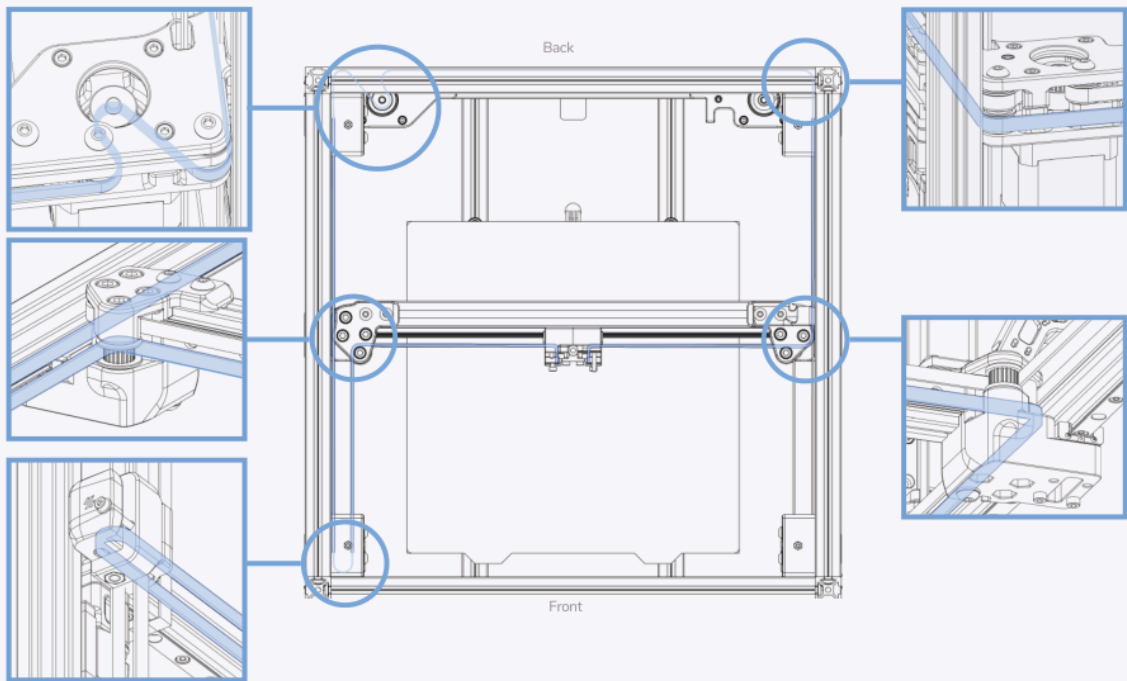


1. Route each nine-millimetre Z belt from its confirmed lower capture around the lower Z-drive pulley and upper purchased tensioner idler, then back to its confirmed upper capture. Keep **teeth inward** around the toothed elements, no twists, and the two vertical runs parallel. The rendered diagrams on pp.118–119 show the loop. Check alignment before securing the upper end; retain service slack, fold excess clear of travel and secure it as p.120 depicts. Repeat four times.
2. Bring all four supports to equal height and manually square the gantry. Stock p.122 (index 121) uses the rear drive stops to square X relative to Y; CNC housings must first provide matched contact geometry. Check with a square and diagonals as well. Adjust mounting alignment rather than using rigid joints or high belt tension to force the frame square.
3. **Gate M9, blocks powered Z motion and quad gantry leveling.** With the gantry retained against falling and motors disconnected from powered control, slowly inspect its usable Z travel by supporting it at all four corners. Move without forcing a corner to lead. Stop for binding, carriage lift, unequal belt tracking or a twisted rail. Recheck at low, middle and high positions. Re-align rails/joints and recheck; a QGL macro cannot repair binding from rigid joints. Keep temporary support until the belts safely support the gantry.
4. Set the front CNC XY tensioners within their confirmed adjustment range. **OMIT** stock p.128's printed-adjuster four-turn preparation. Capture the first ends of the two six-millimetre A/B belts using the **confirmed CNC carriage** procedure; **REPLACE** stock pp.129–131 and pp.139–141's printed carriage hardware with that procedure. Do not assemble stock printed belt-capture halves first.
5. Trace belt **A** through the full upper/lower plane shown on pp.126 and 132–135, and belt **B** through its separate plane on pp.127 and 136–138. At every smooth F695 pair the smooth back contacts the idler; the toothed surface contacts the motor pulley/live toothed idler. Follow both paths with a finger or pointer before cutting. The manual's topology avoids a crossed-belt arrangement; do not add a crossing to imitate another CoreXY design.

OVERVIEW - A BELT



OVERVIEW - B BELT



6. Do not remove a CNC structural pin or bearing retainer to follow the stock tip about removing M3x40. Use tweezers and leave accessible covers off until belting is complete. Verify belt-plane heights at A/B pulleys, front idlers, live joints and carriage captures **as one system**. No numeric CNC pulley height was documented. Cut only after adequate carriage engagement and tensioner reserve are established; equal stock path lengths do not prove identical CNC path lengths without this check.
7. Secure carriage captures with their verified hardware and bring A/B tension into balance without bearing drag. Retain excess until final travel is confirmed. Follow p.142's inspection purpose: neither belt may rub a housing, backer, screw or extrusion.
8. **Gate M10, blocks XY powered motion and setting software travel limits.** By hand, sweep the bare carriage to all four XY corners with gantry low, middle and high; then repeat with the completed UHF toolhead, bed, probe and routed cable installed. Check front-idler/toolhead clearance, live-joint/endstop bump engagement, rear backer and PCB clearance, shaft-support clearances, nine-millimetre Z runs and top adjuster access. Record the physically safe travel envelope; "350 class" does not establish usable 350 × 350 travel after these modifications.

Completion check: all four Z loops track without twist, four upper/lower captures are correctly assigned, both A/B paths stay in their separate planes, all axes move smoothly by hand, and every interference/travel boundary is recorded for commissioning. **Do not issue motion commands while using this documentation check.**

Evidence and readiness of the combined motion system

Interaction	Status for this actual build	Why
6 mm XY joints / 6 mm A/B PDF / 9 mm Z tensioners	Conditionally compatible	Documented widths agree with separate stock paths; actual kit belts and carriage variant still require M1.
A/B mounts / purchased LDO motor shafts	Unresolved	Actual shaft length, support engagement and CNC motor-fastener specification absent; M4.
A/B + front + live XY + CNC carriage + backers	Unresolved	Subassemblies have official PDFs, but no evidence validates this exact combined belt-plane and full-travel geometry; M5/M10.
Rigid joints / separate lower CNC clamps	Incompatible at the same lower interface	Rigid joints replace the lower clamp; #10 bottom pair is surplus.
Proposed two rear + two front upper clamps	Conditionally compatible	Exact allocation preserves #10 top pair but hole/tooth fit and #11 package contents need M8.
Backers / front MGN12 X + underside MGN9 Y	Conditionally compatible	Correct documented compensating sides; ordered length, nuts and obstruction clearance remain M3.
Rigid joints / four Z rails	Unresolved	Product advertises printed/CNC gantry compatibility; actual orientation, fastener assignment and freedom through travel need M7/M9.
Live XY / actual LDO endstop board	Unresolved	Supporting archive offers CAD candidates, but actual board/printed exports/hardware and physical trigger position need M6.

Manufacturer subassembly diagrams were rendered and visually inspected in research. Their reuse terms were not established, so this guide links to them and uses clearly labelled original schematics. Static research and documentation checks do not validate physical assembly or prove the combined machine safe to operate.

05 · UltraFlat bed, kinematic supports and Wings

Build the purchased bed system once, as a unit. **REPLACE** the stock plate and four screw/spacer supports in the V2.4R2 manual, printed pp. 52–60 (PDF indices 51–59), with purchases **06**, **07** and **08**. **KEEP** independent protective earth, thermal protection and cable passthrough requirements. **OMIT** the stock M4 thumb-nut spacers and four plate-mount screws. **ADD** three matched-height kinematic contacts and the two-piece Wings arrangement after its mounting map is verified.

This chapter has a usable mechanical preparation path, with deliberate stops at unresolved hardware interfaces. The purchased bed variant and the Wings support map have not been physically verified. The frame and unpowered gantry may proceed independently while these gates remain open.

Prerequisites, parts and tools

For the early B1–B3 dry fit in chapter 03, the frame is square and both bed support extrusions are **loosely positioned**; leave the deck unfastened. A gantry or toolhead is not required for that support/frame/deck mapping. For later bed installation, preserve deck access and support any installed gantry away from the bed area. Secure the bed support extrusions only after the three-point mounting map is verified. Mains and DC power are disconnected. Keep the purchased stock LDO plate intact as surplus if it arrives with bonded components.

Part	Purchased quantity	Planned installed quantity / disposition
06 MRW Voron 350 UltraFlat plate	1 plate	1; exact standard/magbed/heater variant unverified
07 matched-height kinematic kit	1 set	3 balls and 3 different seats; verify package below
08 MRW Voron Wings	Reported “2x”, unit unresolved	Plan 2 individual Wings , equivalent to 1 current retail set; actual surplus remains unknown
Wings-to-frame hardware	Current retail set lists 4 M4 bolts and 4 M4 T-nuts for 2020	4 of each if the delivered set agrees; bolt length is not stated on the product page
Heater, bed sensor, thermal fuse, suitable build surface	Variants/ownership unverified	Required; match the final plate and mains voltage before acquisition or installation
Dedicated PE lead, terminal, bed cable restraint, protective sleeve/grommet	Kit inclusion and fit unverified	Required; inventory separately from purchases
Additional support for the third kinematic seat	Not reported purchased	Confirm whether Wings layout uses the ~130 mm center extrusion explicitly excluded from the kinematic kit

Use metric hex drivers matched to the actual fasteners, calipers, ruler, square, lint-free cloth, the adhesive maker's specified cleaner, a soft nonmagnetic bed support, appropriate crimp tools and a multimeter. A low-range torque tool is needed only if the actual manufacturers specify a torque; none is invented here. Do not drill the purchased plate to make an assumed mounting map fit.

B1 · Identify the plate before bonding anything

Verification gate B1 — blocks bed surface installation and scan commissioning. Read the label and inspect the top, underside and mounting holes. Record the exact plate variant, installed magnet arrangement, heater option and sensor. The current MRW350 listing describes a **355 × 355 × 9 mm** plate, with standard and embedded-magnet versions; these are product specifications, not measurements of the delivered item. A 350 bed does not establish the kit's extrusion size.

If this is the embedded-magnet Magbed, Cartographer scan meshing is manufacturer-incompatible. Cartographer explicitly warns that localized embedded magnets distort a mesh; it accepts a uniform magnetic sheet. Stop that probing combination and obtain a documented resolution for the purchased hardware. Do not treat Touch as an established workaround, remove magnets, replace the bed, or change probes silently. If it is a plain plate with a continuous suitable sheet, that combination is **conditionally compatible**, pending physical checks and calibration. [MRW350](#), [Cartographer FAQ](#).

1. Photograph only technical labels and hole layouts for local verification. Record measurements in the manifest without publishing order or address material.
2. Mark the working top and underside. Identify the **Trident side-mount holes used by Wings**, the three ball attachment holes, PE attachment and sensor/fuse attachment points from the delivered drawing. The MRW listing identifies M5×0.8 sensor/fuse holes; the stock manual's M3 fuse/PE hardware cannot be copied into those holes.
3. Lay the unmodified LDO bed aside. If its heater or magnetic sheet is already bonded, classify it as surplus with bonded components. A new compatible heater/surface is **required-but-not-confirmed-owned** until the inventory proves otherwise; this guide does not instruct adhesive removal and reuse.

Completion check: the plate's actual variant, hole map, heater, surface and sensor are individually identified; B1 is closed only with that evidence.

B2 · Inventory and protect the kinematic parts

Manufacturer warning before unpacking: keep magnets separated. Letting them snap together can shatter them. Keep fingers and nearby ferrous tools clear. The manufacturer PDF's exploded figures were inspected; they show that the **bases attach before the magnets and covers**. [Matched-height installation PDF, first and second unnumbered pages](#).

The linked PDF, created April 8, 2022, lists the following. Count the delivered parts against this list; the product page differs on M3 screw length and M2 head style.

Kinematic set contents in the PDF	Pieces per set
Flat-magnet base / chamfered-magnet base / pin base	1 / 1 / 1
Internally threaded balls / pins / magnet covers	3 / 2 / 2
Flat magnets / chamfered magnet, rated 150°C in the source	2 / 1

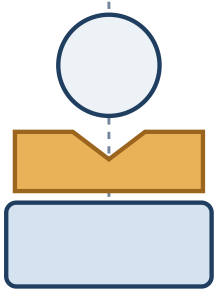
Kinematic set contents in the PDF	Pieces per set
M3×10 ball bolts	3
M3×6 BHCS / M3×8 BHCS	8 / 2
M4 T-nuts / M4×8 SHCS	3 / 3
Hammerhead M5 T-nut / M2×4 cap screws	1 / 2

Verification gate B2 — blocks seat assembly. Match the supplied paper instructions to the delivered set. Resolve any M3 length or M2 head-style mismatch before fastening; use screw engagement and recess fit to confirm the actual variant. Count Wings as pieces **and** sets. A current retail Wings set contains two pieces. “2x” could mean two pieces or four pieces: do not mark two leftovers until packaging proves two full sets were delivered.

ADD · Three different contacts, one matched-height set

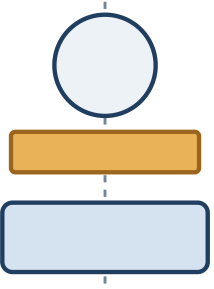
Original schematic · no printer coordinates or fastener dimensions implied

Chamfered seat



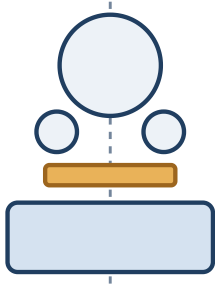
15 mm base in source
Chamfered magnet + cover

Flat seat



12 mm base in source
Flat magnet + cover

Pin seat



Two pins + flat magnet
Confirm location-block direction

B2 / B3: bases attach BEFORE magnets. Confirm Wings support map before assigning contacts.

The original schematic explains contact types, not a Wings installation drawing. The source's stock V2.4 diagram places pin, chamfered and flat seats differently from its Trident diagram. Wings changes the attachment holes, so the stock V2.4 support coordinates are not a substitute for the next gate.

B3 · Dry-fit Wings and confirm the three-point map

Verification gate B3 — blocks final frame attachments and ball installation. Obtain the delivered Wings size, its instructions or manufacturer confirmation of the three seat positions, the pin-block orientation, the third seat attachment, and seat-to-Wing screw/thread engagement for this 350-class frame. Check the actual Trident hole positions in this plate against both outriggers. The product photographs show a chamfered seat on one Wing and a pin seat on the other, but do not provide a dimensioned installation map or fastener lengths. They cannot establish which direction is the machine's front. [Wings product and installation photographs](#).

1. Place the two Wings on the support extrusions with their attachment faces seated squarely; point the arms outward beneath the matching side mounting holes. Keep the four supplied M4 bolts loose during this **dry fit**. Do not add a fourth kinematic contact.
2. Place the three **different** seat bases in the confirmed layout. The chamfered assembly uses the source's 15 mm base; the simple flat assembly uses its 12 mm base. The pin assembly uses two pins, a flat magnet and its location blocks. Do not exchange base heights to correct alignment.
3. Identify whether the third seat requires the excluded center extrusion, its end attachments or another delivered manufacturer-supported feature. Add the actual requirement to the missing-parts list. Do not print the temporary extrusion suggested by the old product page as an assumed permanent heated structural member.
4. Rest the plate on temporary supports, not bare magnets. Check all three ball locations, heater coverage, wire exits and deck access together. Check that the installed bed can expand in plane through the designed contacts; cable restraint must not tether it rigidly to the frame.
5. The stock manual printed p. 60 (PDF index 59) gives a **stock** 38 mm front setback. Use it only as a reference comparison. Record the support position and proposed plate setback from the verified mounting map. Confirm that the deck/DIN attachment holes agree before returning to chapter 03 to fasten them. Finished gantry, belt and nozzle/probe clearance checks occur later at B5/M10; the early dry fit does not establish the final usable envelope.

The Wings listing warns that 350-size arms can twist weaker support extrusions under edge loading and discusses a crossbrace/1515 reinforcement. That is a check for this assembly, **not proof that an unpurchased brace is required**. Record deflection during later controlled probing; do not add an unrelated bed modification by default.

Early completion check: three support positions and attachments are recorded, frame/Wing/plate holes align without force, the deck/DIN attachment map agrees, and additional support hardware is present. This closes the support-mapping prerequisite for chapter 03. Belt and toolhead clearances remain open under B5/M10 until those assemblies exist.

Install the supports and balls

Proceed only after B2 and B3 close. Use the delivered manufacturer's diagram for screw locations; no torque or unspecified bolt length is supplied by this guide.

1. Fasten the confirmed seat bases to their frame/Wing attachments **before inserting magnets**. Confirm each T-nut rotates/seats correctly or each threaded Wing hole has the confirmed engagement. Keep all three bases in their matched orientation.
2. Assemble the chamfered magnet and cover on the 15 mm base; the flat magnet and cover on the 12 mm base; the two pins, location blocks and remaining flat magnet on the pin base. Keep the magnets apart until retained. The PDF shows four M3×6 cover screws per covered seat and the pin assembly's two M3×8 and two M2×4 fasteners; compare actual instructions before using that older hardware callout.
3. Fit one internally threaded ball at each of the three confirmed plate holes, with its attachment screw seated in the plate's top recess. In the PDF these are three M3×10 bolts. Verify that no head stands above the printing plane and that screws do not bottom before retaining the balls.
4. Lower the supported plate onto all three contacts without impact. Each ball must land on its assigned contact; the plate must not rest on the Wings, extrusions, cables or a fourth unintended point. The three contact types provide location while allowing thermal expansion; do not add stock "tighten one / leave three loose" screws across them.

Completion check: all three contacts seat and the plate is stable, no magnet is chipped, covers and pins are retained, and the three planned balls and seats are accounted for. Actual unused T-nuts/bolts from the kit remain documented surplus rather than hidden in the BOM.

B4 · Heater, sensor, fuse, earth and surface

Verification gate B4 — blocks adhesive application and heater commissioning. Identify heater manufacturer, voltage, rated power, dimensions, adhesive, sensor type and lead insulation; establish a fuse compatible with that heater and wiring; identify a suitable thermal-fuse temperature/rating and attachment method. Confirm PE terminal/thread/hardware for the MRW plate. The stock manual's thermal-fuse and PE drawings are **principles**, not authorization to reuse its fasteners on an M5-tapped plate. Electrical approval belongs to the wiring chapter; software cannot certify mains safety.

1. Dry-position the heater on the underside with all ball, PE and fuse holes accessible. Keep leads clear of all contact seats and frame attachments. Apply only after its own installation instructions and B4 establish bonding preparation, position and cure. Do not cover the attachment features or trap a wire beneath a support.
2. Install the sensor in the heater or plate location documented for the delivered combination. Mount the thermal fuse with the intended thermal contact and electrical insulation, in series with the bed-heater circuit. Preserve an independent protective-earth bond to the metal plate; a magnetic ball contact is not a PE conductor.
3. Fit a dedicated PE lead and appropriate ring terminal at the verified attachment; use bare-metal contact and retention according to the actual electrical hardware. The stock manual p. 57 (PDF index 56) calls for **1 mm²/AWG18 or larger PE**. Treat that as a minimum reference while checking local requirements and the actual circuit. Conduct continuity testing in the pre-power checklist.
4. For a plain plate, fit only the confirmed suitable continuous magnetic sheet/build surface, after cleaning and adhesive preparation. For an embedded-magnet plate, preserve the manufacturer's warning that magnet screws can shatter magnets when overtightened. Do not remove or retighten them as a routine assembly step.
5. Route heater, sensor and PE leads through the deck opening with edge protection and restraint. Provide a gentle free section from the plate to the stationary restraint so it can expand; no tension may reach the sensor/fuse or ball contacts. Keep mains/heater wiring appropriately isolated from signal wiring.

Completion check: correct heater/sensor labels are recorded; fuse and PE attachments are independent; all wires have mechanical protection; adhesive cure is complete; B1 compatibility remains satisfied. Do not energize the heater here.

B5 · Establish the actual bed and travel envelope

After completing the mechanical bed installation and cable protection above, continue to [toolhead assembly](#), then return here for B5 and chapter 04's M10 full-travel check **before any powered motion**. Record B5 as open while the toolhead is absent; do not hold up independent toolhead assembly.

Verification gate B5 — blocks powered Z movement and final travel limits. Measure the actual top-of-bed height with the intended removable print surface installed and the finished UHF toolhead in place. Combined matched-height seats, 9 mm nominal plate, Wings and UHF nozzle change the envelope. A purchased “350”

plate does not guarantee 350 mm X/Y/Z travel.

With motors disconnected and the gantry safely supported, inspect reachable corners, Z belts, bed leads, carriage, probe, nozzle and lower gantry components throughout the conservative mechanical envelope. Move only as the unpowered gantry procedure permits. Record nozzle and probe access to the bed edges, the lowest safe gantry position and lost top/bottom Z clearance. Keep measured machine limits unset until later commissioning verifies them.

Bed installation is complete when the actual plate variant, installed Wings count, three-contact map and heater/surface disposition are recorded, and the assembly is secure and free to expand. The final envelope check is complete only after returning from chapter 06 with the finished toolhead and recording the measured bed height and B5/M10 clearances. Every unresolved item retains its gate.

06 · Rapido 2 Fiber UHF, StealthBurner and Cartographer

KEEP the separate StealthBurner/Clockwork 2 assembly sequence when the delivered extruder hardware confirms that baseline. **REPLACE** the stock plastic X carriage with purchase **04**, the stock hotend/cartridge with purchase **03** and a validated UHF cartridge, and the stock Omron probe with purchase **05**. **OMIT** TAP, Klicky, Omron mounting and their macros from this path. **ADD** the Cartographer mounting-height, coil-clearance and cable checks below.

The V2.4R2 manual printed pp. 146–147 (PDF indices 145–146) directs toolhead assembly to the separate manual. This chapter follows [StealthBurner commit 8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b](#), [assembly manual version 2023-07-07](#). Its printed page equals zero-based PDF index plus one. UHF-specific interfaces are gated rather than represented as tested.

Prerequisites, parts and tools

For the **bare-carriage procedure**, begin with the X rail and backer installed and the gantry's belt routes accessible. Perform that procedure when the gantry chapter calls for it, before the bed or complete toolhead exists. For **final toolhead/probe installation and the full travel envelope**, the gantry and identified bed geometry must be available. Power is disconnected throughout. No printed stock hotend cartridge is assembled for later replacement.

Item	Quantity / disposition
03 Rapido 2 Fiber / 2F, UHF/PT1000	1 hotend; exact hardware revision, voltage, heater rating, sensor and nozzle remain to verify
04 Cartographer CNC mount	1 carriage, 2 belt clamps/bolts, 2 M3×6 BHCS for probe listed in current package; ordered belt width and extenders unknown
05 Cartographer V4 AIO Standard	1 probe; confirm complete assembly, revision and interface
Clockwork 2 NEMA14 motor and BMG-type drive/idler/tension hardware	1 set from the kit if verified ; do not assume kit receipt established it
CW2 printed main body, motor plate, guidler A/B, latch, latch shuttle	1 each; exact filenames below
UHF front and rear cartridge	1 each conditional on T2; no HF substitute
StealthBurner main body, 4010 heatsink fan, 5015 blower	1 each, voltage/version verified before connection
PTFE filament guide	1 cut piece, measured against the finished CW2/cartridge
Nitehawk board mounts, cable door and strain relief	Actual SB or SB V2-specific parts; resolve in wiring chapter
UHF probe extenders and appropriate longer fasteners	Conditional requirement after measurement; not confirmed purchased

Tools: soldering iron/heatset tip, suitable metric hex drivers, calipers, flush cutters/file for built-in supports and fan preparation, light bearing grease per Voron sourcing guidance, and a probe-height gauge. For any hotend support-screw disturbance, have the manufacturer's actual torque tool and instructions first. Hot-tightening is deferred until electrical and heater gates close.

T1 · Confirm the extruder, carriage and endstop interfaces

Verification gate T1 — blocks final printed parts and rail/belt fastening. Identify the delivered extruder motor/gears, X rail/block, CNC carriage revision and belt-width variant. Purchase 14 is documented as **6 mm XY**; the carriage must capture that same belt width. The listing supports the 6 mm carriage on MGN9 or MGN12, while its 9 mm carriage is incompatible with MGN9. Neither fact proves this delivery. Confirm the conventional Voron belt path with all Vitalii subassemblies; do not reverse it to resemble another gantry. [CNC product specifications](#).

For the bare-carriage operation, close the carriage/rail/belt portion of T1 independently; extruder hardware identification is required before the later CW2 core, rather than before this rail attachment.

The [CNC installation figures](#) were visually inspected. They show **four M3×8 SHCS** to the rail, **one M3×6 BHCS per belt clamp**, **two M2×10 BHCS/SHCS** for the X switch and **two M3×6 BHCS** for a standard probe without extenders. The text warns that original carriage versions instead required **M3×8 BHCS at the rail**. Match the delivered countersinks/counterbores before selecting head style or length. Do not transfer stock plastic-carriage M3×30 bolts or printed clamp screws to the CNC body.

Bare CNC carriage and belt capture — perform during the gantry chapter

Parts for the illustrated modern revision: **1 CNC carriage**, **4 M3×8 SHCS**, **2 supplied belt clamps** and **2 M3×6 BHCS**, the verified X rail block and the two routed 6 mm XY belts. The switch, probe, extenders and full toolhead are fitted later. Use hex drivers and calipers; no generic screw torque is specified.

1. Support the gantry and position the rail block away from the rail ends. Clean the mating faces and present the CNC carriage with its two probe legs downward and its toolhead attachment bosses facing forward, matching the official exploded rail drawing.
2. Start the **four verified rail screws** by hand. For the modern illustrated carriage these are M3×8 SHCS; use the early BHCS revision only when identified. Verify heads clear both belt paths, all screws engage the block without bottoming, and the carriage seats flush without rocking. The manufacturer suggests suitable threadlocker; keep it away from rail lubrication and use its own material/retention instructions. Retain evenly without forcing a misaligned part.
3. Route the two belts through the carriage captures as shown in the official drawing and the gantry routing procedure. Preserve their separate upper/lower planes and the normal Voron return directions. Seat each belt's toothed return fully in its matching machined capture; no tail may sit under a rail screw or foul the opposite belt.
4. Fit **one supplied CNC clamp per capture**, starting its confirmed clamp screw by hand. The modern drawing uses one M3×6 BHCS per clamp. Confirm each clamp lies flat and retains the belt without its screw bottoming. Do not substitute stock printed belt clamps or their longer screws. Leave tail length for the gantry procedure's tension/alignment checks before trimming.
5. Move the supported carriage gently by hand across the conservative accessible X range and inspect belt engagement, rail-screw head clearance, tooth alignment and both belt planes. Stop for twisting, rubbing, a slipping capture or binding. Final belt tension and complete XY-envelope checks remain in the gantry chapter.

Completion check: the bare CNC body is flush on the rail block, four correct rail screws are retained, two separate belt captures hold their tails, and manual X travel has no hardware or belt interference. Record actual carriage revision, belt width and screw head types. Later attach the physical X switch with its two confirmed M2×10 fasteners before closing wiring access.

KEEP a physical X switch on the carriage when its provision matches the source figure; it is separate from Z probing. Confirm Y switch and any Vitalii replacement mounting in the gantry chapter. **OMIT the stock nozzle Z-pin and Omron probe from the planned Cartographer virtual Z-homing path**; that omission becomes final only when the software generation and safe homing workflow in the software chapter are verified. It does not create permission for unconfigured homing.

T2 · Select an actual UHF cartridge before printing it

Verification gate T2 — blocks hotend cartridge printing and assembly. The pinned official Voron phaetus_rapido_v2/README.md explicitly describes a **High Flow** mount. Its front/rear files are not proof of UHF compatibility. The purchased Fiber UHF must retain its UHF length, nozzle and cooling arrangement.

Phaetus's exact [Rapido-2F source at commit 7401934e28d56844ba4f311e9d588b1e4d0510dd](#) contains Product Adaptor Models/voron 2.4-Rapido 2.zip, including voron 2.4-Rapido 2/uhf前.stl and voron 2.4-Rapido 2/uhf 后.stl (front and rear). These are a manufacturer-supplied **candidate**, not a verified combined CW2/CNC/V4 installation. Confirm with the actual 2F CAD and delivered nozzle that they align the filament inlet, fan airflow, nozzle plane, CW2 attachments, SB shell and sensor keep-out. Obtain the exact supported combination or record a physical dry-fit comparison before using them. They are linked, not redistributed: the Phaetus README limits models to personal use.

A binary-STL comparison of the candidate archive finds its UHF front extends approximately **8.49 mm** farther downward than its HF front in the archive's own coordinates; its two rear meshes have the same bounding dimensions, with different hashes. This explains why a UHF-specific front is needed. It does **not** establish cooling alignment or fit with the purchased nozzle. The CNC listing's **8.5 mm probe extenders lower the probe**; they do not turn an HF duct into a UHF cooling duct.

Exact printed path in pinned StealthBurner repo	Qty	Status
STLs/Clockwork2/Direct_Drive/main_body.stl	1	KEEP if CW2 confirmed
STLs/Clockwork2/Direct_Drive/motor_plate.stl	1	KEEP if CW2 confirmed
STLs/Clockwork2/Direct_Drive/[a]_guidler_a.stl / [a]_guidler_b.stl	1 each	KEEP if CW2 confirmed
STLs/Clockwork2/Direct_Drive/[a]_latch.stl / [a]_latch_shuttle.stl	1 each	KEEP if CW2 confirmed
STLs/Stealthburner/[a]_stealthburner_main_body.stl	1	Conditional until the UHF cartridge/shell fan paths clear
STLs/Clockwork2/cable_door_for_pcb.stl, [a]_pcb_spacer.stl, chain_anchor_2hole.stl or chain_anchor_3hole.stl	Variant-dependent	Unresolved: Nitehawk revision and umbilical mounting may replace these

Exact printed path in pinned StealthBurner repo	Qty	Status
STLs/Stealthburner/Printheads/phaetus_rapido_v2/stealthburner_printhead_rapido_v2_front.stl and stealthburner_printhead_rapido_v2_rear_cw2.stl	0 in main path	OMIT as unverified HF candidates
STLs/Stealthburner/[o]_stealthburner_LED_carrier.stl, [c]_stealthburner_LED_diffuser.stl, [o]_stealthburner_LED_diffuser_mask.stl	1 each if installed	Optional lighting / verify kit inclusion

Voron print guidance is ABS, 0.2 mm layers, forced 0.4 mm extrusion width, 40% infill, four walls and five top/bottom layers (SB p. 4 / PDF index 3). Use the documented settings for applicable functional parts; LEDs require the indicated opaque/translucent material. Do not label unverified LDO printed-part inclusion as owned. Archive filenames have legacy character encoding: if an extractor produces garbled names, identify the UHF members by the source archive and record the extraction mapping rather than rename an HF file.

Assemble the confirmed Clockwork 2 core

This independent core can proceed after T1 confirms CW2 even while the UHF interface remains unresolved. Have 1 BMG-type drive assembly, 1 idler assembly, 1 spring/thumbscrew tension set, **2 MR85 bearings**, 1 NEMA14 motor and the six direct-drive printed parts.

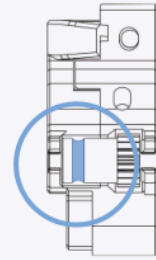
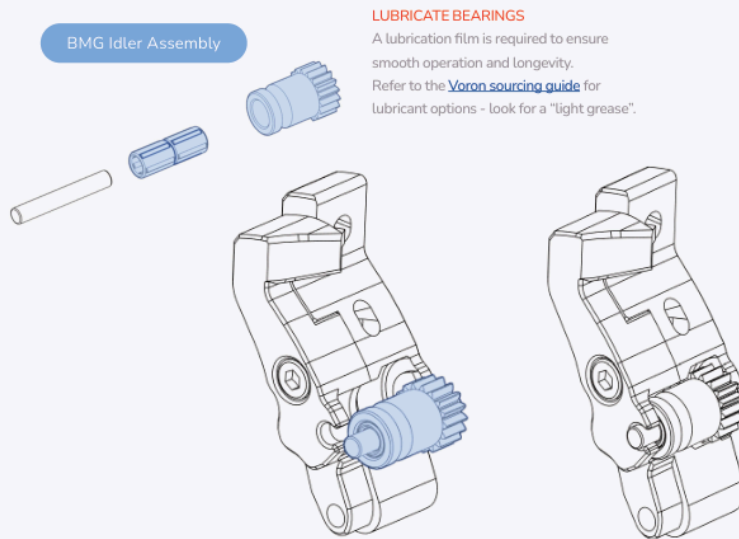
The core drawings call for **1 M3×6 FHCS**, **4 M3×25 SHCS** (two body bolts, one guidler pivot, one latch pivot), **1 M3×16 SHCS** for the two-piece guidler, **1 M3×30 SHCS**, **2 M3×8 SHCS** and **2 M3 washers** for the motor. Heatset locations are illustrated in SB pp. 11–15 (indices 10–14); count those positions against the actual matched printed parts, with additional PCB positions governed by the actual Nitehawk mount. Cable-cover/anchor hardware is separate and conditional.

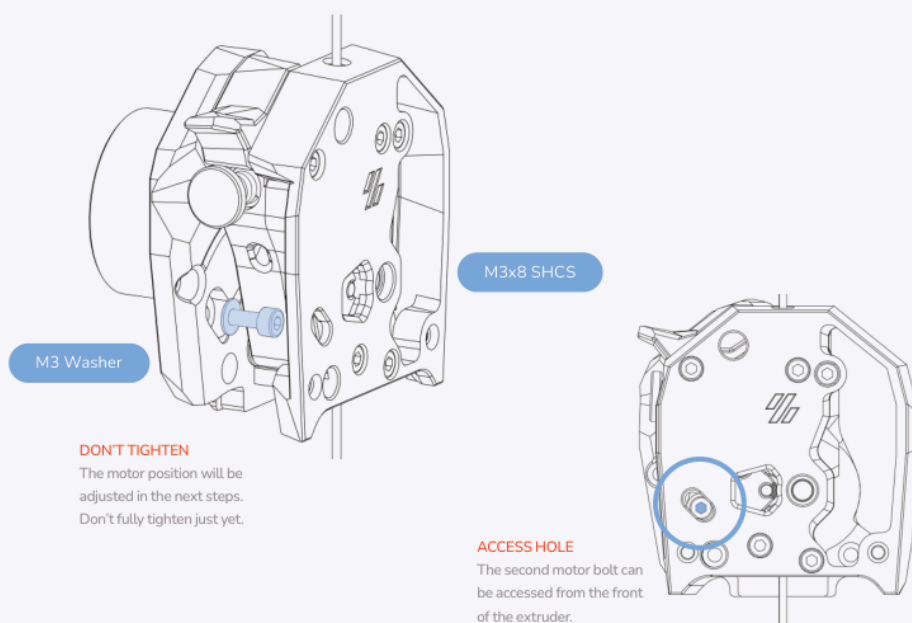
1. Install heatsets in the six core parts according to those illustrations. Keep motor-plate inserts below the surface where shown on p. 12 and the guidler/shuttle inserts flush where shown on p. 15. Let each cool before fastening; verify no insert intrudes into a bearing pocket or moving joint.
2. Join guidler A/B with its M3×16 screw, fit the idler shaft/bearing/gear in the orientation shown on p. 17, and fit its spring/thumbscrew. Apply light bearing grease to the idler's working bearing surface, not the filament-gripping teeth.
3. Fit the two MR85 bearings to motor plate and main body. Bearings must slide on/off the drive shaft; fully seat their **outer rings** in the pockets without pressing through the inner ring. The p. 19–20 warning against a forced bearing fit applies here.
4. Install the M3×6 flat-head anti-squish screw as shown on p. 20. Set the drive-gear grub screw against the shaft notch; initially leave drive position adjustable. Assemble the body with its two M3×25 bolts, align the gear teeth to a sample of 1.75 mm filament, then secure the drive gear (pp. 21–24 / indices 20–23). Verify the shaft does not touch the motor housing.
5. Install the guidler and latch with the remaining two M3×25 pivots (pp. 25–27). They must move freely; tightening a pivot until it clamps the arm defeats the mechanism. Adjust anti-squish/tension only enough to allow the gears and filament path to work without binding.

6. Fit the NEMA14 motor and the p. 28 M3×30 screw. Start the two M3×8 motor screws/washers loose; adjust mesh through the access hole, leaving faint gear play and full tooth overlap; then retain the motor (pp. 28–30 / indices 27–29).

GUIDLER ARM

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These retained CW2 diagrams apply only after CW2 identification. They do not depict or verify the custom UHF hotend, CNC carriage or V4 probe. © Voron Design, GPL-3.0; [pinned manual](#).

Completion check: latch opens and closes, idler pivots freely, shaft has clearance, gears turn smoothly, and a sample filament follows the centered path. Do not connect the motor yet.

T3 · Install the exact 2F UHF hotend and cooling cartridge

Verification gate T3 — blocks support-screw changes and hotend/cartridge installation. Read the delivered 2F label and instruction sheet: revision, voltage, heater power/current, PT1000, nozzle type/diameter and installed UHF adapter. The [manual shipped in Phaetus's pinned Rapido-2F repo](#) is titled “Rapido Hotend 2,” ©2023 and covers several assemblies. It does not verify this unit's electrical ratings. Its internal support-screw specifications cannot be copied blindly to a revised 2F.

Do not loosen the thin heatbreak/support screws merely to fit a guessed cartridge. The Phaetus manual warns against bending the thin heatbreak and calls for even tightening with a low torque tool when its matching titanium-screw version is assembled (printed p. 07 / PDF index 7). The hot nozzle burns; hot-tightening is essential to prevent leaks for the documented nozzle design but belongs **after controlled heater commissioning**, not this cold assembly stage.

1. After T2/T3 close, remove only the top groove-mount adapter/collet where the confirmed cartridge requires four heatsink attachment screws. The Voron HF README describes two M2.5×5 adapter screws and four M2.5×8 heatsink screws; use that hardware only when the exact UHF mounting design and delivered heatsink confirm it.
2. Dry-fit the exact UHF front and rear around the heatsink. Confirm the hotend's fastener heads, heater/sensor exits and support screws clear every boss. The official Rapido V2 reference figures place heater wires **below** the mounting boss and sensor wires **above** it. Apply that routing only if the exact 2F geometry agrees; do not strain its heater tab or sensor to mimic another revision. [Pinned orientation figures](#).
3. Fasten the validated front/rear and heatsink using that cartridge's documented screws/inserts. The HF reference uses two M3×12 join screws and four M2.5×8 heatsink screws; these remain a comparison, not an automatically approved UHF stack. Verify straight cold filament passage; do not use a screw to force the heatbreak into alignment.
4. Cut/seat PTFE only after the UHF cartridge and CW2 are dry-assembled. SB p. 43 (index 42) uses **11 mm above the cartridge top** for its CW2 baseline. Verify the actual path joins without a gap or compressed/bowed tube before final cutting; do not carry over a Revo cartridge's total tube length.
5. Install the silicone sock matching the actual UHF block/nozzle. Fit the fan shell only after confirming the longer cartridge's outlets aim around the nozzle without touching the sock/block or obstructing heatsink airflow.

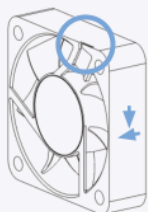
Completion check: the hotend is rigid without heatbreak stress, fan airflow reaches the heatsink, part ducts and socket clear the UHF block, filament passage is straight, and actual nozzle geometry is recorded. Leave electrical load confirmation to the wiring chapter.

Assemble fans, lighting and attach the toolhead

With a validated UHF shell/cartridge, follow the illustrated SB pp. 45–56 (indices 44–55): remove the built-in supports; add the optional lighting carrier/diffuser/mask before fans; route the lighting to the right side; clip the heatsink fan with wires at the top and airflow **inward**; then fit the blower. Confirm fan voltage before wiring. Trim the 5015 front/ears only if this delivered fan matches the documented design; the optional `STLs/Tools/SB_5015_Cutting_Tool_A.stl` and `_B.stl` are one each. The source drawing uses one M3×6 FHCS for the blower.

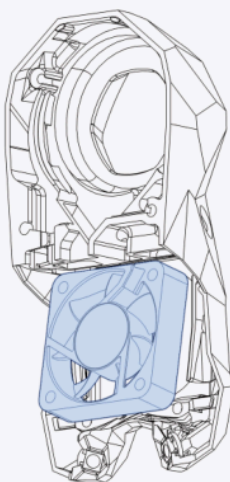
HOTEND FAN

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FAN ORIENTATION

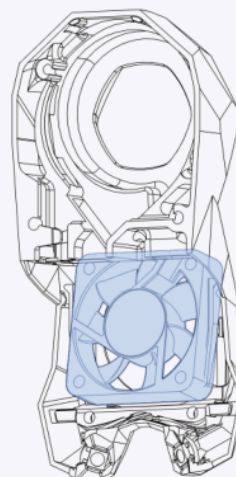
Rotate the fan so that the wires exit on the top and the air is pushed "inwards".



FAN INSERTION

Insert the fan at a slight angle and clip it into place.

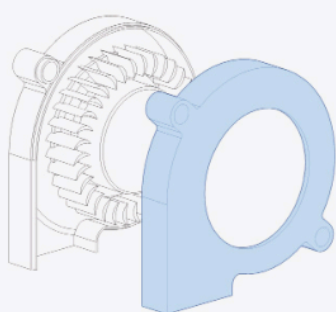
Mind the fan orientation.



53

PART COOLING FAN

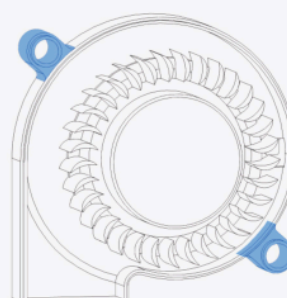
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FAN PREPARATION

Remove the front of the 5015 fan. Clip off and file down the stock mounting ears.

There's a trimming jig STL included in the release to make this task easier.



55

These fan diagrams show the standard SB reference assembly. Validate the UHF cartridge and exact fan first; the drawings establish fan preparation and orientation, not UHF duct clearance. © Voron Design, GPL-3.0; [pinned manual](#).

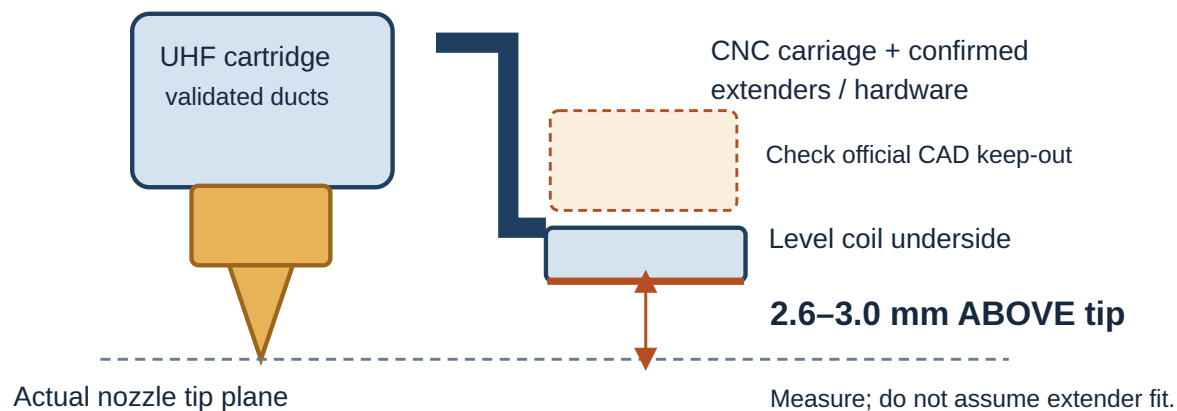
After T1 and T3 close, retain CW2 to the confirmed CNC mount/cartridge interface. SB pp. 64–67 use two M3×8 for CW2, then two M3×25 and two M3×50 for the standard shell. These are **standard reference fasteners**: validate their engagement, shoulder clearance and bottoming in this CNC/UHF stack before use. Do not press a printed shell into position with a longer screw. Fit the correct Nitehawk board mount/door and cable strain relief from the wiring chapter before closing access.

T4 · Mount and measure Cartographer V4

Verification gate T4 — blocks final probe fastening and probing. Confirm this is the **V4 AIO Standard**, the carriage version, extender inclusion, screw engagement and coil keep-out. The carriage is sold as 6 mm, 6 mm with **8.5 mm UHF extenders**, or 9 mm with extenders; the purchased variant is unknown. The example installation figures show a standard probe, so the actual V4 CAD and dimensions must match before use. Recheck B1: embedded bed magnets remain incompatible with scan meshing.

T4 · Measure the finished UHF nozzle and probe together

Original explanatory schematic · not to scale · lateral offset deliberately unspecified



Probe extenders lower the sensor; they do not make an HF cooling cartridge suitable for UHF.

1. Mount the probe coil flat beneath the CNC carriage in the demonstrated orientation, using its two confirmed M3 holes. Standard illustrations specify two M3×6 BHCS without extenders; extender assemblies need their own confirmed longer screws. Check the mounting parts bear on the intended surfaces without bending the PCB.
2. If the longer UHF nozzle leaves the coil too high, fit **confirmed** UHF extenders and hardware; measure again. Do not stack arbitrary washers or assume 8.5 mm extenders automatically produce the correct final height.
3. Measure the coil underside relative to the **actual installed nozzle tip**. Current manufacturer hardware setup specifies **2.6–3.0 mm above the nozzle tip plane**, with the probe level. Use a gauge, then record measured X/Y offsets rather than the listing's nominal coordinates.
4. Check the manufacturer's CAD keep-out above the coil against carriage, extenders, fastener heads and wires. Keep unintended metal out of that region. Sweep the toolhead manually across the conservative accessible envelope; the coil must never touch plate clips, side mounts, frame, brush or hotend wiring.

5. Route the probe cable along the supported umbilical/Bowden route with strain relief, free bend clearance and no snag. Manufacturer warning: the supplied USB cable is **not rated for cable chains**. Select USB/CAN only in the wiring chapter after exact pins/power are verified; an original Nitehawk SB is not a CAN host.

[Cartographer hardware setup.](#)

Nozzle cleaning must avoid sweeping the Cartographer coil over a metal brush. Use a documented silicone cleaning arrangement or manual cleaning during initial calibration; ownership/location of a silicone brush is unverified, so no automatic brush macro is enabled. Clean the nozzle before any later Touch workflow. The firmware generation, safe Z homing, QGL, scanning mesh and measured boundaries are established in the software chapter; do not use the older FAQ macros as configuration authority.

Completion check: purchases 03/04/05 each have an identified variant and planned disposition; the probe is retained and level at the measured height, no metal violates keep-out, all belt clamps and switch mounts clear travel, and cable relief has full bend clearance. The bed/nozzle envelope is recorded with the finished UHF toolhead. Any open T1–T4 gate remains a block on its affected operation. Continue to the wiring chapter without powering the toolhead.

07 — Wiring the modified printer

Readiness: conditional instructions; heater circuits remain blocked. Complete independent cable preparation while the gates below are open. No pinout in this chapter establishes what was shipped in purchase **01**.

Prerequisites: frame and gantry assembled and manually checked; modified bed and toolhead fitted; electronics compartment accessible; purchase **03** heater and PT1000 leads individually identified; purchase **05** probe identified as V4 Standard. **Parts:** one actual mainboard; one actual toolboard, its matching fan adapter and USB/power adapter; one host; six motion motors and one extruder motor; one PSU arrangement; one bed heater/sensor/fuse/earth harness; one toolhead cable; one probe cable. These kit component quantities describe the intended architecture, not verified box contents. Terminal, fuse, cable, bed-heater and adapter ownership is unconfirmed. **Tools:** multimeter, ferrule crimper, wire stripper, connector crimper appropriate to the actual contacts, PH2 driver and hex drivers. LDO specifies E0508 heater ferrules; verify conductor fit before selecting them.

KEEP: protective earth, separation of mains and low-voltage wiring, securely mounted electronics, individual motor labeling and strain relief. **REPLACE:** stock Octopus/toolhead/probe wiring with the verified board route below. **OMIT:** stock Omron probe, Klicky/TAP additions and stock nozzle Z-switch wiring in the selected Cartographer path. Keep omitted parts as surplus; do not install then remove them. **ADD:** Cartographer USB wiring and the measured PT1000 circuit.

Base manual references are printed pp. 180–209 (zero-based PDF indices 179–208), a **stock reference**, not the pinout for this modified printer. The [LDO Rev D wiring guide](#) supersedes stock kit wiring only when the box and cables match it. Its toolboard discussion still links the original Nitehawk SB; the [SB V2 guide](#) governs a confirmed V2 board.

E1 — Identify boards and the power architecture before connecting cables

Record PCB silkscreen revision, MCU marking, matching fan adapter revision, PSU model/rated outputs, supply-voltage selectors, fuse markings and cable labels in the build manifest. Photograph each connector with its latch/key visible, without private order material. Resolve kit revision independently from Voron design revision.

Confirmed hardware only	Firmware distinction	Wiring evidence
Leviathan V1.3	STM32H743, 25 MHz; manufacturer USB example uses 128 KiB Katapult	Pinned V1.3 manual printed pp. 5–9, indices 4–8
Earlier Leviathan	Do not apply V1.3 target or assume F446 until PCB is identified	Obtain exact-revision manual and schematic
Nitehawk SB V2.0.0	STM32G0B1, 12 MHz, 8 KiB Katapult; USB PA11/PA12	Pinned V2.0.0 schematic and V2 connector diagram
Original Nitehawk SB	RP2040, separate firmware/configuration; manufacturer Katapult instruction uses 16 KiB	Original SB documentation and pinned original configuration

The generic `voron2_leviathan.cfg` header and the Rev D webpage describe an F446, while the V1.3 manual describes H743. This is a documented source disagreement; **do not flash from that generic header**. Original SB and SB V2 fan adapter headers also differ. The V2 manufacturer FAQ says the old fan adapter cannot be reused.

1. Mount electronics on their verified brackets; confirm no conductive PCB underside contacts a DIN rail or fastener. Prepare PSU input selection using its label/manual. The LDO note about 115/230 V selectors does not apply to its universal-input alternative.
2. Label six motion-motor harnesses at both ends: A, B, front-left Z, rear-left Z, rear-right Z and front-right Z. Retain actual connector numbering in a separate record. Verify paired windings and the manufacturer's motor connector order with the power disconnected.
3. Dry-route mains, bed-temperature, motor, host and toolhead harnesses with ducts open. Ensure screw terminals and connectors remain accessible. Check that the MRW support freedom is not restrained by a tight cable.

Completion check: every board, fuse, PSU input and output, connector, motor and cable has a known identity. Unknown revision blocks board-dependent connections and firmware selection, while cable labeling and independent mechanical work can proceed.

E2 — Resolve the Rapido electrical load before connecting HE0

Purchase **03** is reported as **Rapido 2 Fiber UHF/PT1000**, not an established heater rating. Obtain its exact revision/label, voltage, rated operating load, cold resistance/startup or peak-current information and heater lead specification from Phaetus or the actual package. The linked Phaetus Rapido-2F manual is titled Rapido Hotend2; it does not establish the exact purchased Fiber heater's startup load. Do not substitute Rapido HF, Plus or X specifications.

The [Nitehawk SB V2 electrical table](#) specifies **4.5 A maximum continuous hotend current**, limited by the MOSFET. At 24 V this is 108 W electrically; that arithmetic is not a Rapido rating or permission to run a 108 W heater. Startup current, pulsed MOSFET capability, terminal/umbilical contacts, cable gauge/temperature, branch fuse and shared PSU headroom all require verification. The pinned V2 schematic labels the heater MOSFET AON7524; its part name alone does not rate the assembled board for a larger heater.

Circuit	Confirmed manufacturer fact	Required decision
V2.0.0 HE0 → Rapido	HE0 control PA7; 4.5 A continuous board limit	Exact heater startup and operating current must fit the entire path
V1.3 mainboard heater output	V1.3 manual: 180 W/7.5 A maximum	Not a preapproved alternate route; additional cable/fuse/PSU evidence and wiring review required
Original SB HE0	Different RP2040 pin mapping; separate documentation	Reassess ratings for actual revision; V2 pins do not apply
Bed-heater signal output	Controls a rated SSR only for an AC heater design	Never put mains on a DC heater terminal or probe connector

Unresolved is a hard heater block. Leave the Rapido heater unplugged and insulated. Do not bypass a fuse or enlarge it to obtain a first heat-up. A software `max_power` setting limits average duty cycle and cannot establish safe instantaneous current. The [historical LDO Rapido erratum](#) concerns swapping **Octopus** outputs; neither its wiring swap nor its PA1/PA2 assignments applies to this build.

Once E2 is signed off with ratings and exact PCB revision:

1. Prepare the heater ends using correctly sized ferrules and the board's documented terminal procedure; never clamp solder-tinned strands under a screw terminal. Do not invent terminal torque. The resistive heater itself is nonpolarized, but the PSU/adaptor path is polarized.

2. Connect the PT1000 to confirmed TH0, separate from heater leads. For **V2.0.0 only**, TH0 is a 2-pin JST-PH connector, ADC **PB12**, documented pull-up **2200 Ω**. It is a resistance sensor circuit, not a supply output. Use Klipper `sensor_type: PT1000`; do not retain a stock thermistor model.
3. Connect extruder motor and fans to the matching toolboard/fan adapter. V2 fan voltage selection is modified by PCB traces/pads, rather than a universal jumper. Check the actual fan ratings; do not perform that PCB modification unless the exact manufacturer's procedure is required for the purchased fans.
4. Install the toolboard-to-extruder-body and USB-adapter-to-earthed-frame grounding links described by LDO for the actual board revision. These static-discharge links do not replace protective earth to bed/frame/PSU.

Completion check: the heater-path record includes all current limits, cable/connector/fuse identity and startup evidence; PT1000 is individually labeled and cannot be confused with CT or HE0; connector covers close without pulling conductors.

E3 — Verify the modified bed circuit before mains work

The stock LDO bed's prebonded heater, magnet layer, thermal sensor and fastening pattern do not establish the MRW bed's installed components. Purchase **06** heater options remain unknown. Do not detach and reuse a bonded stock heater or magnetic layer on the UltraFlat plate without explicit manufacturer instructions.

LDO's manufacturer warning requires mains work by certified personnel trained in local regulations and safety standards. Keep the inlet unplugged during assembly. Software and a multimeter continuity check do not certify electrical safety.

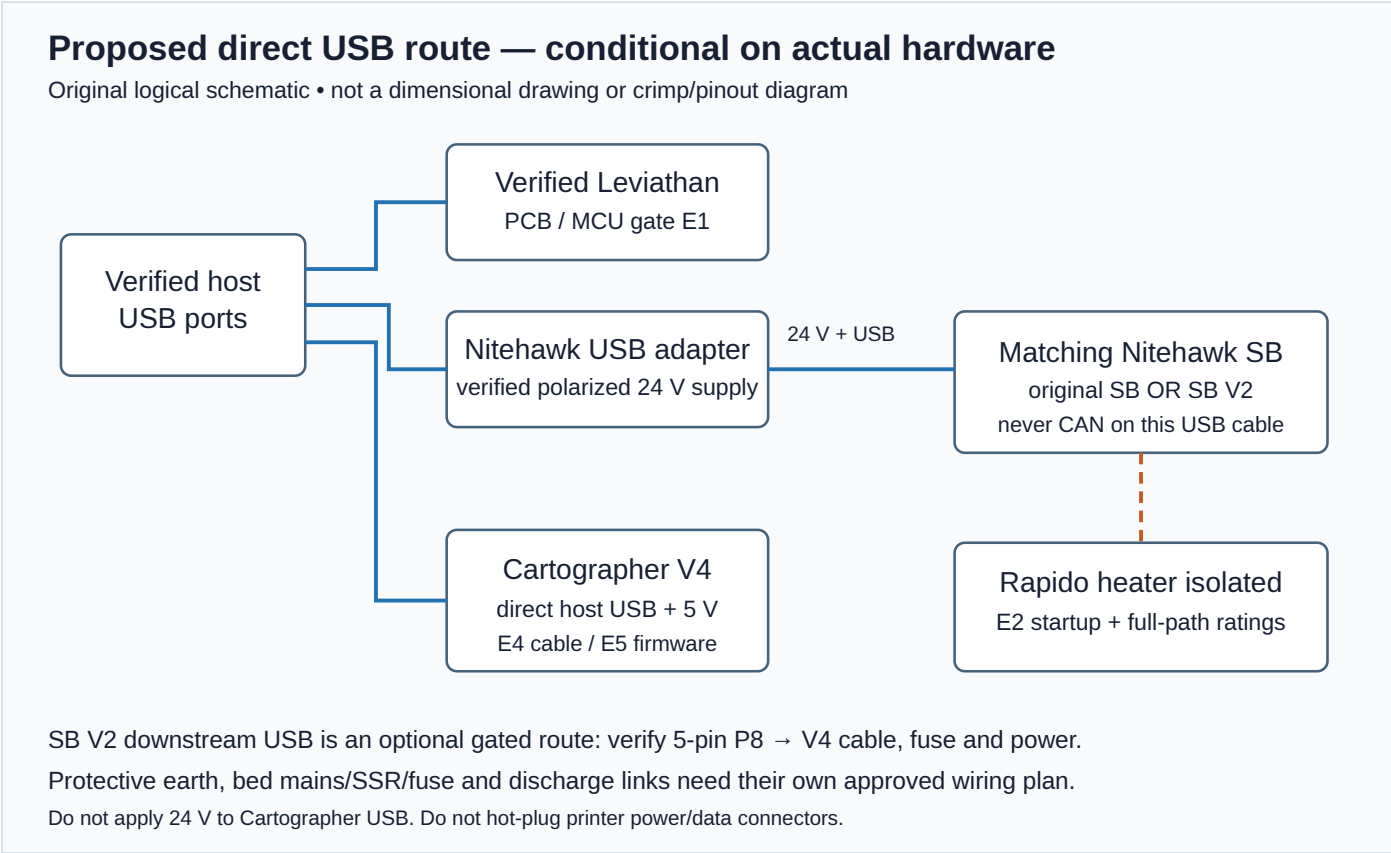
1. Confirm heater voltage/power/physical size, bonded mounting procedure, sensor model, leads, thermal fuse temperature/current/attachment method, SSR make/rating/derating/heatsink and inlet/branch fuses for the actual bed system. If any is missing, classify it as **required but not confirmed owned** and keep bed wiring and heating blocked.
2. Have the qualified installer wire inlet, isolating switch, PSU, SSR and bed using the actual circuit and local requirements. Keep mains terminals covered; retain a separately identifiable protective-earth conductor. Follow terminal ratings and manufacturer's tightening instructions. The Leviathan's DC bed-output rating is not a mains rating.
3. Connect a protective-earth lug directly to the MRW plate at its approved location with hardware providing reliable metal contact, strain relief and slack for thermal expansion. Do not rely on anodized Wings, kinematic points, magnets or a linear rail as the earth path. Ground PSU enclosure and conductive frame as specified by the qualified installer.
4. Fit an independently acting thermal fuse in the heater power path according to heater/bed instructions. Verify its attachment and wiring; Klipper heater verification is an additional software safeguard, not a substitute.
5. Route bed leads so that gantry/belts, kinematic motion and Z travel cannot rub or pinch them. Keep thermistor leads away from mains terminals. A stock Rev D bed-WAGO/thermistor splice arrangement is reusable only after connector ratings, approved wire sizes and MRW geometry have been verified.

Completion check: the bed circuit has a documented approved diagram, fuse and sensor identity, protective-earth connection and full travel strain-relief inspection. Bed power remains disconnected until the pre-power checklist passes.

E4 — Choose the Cartographer interface and cable before

connection

Proposed primary route: direct host USB to Cartographer V4 Standard, with its documented **5 V USB power**, after the actual USB firmware and cable have been identified. Retain the Nitehawk’s independent host USB path. This avoids an unverified custom downstream adapter; it does not prove cable ownership or cable-chain suitability.



The figure shows logical connections only. The Nitehawk XT30(2+2) combined power/data cable carries **24 V and USB**, not CAN. Its ordinary three-pin PROBE port supplies 24 V and must not be used as Cartographer USB power.

The V2’s downstream USB route is **conditionally compatible**, not automatically wired:

Logical signal	Nitehawk V2.0.0 P8, schematic numbered pins	Cartographer Standard USB documented signal order
5 V VBUS	Pin 1, via F1 BSMD0603-075-6V	Ground, Data+, Data-, 5 V in the manufacturer’s illustrated view
Data-	Pin 2	Match Data- by continuity and illustrated view
Data+	Pin 3	Match Data+ by continuity and illustrated view
Ground	Pins 4 and 5	Match ground; fifth Nitehawk contact needs a documented harness treatment

This table is a net list, not a left-to-right crimp recipe. The inspected V2 photograph reads GND, GND, D+, D-, 5 V from its illustrated side, whereas schematic numbering starts at VBUS. Plug-side views can reverse order. LDO calls P8 JST-ZH1.5 5P; Cartographer V4 Standard is advertised with a four-pin Molex Sherlock connector. The generic Cartographer Standard wiring figure is an older-format reference; match the actual V4 connector documentation rather than transferring its pictured contact positions. Verify exact connector/contact part numbers,

probe pinout orientation, supplied cable and the downstream port's usable current/fuse behavior before fabricating or buying that harness. Do not assume the 5 V converter's total rating is the USB port rating. No such downstream port is established for original SB.

1. For the primary direct USB route, check the keyed probe connector against [Cartographer wiring diagrams](#). Never apply the toolboard's 24 V to USB VBUS.
2. Keep the supplied Cartographer USB cable out of a drag chain unless its exact variant is documented as chain rated. Cartographer warns that its supplied cable is not chain rated. Route an approved umbilical beside the Bowden path with end strain relief and clearance throughout travel; provide a verified chain-rated replacement only if required by the actual route.
3. The Nitehawk toolhead cable has a documented nominal 28 mm bend radius; retain at least that radius without tight bends at the connector. Tie down both ends and retain enough slack. Do not transfer this rating to the Cartographer cable.
4. With power disconnected, verify each cable's end-to-end mapping, no shorts, keyed mating orientation and ground/shield treatment. Check cable sweep by manually moving supported mechanics through the measured envelope.
5. Mate power/data connectors only with the printer powered off. LDO specifically prohibits hot-plugging the Micro-Fit end; Cartographer likewise prohibits powered connector changes.

CAN is a conditional alternate only if actual hardware requires it. Identify a supported CAN interface, V4 CAN firmware/bitrate, bus ends, termination state, actual CAN-port supply voltage and V4 CAN power input specification. The Leviathan V1.3 CAN connector diagram shows **24 V**, so USB's 5 V pinout cannot be copied to it. Do not connect to the Nitehawk USB connector as CAN. Follow the V4 switching and termination instructions after those facts are settled; neither CAN hardware nor cable is confirmed owned. Record the chosen route; do not enable both serial and CAN UUID in one MCU section.

Completion check: the chosen interface, firmware, cable identity, pin-to-pin mapping, supply voltage, rating and full travel routing have been recorded. Keep unknown cables disconnected. Continue to [software preparation](#) without energizing heaters.

Source and figure notes

Manufacturer connector diagrams and schematics were downloaded and visually inspected, including V2 downstream USB F1/P8 and Leviathan V1.3 pin assignment/USB firmware target. Third-party images/PDFs are linked rather than redistributed because their independent reuse terms are not established. The local SVG is an original logical schematic, not an assembly drawing. Exact revisions, hashes and retrieval dates are recorded in [electronics research](#).

08 — Software preparation and the probing workflow

Prerequisites: E1 PCB identification, E4 interface/cable verification and an approved unheated initial configuration; E2/E3 are still required before heaters are enabled. **Parts:** one identified host, one identified mainboard, one identified toolboard and purchase **05** Cartographer V4 Standard. **Tools:** host terminal, version-control tools, editor and Klipper/Moonraker interface. Record versions locally; never publish credentials or private host addresses.

KEEP: CoreXY kinematics, four independently mapped Z motors, physical X/Y endstops where their replacement mounts are proven, and Klipper heater verification. **REPLACE:** stock Z endstop with the Cartographer virtual endstop after E5/E7. **OMIT:** Omron, Klicky, TAP and nozzle-switch macros/configuration in this selected path. **ADD:** matching Cartographer driver and separate measured probe/motion configuration.

E5 — Select one board/driver/firmware generation

The [commented examples](#) are review worksheets, not a `printer.cfg`. Every line is commented, required measurements are explicit placeholders, and none includes a motor movement or heater macro that can run. Do not rename an example and treat it as deployable.

Component	Proposed documentation baseline	Evidence boundary
Klipper host and main/toolboard firmware	Klipper3d/klipper source snapshot 461c4e3722c3a897fba1c6b3f0780a5315043842	Host/MCU builds must match; installed version unknown
Leviathan V1.3 only	H743, 25 MHz, USB; manufacturer Katapult target 128 KiB	Factory firmware/bootloader unverified; generic F446 header conflicts
Nitehawk SB V2.0.0 only	G0B1, 12 MHz, 8 KiB Katapult, USB PA11/PA12	Separate from original SB firmware and pins
Original SB only	RP2040, USB, manufacturer's 16 KiB Katapult offset	Conditional original-board worksheet only; not active beside V2
Cartographer V4 Survey candidate	V4 6.1.0 USB Full , 8 KiB-offset binary in firmware source e5c2b17dbe04ec1f747af5d81b2215949a0a9f8e	Actual installed firmware/host performance unknown
New Cartographer host driver	cartographer3d-plugin snapshot 06e016905cc3d209ceb543afd59a580bcabc44be	Current-generation documentation family; exact candidate combination not tested on this printer

This is a pinned research baseline, not an assertion that those snapshots have been jointly hardware-tested. Confirm a released compatible plugin/firmware pair for the actual hardware with the vendor before probing; record the selected plugin version/commit and firmware filename/hash in the machine manifest. Review release changes before updating. The locked V4 USB firmware binary's SHA-256 is in the research file; it was hashed, **not flashed**.

The current [Cartographer Klipper setup](#) uses the new plugin and `[cartographer]`. The **original-plugin** archive uses a different `[scanner]` workflow; still older classic pages can use `[cartographer]` with different behavior. A section name alone does not prove generation compatibility. Remove legacy includes, old scanner/cartographer calibration

blocks in `SAVE_CONFIG`, and duplicate updater sections before choosing the new Survey workflow. Preserve a private backup first. Avoid copying archived `CARTOGRAPHER_CALIBRATE` / `CARTO_TOUCH` macros into the new configuration.

1. Prepare the host using [Voron software setup](#) and the actual host/OS instructions. Inspect the kit's existing configuration first if it later becomes available. Save an inventory of active includes, devices and versions before changing it.
2. Compare the exact board target and bootloader to the manufacturer's diagrams. If an update is required, follow the identified board's manufacturer procedure. **No flashing is part of documentation validation.** A V1.3 H743 must not receive F446 firmware; an original SB must not receive V2 G0B1 firmware.
3. Use [Leviathan worksheet](#) plus exactly one of [V2 toolboard worksheet](#) or [original SB worksheet](#), only after the matching PCB is confirmed. Record actual USB device paths after supervised first-start; never invent them.
4. Install the matching vendor driver using its reviewed installation procedure at the chosen version. The vendor's installer resolves a released package; a pinned documentation checkout alone does not pin the installed Python package. Record that package version, dependencies and source commit after installation. Do not run arbitrary moving-branch shell pipelines as a substitute for review.
5. Audit the final merged configuration: only one `[mcu nhk]`, one `[extruder]` heater/sensor definition, one authoritative endstop per axis, one `[cartographer]`, one Z virtual-endstop definition and no active stock probe macro. LDO's partial toolboard configuration assumes it overrides mainboard extruder/endstop fields; manually verify the merged values rather than relying on include order to conceal contradictory settings.
6. Keep device IDs, sensor limits, run currents, motor direction, offsets, travel boundaries, QGL positions, mesh bounds, rotation distance, PID constants and tuning results unset until confirmed or measured. Retain Klipper's `[verify_heater]` behavior; do not weaken it to hide heater faults.

Completion check: the source-lock record and installed versions agree; each MCU path maps to the physically identified board; firmware family, bootloader and interface match; a reviewed configuration exists with heaters disabled/disconnected until E2/E3 and the first-start checklist permit them.

E7 — Probing geometry and bed compatibility before Z homing

Purchase **06** magnetic variant remains unknown. Cartographer's [official FAQ](#) identifies embedded-magnet beds as incompatible with normal scanning. A uniform magnetic sheet is a different arrangement. **An embedded-magnet UltraFlat variant blocks this guide's scan Z-home, QGL and mesh path.** Do not assume nozzle touch eliminates electromagnetic effects or use touch mesh as a workaround; current [Touch documentation](#) explicitly marks touch mesh unimplemented. Preserve the purchased bed and register the unresolved interaction rather than silently replacing it.

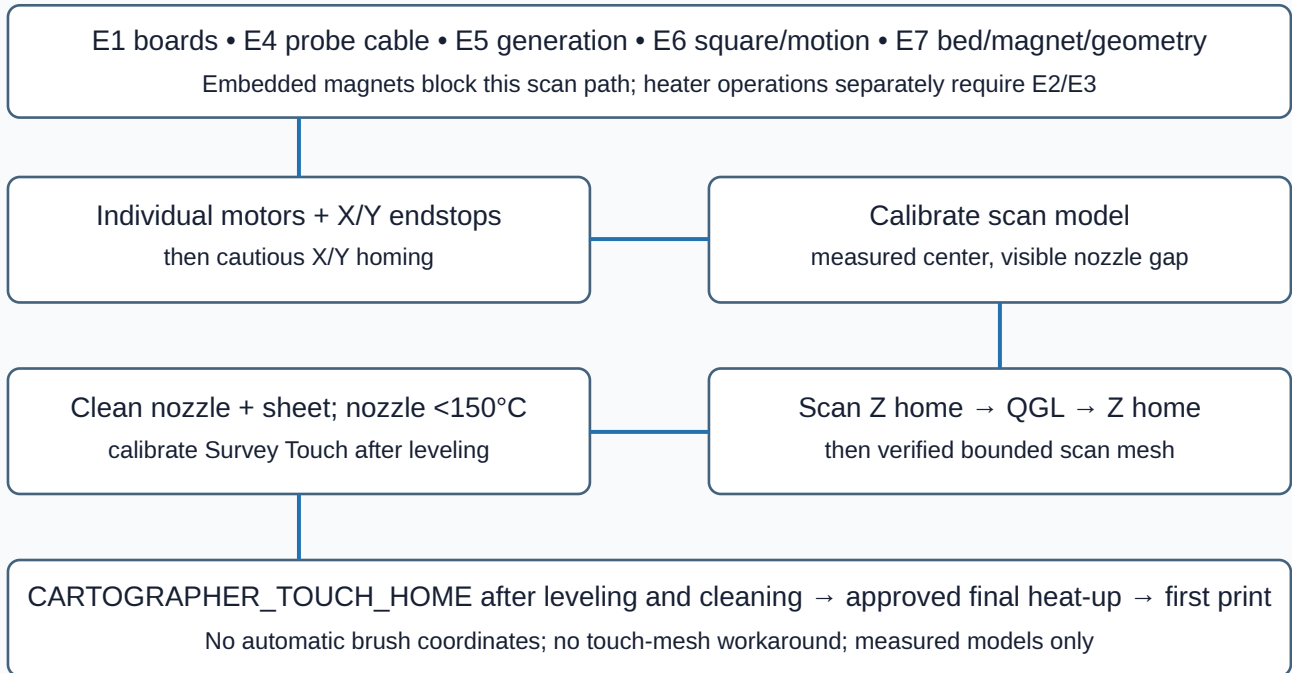
Before any Cartographer motion, establish the bed's magnet arrangement and installed spring-steel sheet, probe height/keepout from chapter 06, measured nozzle-to-coil X/Y offsets, nozzle and coil reachable rectangles, safe Z clearance and the physical center reference point. Never set 175,175 merely because the provisional plate is 350 class. A point is valid only if **both nozzle and coil** remain over the usable surface as required by the procedure.

Configure Z with `probe:z_virtual_endstop`, `homing_retract_dist: 0`, and a verified central safe-home position after E5/E7. Establish QGL gantry coordinates and probing points from the actual four Z support locations and measured usable surface; do not reuse a vendor's stock 350 coordinates blindly. Mesh limits refer to probe coordinates, so apply measured offsets and travel limits consistently. See [Klipper configuration reference](#).

Selected workflow after commissioning gates pass

Selected current-generation workflow

Original process schematic • all physical actions require closed gates and supervision



A successful documentation build is not physical validation of mechanics, firmware, safety or probing.

The selected current-generation workflow separates scanning from nozzle touch. Ordinary Z homing, QGL and bed mesh use the calibrated scan model. Survey Touch establishes nozzle contact **after** leveling, with a clean nozzle below 150 °C. Keep the bed mechanically seated and the same sheet fitted through calibration and printing. The commands below are instructions for a future supervised physical procedure, not actions run during this project.

1. After the individual motor/endstop/travel checks in [commissioning](#), home X and Y separately and verify their direction and stop behavior. Run the vendor's **new plugin** scan-calibration sequence at the measured zero-reference point: `CARTOGRAPHER_SCAN_CALIBRATE`, carefully approach using the specified feeler-gauge procedure, and save the measured model. The manufacturer's procedure targets 0.1 mm and warns to retain a visible gap; do not automatically lower from an unknown Z position.
2. Verify scan Z homing with immediate-stop access, then run QGL at conservative verified clearance and points. The four rigid Z joints must already be squared and free from binding. Recheck scan Z homing after QGL; do not command repeated leveling to force a mechanically skewed gantry.
3. Remove filament residue from the nozzle and debris from the sheet manually while stationary. Use the hotend maker's temperature and handling instructions only after E2 is closed; cool the nozzle below **150 °C** before Survey Touch. No automatic brush/purge bucket is confirmed purchased, so no cleaning coordinates or brushing macro is enabled.
4. Calibrate Survey Touch using `CARTOGRAPHER_TOUCH_CALIBRATE` only with rigid toolhead/bed/gantry and a clean nozzle/sheet. Observe contact and keep immediate-stop access. Save the measured touch model. If contact does not occur where expected, stop and investigate mechanical noise/rigidity rather than blindly increasing thresholds.

5. For a normal start after all calibrations: fit the recorded sheet, clean nozzle, bring bed to the validated condition and nozzle to a verified non-oozing temperature below 150 °C; scan-home; QGL; scan-home Z again; generate the bounded scan mesh; perform `CARTOGRAPHER_TOUCH_HOME` after leveling and just before final print heat-up; heat to the approved print temperatures; begin a supervised first layer. If the exact selected release changes required order, update this workflow and its lock together before use.

Completion check: scan and touch models are measured for the actual geometry/firmware and no contradictory Omron/Klicky/TAP/nozzle-switch configuration remains. Record new offsets or sheet changes as calibration changes, not as fixed values in this public guide. Proceed to [commissioning and first print](#).

09 — Pre-power, first start, calibration and first print

Readiness: this chapter is a supervised procedure to be performed later on the physical machine. The documentation project has not flashed firmware, powered a printer, moved motors or heated anything. Static checks cannot validate assembly, ratings, mains safety or print quality.

Prerequisites: completed mechanical chapters and closed gates for the next action; actual configuration reviewed against hardware; E2/E3 closed before any heater enable. **Parts:** one assembled printer, approved protective-earth and thermal-protection system, fitted build surface and one intended filament sample for the final print. **Tools:** qualified electrical inspector's appropriate test equipment, multimeter, hex drivers, feeler gauge, ruler/calipers, terminal/UI access and direct emergency-stop/power-isolator access. Do not invent a test-equipment or torque specification from the plastic kit manual.

KEEP: Voron startup checks, independent electrical inspection and Klipper thermal safeguards. **REPLACE:** stock probe calibration with chapter 08's gated Cartographer sequence. **OMIT:** stock probe/endstop self-tests for devices that were omitted. **ADD:** rigid-joint binding, modified-bed expansion/clearance, PT1000 and heater-load checks.

Pre-power checklist — all circuits initially isolated

LDO's mains warning requires certified personnel trained in local regulations and safety standards. The checklist records evidence; it does not certify electrical safety.

- [] Inlet unplugged; disconnect/isolator identified; no powered connector changes planned. No loose tools, wires, swarf or fasteners in the electronics compartment.
- [] E1 closed for each actual board/adaptor/PSU: PCB silkscreen and MCU recorded; correct firmware target; input voltage selection correct; polarity checked; no unintended dual host/5 V supplies or backfeed route.
- [] Qualified inspector checks mains isolation/separation, terminal coverage, conductor size, contact/crimp quality, wire retention, fuse and switch ratings, strain relief and local protective requirements. AC switching and SSR orientation match the actual approved diagram.
- [] Protective earth verified to inlet, PSU enclosure, conductive frame and MRW bed using the required inspection method. A dedicated bed connection contacts the approved metal surface; anodized Wings/kinematic points are not the sole earth path. LDO static-discharge links are checked separately.
- [] E3 closed: actual heater, voltage, watts, sensor, SSR/heatsink and independent thermal fuse identified; heater adhesion/attachment and fuse mounting meet their manufacturer's procedure. Stock bonded heater or magnetic layer was not assumed reusable.
- [] E2 closed before hotend enable: exact Rapido2 Fiber voltage/cold/startup/current evidence and complete MOSFET/connector/wire/fuse/PSU path ratings documented. Software power limits are not used to excuse a current mismatch.
- [] Heater leads disconnected and individually insulated for initial board/sensor work. No bare terminal can touch frame/PCB. PT1000 and bed sensor leads are separate from heater/mains wiring.
- [] E4 closed: probe cable power and signal mapping verified; direct USB or documented downstream USB/CAN route selected; no 24 V applied to USB; no hot-plugging. Cable bends and strain relief meet each cable's rating.

- [] Electronics mounted securely and cooled; correct fan voltage/connector mapping; cable lids/covers close without crushing wires.
- [] E6 mechanical prerequisites complete: frame and gantry squared, rigid Z joints installed without forcing alignment, belts routed correctly, pulleys/capture checked, axes manually free over measured full travel and gantry supported against falling while unpowered. Kinematic bed retains its intended thermal freedom.
- [] Full sweep checked for Rapido UHF/CNC carriage/probe collision, backer screws, XY joints, Wings/bed clamps, belts and all wiring. Conservative provisional travel envelope recorded from measurements, with clearance in reserve.
- [] Active config has one endstop authority per axis and one probe generation. Cartographer offsets, mesh/QGL points and safe-home position remain blocked until E7; no automatic home, QGL, brush, heat or print macro runs on connect.

Completion check: record inspector, date and actual findings in a private local commissioning record. Failed items remain blocking; a rendered site or passed link check cannot close them.

First start — sensors and communication before motors/heaters

Only after the qualified inspector permits energization, power the reviewed isolated arrangement with heaters disconnected. Remain at the printer and have immediate isolation available.

1. Confirm expected board status indicators and communications, with no smell, noise, heating connector or unexpected fan/motor motion. Stop if anything differs from the identified circuit.
2. Identify host device paths read-only, one verified device at a time with power off before cable changes. Match paths to mainboard, toolboard and probe; update the private config without fabricated IDs. Confirm E5 version records and host/MCU compatibility before enabling functions.
3. Verify cold hotend, bed and optional chamber readings are plausible for room conditions and consistent with an independent reference. For the reported PT1000, use the actual TH0 pull-up and `sensor_type: PT1000`. Sensor fault, implausible reading or inverted response blocks all heating. Do not bypass `min_temp`, `max_temp` or heater verification to hide an error.
4. Warm only the sensor locally using a safe non-powered method appropriate to its construction and confirm the corresponding channel changes; do not swap sensors while powered. Confirm neither heater can energize at this stage.
5. Check fan identity/direction at a safe setting once fan voltage is confirmed. Verify hotend heatsink cooling starts according to the reviewed heater-fan behavior before any controlled hotend test.

Completion check: stable communications, correctly mapped temperature channels and fans, documented actual IDs and no unintended energization.

E6 — Individual motors/endstops before homing or QGL

The [Voron startup sequence](#) and [Klipper configuration checks](#) govern supervised checks. Ensure adequate clearance, correct current limit from the actual motor/board specification, and no tension/binding from the rigid-joint system. Keep hands clear of powered mechanics.

1. Test each physical X/Y switch by hand with `QUERY_ENDSTOPS` while the machine is stationary. Confirm the intended channel changes state and is otherwise open. Repeat for the exact replacement endstop mounts. Do not require omitted Z-switch or Omron states; the Cartographer virtual Z endstop is calibrated later.

2. With a supported, safe gantry position and a reviewed configuration, test **one motor at a time** using Klipper's `STEPPER_BUZZ` procedure. Identify A/B and each Z corner physically. Stop on wrong identity, direction, roughness or unexpected resistance. Correct wiring/config only with the machine powered off and repeat only the affected check.
3. Interpret the individual A/B motor buzz results using the documented CoreXY arrangement: one motor's movement is not an X or Y axis move. Confirm motor identity and direction before homing; do not bypass Klipper's homing requirement with forced moves or invented position values.
4. Establish homing direction and conservative travel limits from measurements. Perform X and Y homing individually with immediate-stop access, then conservative interior moves. If a switch does not stop the intended motion, isolate power and correct the fault before proceeding.
5. Verify the four Z motor signs/identities, unobstructed travel and documented gantry square. Do not perform Z homing or QGL until E5/E7 and the scan-calibration procedure are ready. Fixed joints must not bind at any height; software leveling is not a remedy for forced geometry.

Completion check: each motor and endstop has a physical identity and observed expected behavior; travel boundaries were measured and cautiously verified; no cable, bed, probe or hotend collision occurs. Proceed to scan calibration only if the bed magnetic arrangement is compatible and all E7 measurements are recorded.

Controlled heater tests — only after E2 and E3 close

1. Isolate power, reconnect **one approved heater circuit** at a time and recheck its terminals and sensor. The qualified installer approves any mains circuit changes. Maintain independent thermal protection, direct attendance and power-isolator access.
2. Apply a modest initial setpoint appropriate to that heater and material, observe the correct sensor rising and the other channels staying consistent. Stop immediately for uncontrolled rise, incorrect-channel heating, smell, discolored/hot connectors, cable heating, communication loss or heater verification fault. Never continue to PID tuning to diagnose an unsafe electrical path.
3. Confirm a commanded off-state stops electrical heating and the temperature trends as expected; residual stored heat is normal and must be distinguished from continued energization. Verify the physical isolator stops the system as intended; do not short outputs or deliberately defeat a sensor/fuse to test protection.
4. After the hotend electrical test passes, return to the exact delivered Phaetus revision/nozzle instructions. If those instructions require nozzle hot-tightening, perform it before first extrusion using the documented temperature, torque, support method and burn precautions. Do not disturb a factory assembly when its instructions do not require it, and do not copy another Rapido revision's values. Reinspect the leads and cartridge after this handling.
5. Only after basic behavior passes, follow the hotend/bed limits and Klipper PID-calibration procedure at intended operating conditions. Save **measured** results. Check hotend cooling, thermal expansion, bed seating and cable slack through a full stabilized heating cycle.

Completion check: electrical load/path ratings remain within approval, temperatures behave correctly, independent protections remain fitted, and measured PID/limits are saved privately. Both heater circuits must pass independently. A hot connector or unexplained overshoot is a failure, not a tuning target.

Calibration and the first print

1. Follow [chapter 08](#) for scan model, conservative Z homing, QGL, rehoming, nozzle cleaning, Survey Touch calibration and bounded mesh. Keep the verified magnetic arrangement and surface fitted. Current touch-mesh support is not a fallback for an embedded-magnet bed.
2. Measure extruder rotation distance, then tune first layer, pressure advance and input shaping using the [Voron setup documentation](#) and exact hotend/nozzle/filament limits. Record the actual nozzle included with purchase 03 before selecting slicer parameters. Do not insert someone else's PID, offset, acceleration or pressure-advance numbers.
3. Generate a small first-layer test and modest calibration part entirely inside the verified envelope. Review the slicer's machine size, start sequence and temperature limits. Use the validated scan-home/QGL/mesh/clean-nozzle/Survey-touch sequence; no automatic brush is enabled without purchased hardware and measured coordinates.
4. Attend the whole first print. Stop for scraping, skipped motion, loose joints, warped/rocking bed, unstable temperatures, cable rubbing or changed probe behavior. Inspect the rigid joints, belt capture, UHF/probe mount and bed freedom again after cooling.
5. Record actual results, unresolved faults and measured configuration privately. Update only the technical manifest and evidence statuses in the public repository. Recheck affected clearances and recalibrate when hardware, firmware generation, nozzle/sheet geometry or cable route changes.

Completion check: an observed successful first print and recorded configuration are physical commissioning evidence. Until that occurs, this repository remains an illustrated, gated assembly guide with a reproducible website—not a verified build-ready machine configuration.

10. Enclosure, filament path and first print

Prerequisites: mechanical travel checks complete; all electrical, probing, heater and calibration gates in chapters 7–9 closed for the operations below. Never treat completion of this chapter as a waiver for an open gate. **Tools:** hex drivers, scissors, ruler, calipers and appropriate adhesive. **Parts:** one each back/top/bottom panel, two side panels and two doors; retained 350 skirts (front A/B one each, side A/B two each, rear centre one), panel clips selected by actual thickness, six stock hinges, two each handle A/handle B/latch, 1 mm foam and VHB tape, two 60×20 bay fans if confirmed in the kit, four belt covers, one spool holder, one Bowden retainer, PTFE tube, screws/nuts in the specific retained figures. LDO-provided options still require a carton audit.

Enclose only after cable and motion checks

KEEP retained skirts and panel mounting pp211–213,217–219,222–249. **REPLACE** display/inlet instructions with the actual LDO electronics branch. **OMIT** stock mini12864/EXP cabling when the verified kit uses the LDO DSI touchscreen; do not build both display systems.

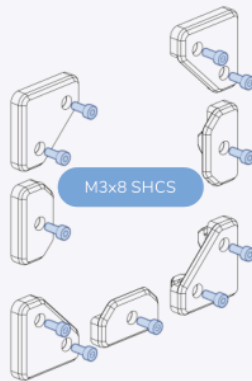
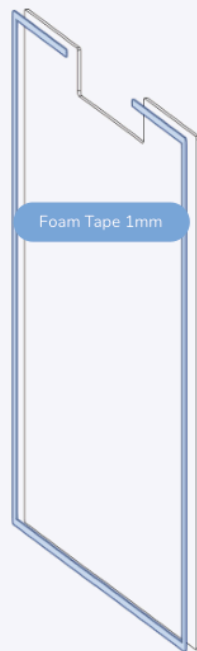
P2 — panel, inlet and display fit before closing the electronics bay

Measure the actual deck/bottom/side/top thickness and inlet body. Confirm skirt size, bay-fan voltage, touchscreen type and mount before installing. The provisional LDO BOM and supplement disagree on deck thickness, and the display may be a purchased kit option. Select the matching parts instead of adapting by force.

1. Clear built-in skirt supports and install the pictured inserts, pp211–213. Fit the two bay fans in the retained side supports with airflow arranged to ventilate the electronics bay. Fit the retained front and side skirt pieces with the M3x8/appropriate M3 nuts in pp217–230. The drawings distinguish the frame nuts from print inserts; do not screw into an extrusion slot without its nut. Preserve service access to the inlet and fuse.
2. Fit the LDO combined inlet skirt only after its dimension/clearance check in wiring. Fit the confirmed touchscreen mount, or leave that display area safely blank until the kit option is identified. Close all unused openings exposing mains; do not operate with a reachable uninsulated terminal.
3. Apply VHB to the bottom-panel clips/hinges, p232, then use M3x8 SHCS in the retained bottom-panel fittings, p233. Ensure opening the panel does not pull a PE conductor, strain a connector, or contact a live terminal.
4. Fit the four belt covers with the confirmed M3 hardware, pp234–236. If LED wires use the Z-motor opening, the LDO `z_belt_cover_a_led.stl` is a conditional alternative. Keep wires clear of moving belts. The CNC upper tensioners need access from above; do not permanently block their adjustment.

BACK PANEL

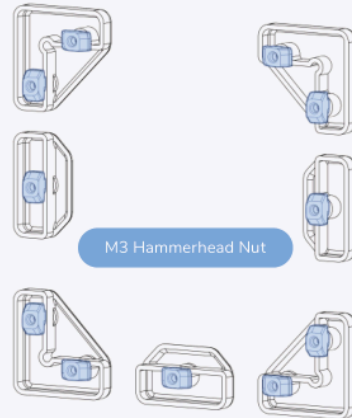
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APPLY FOAM TAPE
Use foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.

HAMMERHEAD NUTS?

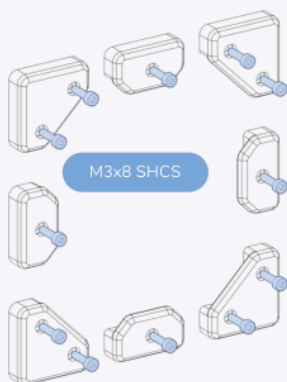
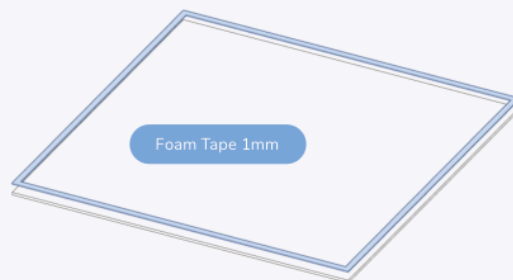
A drop of thread locker will turn the hammerhead nuts into a 1/4 turn quick release for the panels. Best done once the assembly is finished.



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TOP PANEL

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APPLY FOAM TAPE
Use foam tape on the contact areas between the panels and the frame to mitigate noise from vibrations.



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5. Apply 1 mm foam at back/side/top panel-frame contact areas, pp239–244. Use the matching corner/midspan clips with M3x8 SHCS and M3 hammerhead nuts, as the figures show. Confirm each nut has rotated into capture. Check that panel edges do not touch backers, rear belts or electronics wiring.
6. Prepare the two sets of stock door handles/latches with their magnets, p245. Test magnet polarity **before** gluing. Fit three hinges per door and the shown M3x8/hammerhead/VHB interfaces, pp247–249. If the actual kit supplied LDO alternate metal/printed hinge hardware, use its matched supplement and matrix instead; do not mix stock hinge lengths into it.
7. **OMIT** stock exhaust-fan assembly pp250–256 from the primary path: the Rev D supplement says its fan is not included. Close the opening using the documented LDO `exhaust_cover.stl` with stock `exhaust_filter_grill.stl` after verifying the provided fasteners. No additional filter modification is installed by this guide. Keep the chosen filament entry unobstructed.

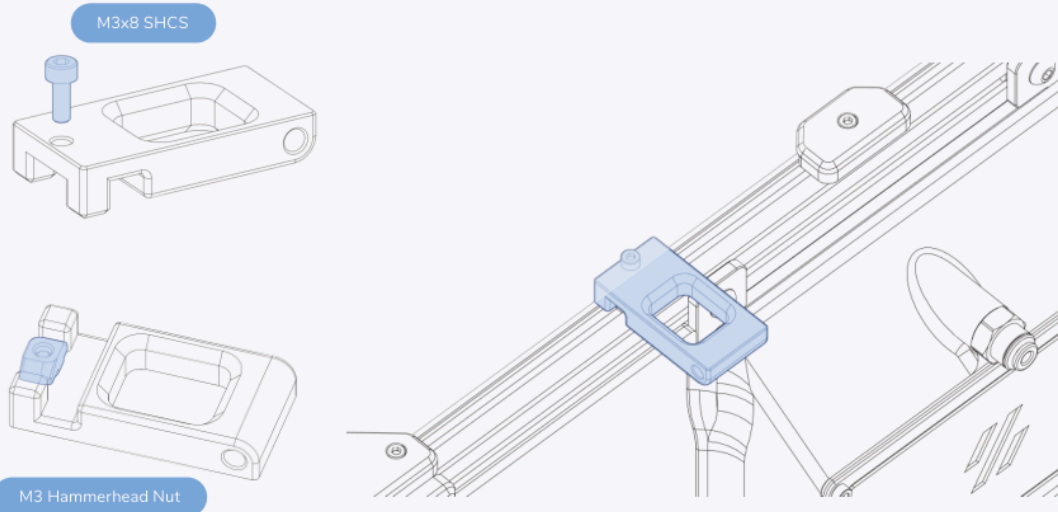
Completion: panels secure and removable, electronics guarded, bay ventilation clear, upper tensioners serviceable, full manual travel still clears all panels.

Filament path — KEEP

1. Assemble the stock spool holder and its M3x8/hammerhead attachments, p257. Mount it where the spool clears panels and cannot pull the printer over.
2. Fit the Bowden retainer with its M5x16 BHCS/M5 T-nut and the PTFE route, p259. With an omitted exhaust housing, confirm a compatible inlet/retainer route before fixing the tube; the old exhaust BSPP holder is not automatically present. Keep the tube outside all belt loops and without a sharp toolhead bend.
3. Manually sweep the toolhead with the tube and spool in place. Confirm CW2 latch access, filament insertion, cable bend room and the full safe envelope.

SPOOL HOLDER

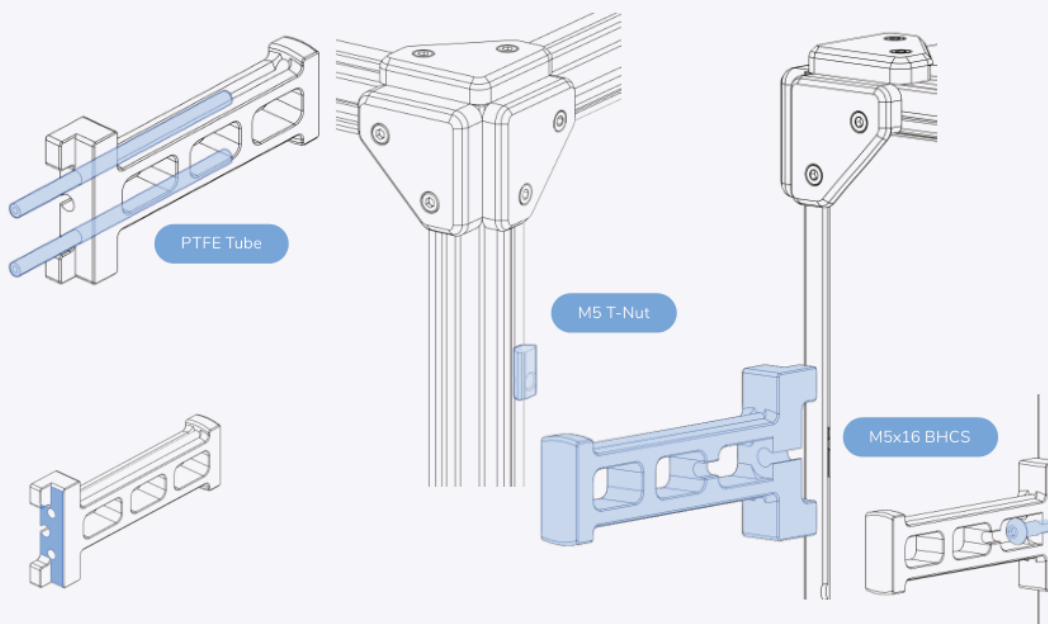
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SPOOL HOLDER

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Controlled first print after the commissioning checklist

1. Review every open gate in [the register](#). An open heater, movement or probe gate prohibits a print. Confirm the actual successful checks and tuned values are recorded privately or as technical measurements.
2. After the final enclosure/routing changes, repeat conservative travel checks and the scan/QGL/mesh sequence in chapters 8–9. Perform temperature-specific probe calibration/compensation only as the selected current driver documents. Do not reuse a cold bare-bed calibration as an assumed heat-soaked result.
3. Use a small simple first model such as the retained `STLs/Test_Prints/Voron_Design_Cube_v7.stl`. Set the slicer to the measured safe XY/Z envelope, actual nozzle diameter, filament and verified temperature limits. Do not copy the upstream legacy slicer profile onto this UHF build.
4. Begin with conservative speed/acceleration and watch the complete first layer. Stop if the nozzle contacts the surface, filament does not feed, temperature becomes implausible, belt rub appears or a cable stretches. Correct the cause before resuming. Do not leave the first controlled print unattended.
5. After cooling, inspect the cube, first-layer consistency and all cable/mount clearances. Record measured results. Tune extrusion, pressure advance and input shaping later using the selected Klipper/Voron procedure; no results or ready-to-run values are supplied by this documentation project.

Completion: one observed first print within measured boundaries, no open blocking gates and a recorded calibration baseline. This is a builder completion check, not a claim that this project performed a print or electrical certification.

Sources: original manual pp210–260; [LDO printed supplement](#); [LDO build FAQ](#); [Voron initial checks](#); [Klipper configuration reference](#).

Compatibility register

Statuses apply to the exact named combination. Advertised nominal fit is not physical validation. All actual delivered PCB/product revisions remain unverified unless explicitly identified. No combined machine arrangement is claimed physically tested.

Combination	Status	Hardware revision	Evidence / reason	Gates
bed-system	conditionally compatible	MRW350 standard/magbed unverified; matched-height kit delivered revision and 350 Wings unverified	Manufacturer designed Wings for Ultraflat+matched-height. Physical support map, third-seat attachment and fasteners unresolved. Sources: mrw350, mrw-kinematic-product, mrw-wings	B1; B2; B3; B5
embedded-magnet-bed-cartographer-scan	incompatible	MRW embedded top Magbed if delivered, Cartographer V4 scan	Manufacturer states embedded magnets cause mesh issues. Actual purchased bed variant unknown. Sources: mrw350, cartographer-faq	B1; T4
plain-bed-uniform-sheet-cartographer	conditionally compatible	MRW plain350 if delivered, uniform magnetic sheet, Cartographer V4	Manufacturer accepts uniform magnetic sheets; installed surface and repeatable probing not physically validated. Sources: cartographer-faq	B1; B5; T4
cnc6mm-mgn12-liveidler6mm	conditionally compatible	Cartographer CNC6mm purchased variant unknown, provisional V2.4R2 MGN12, Vitalii live-idler6mm purchase verified	Nominal rail/belt compatibility advertised; complete mixed gantry/shaft/clearance validation remains unresolved. Sources: cartographer-cnc-product, cartographer-cnc-6mm-cad	T1
cnc9mm-mgn9	incompatible	Cartographer CNC9mm with MGN9	Explicit manufacturer exclusion; not selected main build path. Sources: cartographer-cnc-product	T1
rapido2fuhf-sb-cw2-cnc-v4	unresolved	Purchased Rapido2F UHFPT1000 actual revision unknown, CW2 if confirmed, CNC variant/extendors unknown, V4Standard	Exact manufacturer UHF candidate exists but no verified combined duct, fan shell, CW2, nozzle/coil/extender fit. HF STL excluded. Sources: phaetus-rapido2f-adaptors, stealthburner-rapido-v2, cartographer-cnc-install, cartographer-v4-standard-cad	T1; T2; T3; T4; B5

Combination	Status	Hardware revision	Evidence / reason	Gates
hotend-load-temperature-input	unresolved	Rapido2F heater revision, Leviathan/Nitehawk revision unknown	Purchased title confirms reported PT1000/UHF only; load/startup current and board route must be resolved by electronics research. No heat commissioned. Sources: phaetus-rapido2f-manual, west3d-rapido2f	T3
F-USB-NH	conditionally compatible	Nitehawk SB V2.0.0 → Cartographer V4 Standard USB	Manufacturer provides downstream USB device capability, 5V P8/F1 path and signals; exact 5-to-4 pin harness/power rating not confirmed. Direct host USB is proposed primary. Sources: E-NHV2-DOC, E-NHV2-SCHEM, E-NHV2-PINOUT, E-CARTO-WIRING	E4
F-ORIGINAL-NH-USB	incompatible	Original Nitehawk SB family → downstream USB assumption	V2 downstream-USB hardware and pin map cannot be applied to original SB; use independently verified direct host USB. Sources: E-NH-DOC, E-NH-PINOUT	E1
F-NH-CAN	incompatible	Original SB and SB V2	Nitehawk protocol is USB, not CAN. Combined power/data connector is not a CAN port. Sources: E-NH-DOC, E-NHV2-DOC	E4
G-HEATER	unresolved	Reported Rapido2 Fiber UHF PT1000, actual revision unknown → NH SB V2.0.0	4.5A maximum continuous HE0 limit is known, purchased heater startup/load and complete current path are not; no heater enable permitted. Historical Octopus swap does not apply. Sources: E-NHV2-DOC, E-NHV2-SCHEM, E-LDO-ERRATA	E2
G-PT1000	conditionally compatible	Reported Rapido2 Fiber PT1000 → confirmed NH V2.0.0 TH0	Documented PB12 ADC/2200 ohm pullup can be configured as PT1000; actual sensor, connector and cold readings must be verified. Sources: E-NHV2-DOC, E-NHV2-CFG, E-KLIPPER-CONFIG	E1

Combination	Status	Hardware revision	Evidence / reason	Gates
F-LEV-FIRMWARE	incompatible	Leviathan V1.3 H743 versus older generic F446 cfg	Generic F446/12MHz/no- bootloader firmware header does not apply to V1.3 H743/25MHz/128KiB manufacturer USB diagram. Sources: E-LEV- MANUAL, E-LEV-FW, E- LEV-CFG-CONFLICT	E1
H-GENERATION	conditionally compatible	Cartographer V4 Standard Survey6.1.0/new plugin candidate	Current documentation family matches Survey/new-plugin workflow; actual installed version/interface and exact released pair require validation, no physical test. Classic/original-plugin [scanner] workflow remains omitted. Sources: E-CARTO-SETUP, E- CARTO-FW-DOC, E- CARTO-PLUGIN, E- CARTO-FW-BINARY	E5
H-EMBEDDED- MAGNETS	unresolved	MRW350 unknown magnet variant → Cartographer V4 scan	If MRW plate has embedded magnets, current scan path is incompatible. Touch mesh is marked unimplemented and is not a workaround. Uniform magnetic sheet differs; actual bed variant unknown. Sources: E- CARTO-FAQ, E-CARTO- TOUCH	E7
Independent 6mm XY / 9mm Z paths	conditionally compatible	AB PDF1.0; XY PDF1.1; Z tensioner unversioned; actual kit unverified	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-ab- assembly, vitalii-xy- assembly, vitalii-z-tension- product, motion-upstream- manual-reference	M1
A/B double-shear mounts and purchased LDO motor shafts	unresolved	Actual motors unknown; AB document1.0	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-ab- assembly, vitalii-ab- product	M4
A/B, front, live XY, backers and CNC carriage belt geometry	unresolved	AB1.0/front1.0/XY1.1 documents; carriage variant unknown	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-ab- assembly, vitalii-front- assembly, vitalii-xy- assembly, backer-design- readme	M5; M10

Combination	Status	Hardware revision	Evidence / reason	Gates
Rigid joints plus separate bottom CNC clamps at same interface	incompatible	Rigid-joint product unversioned; CNC clamp product unversioned	Rigid product replaces entire joint and lower clamp; two separately purchased bottom pieces surplus. Sources: vitalii-z-joint-product, vitalii-z-clamp-product	M7; M8
Two #10 rear top clamps and two #11 front upper clamps	conditionally compatible	Actual clamp geometry/packages unknown	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-z-clamp-product, vitalii-front-product, vitalii-ab-product	M8
Backers and single-front-MGN12 X / underside-MGN9 Y	conditionally compatible	Base R2 rail arrangement; ordered backers unknown	See the cited chapter and gate for the physical evidence still required. Sources: backer-design-readme, west3d-titanium-product, motion-upstream-manual-reference	M3
Rigid joints and actual four Z rail carriages	unresolved	Product unversioned; actual rail carriage revision unverified	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-z-joint-product	M7; M9
Live XY and actual LDO endstop PCB	unresolved	XY PDF1.1; actual PCB unverified	See the cited chapter and gate for the physical evidence still required. Sources: vitalii-xy-support-cad, vitalii-xy-assembly	M6

Non-deployable configuration examples

Every line is commented. Examples document a specific candidate hardware/firmware generation and deliberately omit unknown IDs, offsets, PID and travel values. They cannot be used as printer.cfg. A board-revision gate comes before selecting a sample; an electrical gate comes before connecting or heating.

- [cartographer-v4-survey-usb.cfg.example](#) — read its target and prerequisites before selecting.
- [leviathan-v1.3-usb.cfg.example](#) — read its target and prerequisites before selecting.
- [nitehawk-sb-original-pt1000.cfg.example](#) — read its target and prerequisites before selecting.
- [nitehawk-sb-v2-pt1000.cfg.example](#) — read its target and prerequisites before selecting.

Do not mix original SB RP2040 pins with SB V2 STM32G0 pins, old Cartographer Classic directives with the new Survey plugin, or F446 firmware headers with Leviathan V1.3 H743. Follow [software](#) after its gates.

Stock-to-custom page and step crosswalk

Pinned manual: 2023-07-04, upstream commit a192410e27ea345644ae5c4b29b4c9c40cbe1a73. Every PDF leaf is covered exactly once. Printed pages and zero-based PDF indices are recorded separately; final index 262 is an unnumbered back cover. REPLACE and OMIT instructions are not active in the custom path.

Printed stock pages	PDF indices	Phase / stock step	Action	Custom replacement
1–11	0–10	Inventory/print preparation; Intro, safety, print settings, hardware naming	KEEP	Procedure ; Identify actual kit before size-dependent printing; no private receipts.
12–19	11–18	Frame; Cube and adjustable bed-support preparation	KEEP	Procedure ; 8 cube A pieces; 4 vertical B; bed supports remain adjustable before F1.
20–20	19–19	Frame/bed layout; Stock 130mm bed support spacing	REPLACE	Procedure ; F1 requires B2/B3 cold Wings map and deck/DIN-hole agreement before fixing supports.
21–21	20–20	Frame; Square frame	KEEP	Procedure ; Compare faces/diagonals before rails and after bed attachments.
22–27	21–26	Z rails; Overview, rail preparation and four Z rails	KEEP	Procedure ; Secure carriage stops; verify actual rail lubrication and spacing.
28–29	27–28	Deck/DIN rails; Stock deck and DIN fastening	KEEP / ADD gate	Procedure ; F1 closes before final deck: DIN hardware anchors into bed supports; F2 panel thickness discrepancy.
30–31	29–30	Lower Z drives; Orientation and heatsets	KEEP	Procedure ; Retained plastic reduction drives only.
32–47	31–46	Lower Z drives; Four stock reduction drives, short loops and feet	KEEP	Procedure ; 4 9mm20T shaft pulleys;4 6mm16T motor pulleys;4 6mm188 loops;8 spacer substitution.
48–50	47–49	Upper Z drives; Plastic upper Z idlers/tensioners	REPLACE	Procedure ; Purchase12,9mm top-adjusted CNC; gateM2 frame hardware.
51–51	50–50	Interlude; Historical page	OMIT	Procedure ; No assembly operation.

Printed stock pages	PDF indices	Phase / stock step	Action	Custom replacement
52–60	51–59	Bed; Stock plate, adhesive heater/fuse, four-point screws and pass-through	REPLACE	Procedure ; 06/07/08 integrated; new heater/surface ownership unconfirmed; B1-B5 before affected step.
61–61	60–60	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
62–63	61–62	A/B geometry; Functional A/B orientation	KEEP	Procedure ; Stock topology reference only, not CNC fastener plan.
64–80	63–79	A/B drives/front idlers; Plastic heatsets, smooth idler stacks, motor mounts	REPLACE	Procedure ; 09/11 official PDF1.0; support-bearing shaft engagementM4; opposite single idler levels.
81–81	80–80	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
82–87	81–86	Gantry rear support; Rear E, A/B attachment and chain supports	REPLACE / ADD	Procedure ; Retain extrusion geometry; CNC screws gated; backers/routing before obstructing ends.
88–88	87–87	Y rails; Underside MGN9 rails	KEEP / ADD	Procedure ; 02 on opposite top surfaces; LDO second-from-end mounting hole note conditional actual rail.
89–95	88–94	Y/front gantry attachments; C extrusions/front plastic idlers and upper clips	REPLACE	Procedure ; CNC fasteners/orientationM5; #10/#11 upper-clamp allocationM8.
96–100	95–99	Live XY joints; Plastic XY stacks	REPLACE	Procedure ; 14 PDF1.1; spring/live-shaft levels; M3x3 versusM3x4 retention conflict.
101–101	100–100	X rail; Single front MGN12 on D	KEEP / ADD	Procedure ; 02 rear backerM3; carriage retained rail type verified.
102–106	101–105	X gantry attachment; Stock XY mounting hardware/endstop nut provisions	REPLACE	Procedure ; Use confirmed CNC mounts; M6 STEP support export/actual PCB; no stock bolt transfer.
107–107	106–106	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
108–109	107–108	Z suspension; Gantry location overview	KEEP	Procedure ; Supported gantry and independent Z belt widths.
110–116	109–115	Z joints/bottom capture; Stock spherical joints, lower clips, Hall magnet	REPLACE / OMIT	Procedure ; 13 complete rigid joint/bottom capture; separate #10 bottoms surplus; drawingM7.

Printed stock pages	PDF indices	Phase / stock step	Action	Custom replacement
117–117	116–116	Z tensioning preparation; Plastic top adjuster extension	REPLACE	Procedure ; Use12 documented adjusters; no stock four-turn preparation.
118–121	117–120	Z belt routing; Four loops and captures	KEEP / REPLACE	Procedure ; Keep inward teeth topology; replace capture hardwareM8, verify all9mm planes.
122–122	121–121	Gantry square; Square gantry	KEEP / ADD	Procedure ; Manual alignment and supported whole-Z binding checkM9 before power/QGL.
123–123	122–122	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
124–127	123–126	A/B belt topology; A/B path drawings	KEEP	Procedure ; Stock reference topology only; verified CNC belt planes before threading.
128–128	127–127	XY tensioners; Plastic idler four-turn prep	REPLACE	Procedure ; CNC front idler range; not stock adjustment counts.
129–131	128–130	Bare X carriage/belt start; Stock printed carriage halves/capture	REPLACE	Procedure ; T1 bare CNC carriage preparation executed during04 before belts;6mm width and rail screws.
132–138	131–137	A/B belt routing; Separate belt routes and stock access-bolt tip	KEEP / REPLACE	Procedure ; Preserve topology; do not remove CNC pins/retainers to imitate stock access step.
139–141	138–140	Carriage belt capture; Stock clamp/pivot fasteners	REPLACE	Procedure ; T1 two CNC clamps/correct hardware; no stock M3x30 through plastic.
142–142	141–141	Belt inspection; Rubbing check	KEEP / ADD	Procedure ; M10 complete backer/CNC/bed/toolhead/ cable sweep and measured safe envelope.
143–145	142–144	Stock Z probe/Hall magnets; Omron retention, stock probe height, Hall magnet	OMIT	Procedure ; 05 Cartographer V4 measured heightT4, no contradictory stock probing.
146–147	145–146	StealthBurner; Separate toolhead manual	KEEP / REPLACE	Procedure ; Pinned SB/CW2 core; exact03 FiberUHF candidateT2/T3; no HF duct substitution.
148–151	147–150	Electronics/host mounts; Stock electronics location and Pi bracket	REPLACE	Procedure ; Actual Leviathan/NH revisionsE1 determine integrated host/mount architecture.

Printed stock pages	PDF indices	Phase / stock step	Action	Custom replacement
152–152	151–151	5V PSU; Stock separate 5V PSU	OMIT	Procedure ; LDO board-specific host power; prevent dual supply/backfeed.
153–155	152–154	PSU/controller mounting; PSU frame and controller brackets	KEEP / REPLACE	Procedure ; Keep confirmed PSU support; use actual Leviathan bracketE1.
156–157	155–156	Inlet/SSR; Stock separate switch inlet/SSR arrangement	REPLACE	Procedure ; Actual combined LDO inlet and MRW heater circuitE3; qualified mains installation.
158–161	157–160	Stock nozzle Z endstop; Pin/PCB Z endstop assembly	OMIT	Procedure ; Selected Cartographer virtual Z home after E5/E7; parts surplus.
162–164	161–163	X/Y endstops; Stock pod/PCB mounting	REPLACE	Procedure ; Retain physical X/Y functions; T1 X switch +M6 Vitalii compatible Y support.
165–172	164–171	Electronics bay attachment; Stock AC distribution/5V/supports	REPLACE / OMIT	Procedure ; Verified board mounts/circuit; no separate5V PSU assumed; no stock Octopus pinout.
173–173	172–172	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
174–178	173–177	Controller targets/pinout; Octopus MCU/driver/jumpers	REPLACE	Procedure ; E1 exact PCB; LeviathanV1.3H743 differs genericF446; SB and SBV2 separate.
179–179	178–178	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
180–193	179–192	Power/bed/controller wiring; Stock AC/DC/SSR/sensor routes	REPLACE / KEEP safety	Procedure ; E2/E3 exact heaters/PT1000/fuse/PE; original diagram is not this board pinout.
194–204	193–203	Cable routing; Chains, strain relief and stock toolhead harness	KEEP / REPLACE	Procedure ; Preserve loose flex routing; new USB pathsE4 and backer/CNC bridgesM3/M6; nonchain-rated probe cable excluded.
205–209	204–208	Motors/hotend/probe/fans; Octopus direct hotend/thermistor/probe wiring	REPLACE	Procedure ; NH exact revision; USB notCAN;4.5A V2 continuous does not establish Rapido startup fit.

Printed stock pages	PDF indices	Phase / stock step	Action	Custom replacement
210–213	209–212	Skirt preparation; Printed support removal, inserts and fans	KEEP	Procedure ; Actual350 and fan voltagesP2.
214–216	213–215	Stock LCD; Mini12864 assembly	REPLACE / OMIT	Procedure ; Conditional actual LDO DSI touchscreen option; no stock display installed first.
217–219	216–218	Skirts; Frame attachment	KEEP	Procedure ; Use selected350 parts and serviceable fittings.
220–221	219–220	Stock LCD/EXP; LCD placement and ribbon cables	REPLACE / OMIT	Procedure ; Actual displayE1/P2; no EXP assumption forDSI.
222–233	221–232	Skirts and bottom panel; Retained panels/fans/hinges	KEEP	Procedure ; Guard mains/service covers; do not crush PE/cables.
234–236	233–235	Z belt covers; Retained belt covers	KEEP / ADD option	Procedure ; Conditional LDO LED wire cover; maintain top tensioner access.
237–237	236–236	Interlude; Historical page	OMIT	Procedure ; No assembly operation.
238–249	237–248	Panels/doors; Foam clips magnets hinges	KEEP	Procedure ; ThicknessP2 and final travel/cable sweep.
250–256	249–255	Exhaust; Stock fan/filter housing	OMIT / REPLACE	Procedure ; Fan not confirmed owned; LDO cover+stockgrill, no unrelated filter upgrade.
257–259	256–258	Spool and filament; Spool holder/PTFE retainer	KEEP	Procedure ; Check retained inlet route with omitted exhaust and probe/cable clearance.
260–261	259–260	Startup/calibration; Generic next steps/help	REPLACE / ADD	Procedure ; Detailed prepower, individualmotors/endstops, gated controlled heat and currentCartographer sequence.
262–262 + unnumbered cover	261–262	Closing attribution; Closing page and unnumbered back cover	KEEP	Procedure ; Upstream attribution preserved; index262 back cover has no printed page.

Hardware, wiring and missing requirements

Purchase inventory stays separate from additional requirements. Package titles are not a delivered-content audit. Exact verified stacks, orientations and warnings are in the phase chapters; unknown fasteners, connector views and ratings block the affected operation.

System	Parts / quantities	Verification	Ownership	Gates
Frame / rails / lower Z drives	Kit 01: quantities in chapter 03	I1 carton audit; 6 mm XY / 9 mm Z / 6 mm reduction loops identified separately	Kit-reported; actual contents unverified	I1 M1 F1 F2
Titanium backers	02 listing: 22 M3x8 FHCS + 10 M3x6 FHCS; approximately 30 M3 T-nuts separately	Count actual occupied holes; verify counterbores, nut/extrusion fit and clearance	Extra nuts required but not confirmed owned	M3
Double-shear A/B pair	09: 14 F695; 12 M5 1 mm shims; 2 M5 10 mm sleeves; 6 M3 20 mm sleeves; 8 M3x8 FHCS; 6 M3x35 SHCS; 4 ultra-flat M3x4; 6 cup shims	PDF bearing count conflict; actual motor shaft engagement and motor fasteners unresolved	Purchased set; contents unverified	M1 M4 M5
Front-idler pair	11: 4 F695; 4 M5 1 mm shims; 2 M5 10 mm sleeves; 4 M3x5 FHCS; 4 M3x45 SHCS; 8 M3x6 BHCS	Extrusion screws/T-nuts and clamp supply differ between live page and PDF	Purchased set; contents unverified	M1 M5 M8
Live-idler XY pair	14: 8 F695; 4 M5 1 mm shims; 4 M5 8 mm sleeves; 2 M3 20 mm sleeves; 4 springs; 4 M3x8 FHCS; 2 M3x30 FHCS; 6 M5x10 FHCS; 8 M3x8 BHCS	Eight retention screws: PDF M3x3 vs M3x4 conflict; endstop CAD export/hardware unresolved	Purchased set; contents and retention length unverified	M1 M5 M6
Rigid Z joints	13 listing: four joints; 16 M3x8 + 4 M3x14 + 4 M5x14	Assembly drawing, head styles and screw assignments unconfirmed	Purchased set; no separate lower clamps installed	M7 M8 M9
Upper Z tensioners	12: four preassembled units; listing 8 M5x25; PDF internal contents in chapter 04	Frame M5x16/T-nuts and left/right assignment need exact revision instructions	Purchased set; attachment hardware unverified	M2
Upper Z clamp allocation	10: two top clamps at rear; 11: two front clamps, conditional	Four integrated lower captures from 13; 10 bottom pair and two of 11 clamps surplus if fit/contents confirmed	Purchased; no physical fit validation	M8
Bed supports	07: three different contacts; 08: one installed pair of Wings	Resolve 2x packaging; third-seat ~130 mm extrusion excluded; bolt lengths and support map	Third support / attachments conditional requirement, not confirmed owned	B2 B3 F1

System	Parts / quantities	Verification	Ownership	Gates
Bed thermal system	06: one MRW350 plate, purchased options unknown	New approved heater, sensor, independent thermal fuse, surface, PE lug and strain relief	Required but not confirmed owned; stock LDO bonded assembly left intact	B1 B4 E3
CW2 / StealthBurner	Kit 01 baseline conditional; 03 exact UHF hotend	Core hardware in chapter 06; UHF cartridge / duct / shell geometry must match	Validated UHF printed cartridge required but not confirmed owned	T1 T2 T3
CNC carriage / probe	04: one carriage; 05: one V4 Standard	4 M3x8 rail screws (head by revision); 2 M3x6 clamp screws; X switch 2 M2x10; standard probe 2 M3x6	8.5 mm UHF extenders and longer screws conditional requirement, not confirmed owned	T1 T4
Heater output / PT1000	V2.0.0 only: HE0 PA7, 4.5 A continuous; TH0 PB12, 2200 ohm pull-up	Exact 03 cold/startup load versus MOSFET, contacts, wiring, fuse and shared PSU headroom	Rated ferrules / contacts required but not confirmed owned	E1 E2
Primary USB route	Nitehawk USB + 24 V umbilical; 05 direct host USB proposed	V4 USB requires 5 V; PROBE 24 V prohibited; verify cable rating and connector orientation	Approved direct USB cable / strain relief required but not confirmed owned	E4
Downstream USB alternate	Nitehawk V2 P8 JST-ZH1.5 5P to V4 four-pin Molex Sherlock	Schematic pins 1 VBUS, 2 D-, 3 D+, 4/5 GND; plug-side view reversals, fifth contact and port load need verification	Conditional alternate harness; not claimed purchased	E4
CAN alternate	Only after actual port, firmware, power and cable/termination identification	Leviathan CAN supply 24 V differs USB 5 V; Nitehawk uplinks use USB	Conditional alternate; no CAN adapter purchase claimed	E4 E5
Mains / protective earth	Actual kit inlet, PSU and SSR arrangement plus MRW circuit	Isolation, fuse/SSR/heatsink ratings, dedicated PE, covered terminals, approved circuit	Rated parts required but not confirmed owned; manufacturer requires qualified installer	E1 E3
Nozzle cleaning	No automatic brush macros active	Kit brass brush may be present; keep Cartographer coil away from metal brush	Manual cleaning primary; silicone brush optional and not confirmed owned	T4 E7

Additional requirements and optional conveniences

Required but not confirmed owned means inventory or acquire only after its interface gate is resolved. Optional alternates are not active main-path requirements and do not add to the fourteen reported purchases.

Item	Quantity	Classification	Gate
Compatible new bed heater, sensor, thermal fuse, magnetic sheet/buildsurface if plain plate	Unverified / unset	required-but-not-confirmed-owned	B4
Dedicated PElead, appropriate M5attachment terminal/hardware and bed cable strainrelief	Unverified / unset	required-but-not-confirmed-owned	B4

Item	Quantity	Classification	Gate
Thirdseat ~130mm extrusion or manufacturer-supported Wings layout support	Unverified / unset	conditional-required-but-not-confirmed-owned	B3
2F UHF validated printed cartridge/frontrear and matching shell	Unverified / unset	required-but-not-confirmed-owned	T2
CNC8.5mm UHF extenders and suitable fasteners	Unverified / unset	conditional-required-but-not-confirmed-owned	T4
Exact boardrevision mounts and umbilical cable relief	Unverified / unset	required-but-not-confirmed-owned	T1
Silicone nozzle brush/location	Unverified / unset	optional-not-confirmed-owned; manual initial cleaning permitted	T4
Approved MRW-compatible bed heater and magnetic/spring-steel surface	Unverified / unset	required but not confirmed owned	E3
Correct thermal fuse, attachment materials and approved bed PE lug/hardware	Unverified / unset	required but not confirmed owned	E3
Rated SSR/heatsink, inlet/branch fuses and approved mains harness appropriate to actual bed	Unverified / unset	required but not confirmed owned	E3
Approved direct host USB-to-Cartographer V4 Standard cable/strain relief	1	required but not confirmed owned	E4
Exact downstream USB P8-to-Sherlock cable if that route is chosen	1	conditional requirement; not confirmed owned	E4
Heater ferrules/connector contacts fitting actual wires and board	Unverified / unset	required but not confirmed owned	E2
Nitehawk static discharge links and revision-correct fan adapter	Unverified / unset	kit contents to verify; not confirmed owned	E1
CAN interface and approved cable/termination if USB is not chosen	Unverified / unset	conditional alternate; not purchased evidence	E4
Nozzle brush/purge bucket	0	optional convenience, omitted from primary path	Unverified / unset

Purchase disposition — all fourteen entries

Every installed count below is a **conditional plan**. Actual installation and package contents are unset. User-provided purchase verification is independent of installation compatibility.

ID / purchase	Evidence / variant	Units / package pieces	Planned installed / surplus	Disposition and gate	Build location
01 LDO Voron V2.4R2 kit	user-reported purchase; receipt not supplied; 350 provisional; LDO Rev D-family unverified; serial/batch and PCB revisions unknown	1 kit; Unverified / unset pieces/unit	1 / Unverified / unset; stock upgraded subassemblies; exact piece count after carton audit	KEEP frame, rails, four lower Z reduction drives and compatible electronics/CW2/SB; replace upgraded functions; I1 E1 F1 F2	Procedure
02 Titanium Backers three-pack	user-reported purchase; receipt not supplied; Ordered length/size unknown	1 pack; 3 pieces/unit	3 / 0;	ADD opposite X/Y rails before closing gantry; M3	Procedure
03 Phaetus Rapido 2 Fiber UHF / PT1000	user-reported purchase; receipt not supplied; UHF/PT1000 reported; revision/voltage/heater/nozzle unknown	1 hotend; 1 pieces/unit	1 / 0; kit hotend, do not install	REPLACE stock kit hotend with exact Fiber UHF after geometry and load gates; T2 T3 E2	Procedure
04 Cartographer CNC StealthBurner/toolhead mount	user-reported purchase; receipt not supplied; Belt-width and extender inclusion unverified	1 mount assembly; Unverified / unset pieces/unit	1 / Unverified / unset; stock carriage/probe retention hardware	REPLACE stock carriage; belt variant and UHF extenders must match; T1 T2 T4 M1 M5 M10	Procedure
05 Cartographer V4 AIO Standard	user-reported purchase; receipt not supplied; USB/CAN-capable; actual firmware generation and interface unknown	1 probe; 1 pieces/unit	1 / 0; unused kit Z sensing hardware	REPLACE stock Z sensing and homing path; USB route provisional only; T4 E4 E5 E7	Procedure
06 MRW Voron 350 UltraFlat bed	user-reported purchase; receipt not supplied; 350 reported; standard vs embedded magnets/heater option unknown	1 plate; 1 pieces/unit	1 / 0; stock plate/heater/magnet remain bonded	REPLACE whole stock LDO bed assembly, no bonded-pad reuse assumed; B1 B3 B4 E3 E7	Procedure
07 MRW Matched Height Kinematic Kit	user-reported purchase; receipt not supplied; Three support points documented; verify supplied hardware and seats	1 set; 3 pieces/unit	3 / 0; stock thumb nuts/spacers; quantity audit pending	ADD three matched support points integrated with Wings and bed; B2 B3	Procedure

ID / purchase	Evidence / variant	Units / package pieces	Planned installed / surplus	Disposition and gate	Build location
08 MRW Voron Wings (purchase notation 2x)	user-reported purchase; receipt not supplied; A current retail set is two wings; purchase notation does not prove number of sets	2 unreconciled purchase units; Unverified / unset pieces/unit	2 / Unverified / unset; 0 if two pieces delivered; 2 if two full two-piece sets delivered	ADD one installed pair; reconcile 2 pieces vs 2 sets before recording surplus; B2 B3	Procedure
09 Vitalii3D Double Shear v2.4 AB Motors Mounts	user-provided order verification; Black/Red; ABEC7 hybrid ceramic; actual part revision unverified	1 set; 2 pieces/unit	2 / 0; stock unused bearing/fastener quantities pending carton audit	REPLACE two stock A/B drives; verify shafts truly engage support bearings; M1 M4 M5	Procedure
10 Vitalii3D CNC Z Belt Clamps four-piece set	user-provided order verification; Black; product lists 2 top M5+M5 and 2 bottom M5+M3	1 set; 4 pieces/unit	2 / 2; two purchased bottom clamps, integrated rigid joints take their function	REPLACE rear upper captures with top pair conditionally; bottom pair surplus; M8	Procedure
11 Vitalii3D Front Idlers for v2.4	user-provided order verification; Black/Black; ABEC7 hybrid ceramic; shipped revision and clamps unknown	1 set; 2 pieces/unit	2 / 0; two of four supplied clamps if four present and proposed fit verified	REPLACE front-idler pair; supplied clamps tracked as separate package suballocation; M1 M5 M8	Procedure
12 Vitalii3D Z Tensioners for v2.4	user-provided order verification; Black/Orange; product set four preassembled; fit actual frame hardware	1 set; 4 pieces/unit	4 / 0; stock upper printed tensioner/idler hardware	REPLACE upper 9mm Z idlers, retain lower 6mm loop tensioning latch; M1 M2	Procedure
13 Vitalii3D Rigid Fixed Z Joints	user-provided order verification; Black; four-piece manufacturer set; actual screw assignment unresolved	1 set; 4 pieces/unit	4 / 0; stock joints/stock lower clips; #10 bottom pair redundant	REPLACE complete Z joints and all lower belt captures; square before powered Z; M7 M8 M9	Procedure
14 Vitalii3D Live Idler XY Joints 6mm	user-provided order verification; Black/Black; ABEC7 hybrid ceramic; 6mm purchased; current PDF revision1.1	1 set; 2 pieces/unit	2 / 0; unused stock XY parts	REPLACE two XY joints; resolve bearing screw length and endstop mount; M1 M5 M6 M10	Procedure

Wings remain 2x with two possible interpretations (two pieces or two two-piece sets). #10 is two top and two bottom clamps: bottoms surplus under the rigid-joint plan. #11 may include four extra clamps; plan two front uppers and two surplus after package/fit verification M8. The original LDO bonded bed remains intact as surplus. See [machine-readable manifest](#).

Printed-parts matrix

The filenames are exact source members. Current upstream files, including unused sizes and alternatives, are accounted for. Quantity is the planned print/installed-piece count, not number of files. `null` means resolve first; a suffix quantity only applies if that alternative is selected. Included-by-LDO means the supplier BOM says it is provided, not that this carton was counted.

Upstream functional-print guidance: ABS, 0.2 mm layers, forced 0.4 mm extrusion width, 40% infill, 4 walls, and 5 top/bottom layers (manual p4). Honor part-specific translucent/opaque guidance for LEDs and any verified vendor instructions. STEP members are not print-ready STLs. No standard HF cartridge is activated for the UHF purchase.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Clockwork2/Direct_Drive/main_body.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/Direct_Drive/motor_plate.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/Direct_Drive/[a]_guidler_a.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/Direct_Drive/[a]_guidler_b.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/Direct_Drive/[a]_latch.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/Direct_Drive/[a]_latch_shuttle.stl	1	required if CW2 confirmed	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/[a]_stealthburner_main_body.stl	1	conditional UHF fit	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/Printheads/phaetus_rapido_v2/stealthburner_printhead_rapido_v2_front.stl	0	omit; HF mount not verified for purchased UHF	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/Printheads/phaetus_rapido_v2/stealthburner_printhead_rapido_v2_rear_cw2.stl	0	omit; HF mount not verified for purchased UHF	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/cable_door_for_pcb.stl	Unverified / unset	unresolved board/umbilical variant	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/[a]_pcb_spacer.stl	Unverified / unset	unresolved board/umbilical variant	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Clockwork2/chain_anchor_2hole.stl	Unverified / unset	unresolved board/umbilical variant	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Clockwork2/chain_anchor_3hole.stl	Unverified / unset	unresolved board/umbilical variant	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/[o]_stealthburner_LED_carrier.stl	1	optional lighting	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/[c]_stealthburner_LED_diffuser.stl	1	optional lighting	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Stealthburner/[o]_stealthburner_LED_diffuser_mask.stl	1	optional lighting	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Tools/SB_5015_Cutting_Tool_A.stl	1	optional fan tool	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
STLs/Tools/SB_5015_Cutting_Tool_B.stl	1	optional fan tool	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	Use documented source guidance; verify the actual interface before printing.
voron 2.4-Rapido 2/uhf前.stl	1	unresolved exact combined UHF fit	7401934e28d56844ba4f311e9d588b1e4d0510dd	Use documented source guidance; verify the actual interface before printing.
voron 2.4-Rapido 2/uhf 后.stl	1	unresolved exact combined UHF fit	7401934e28d56844ba4f311e9d588b1e4d0510dd	Use documented source guidance; verify the actual interface before printing.
STLs/Gantry/AB_Drive_Units/a_drive_frame_upper.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/AB_Drive_Units/a_drive_frame_lower.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/AB_Drive_Units/b_drive_frame_upper.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/AB_Drive_Units/b_drive_frame_lower.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Front_Idlers/front_idler_left_upper.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Front_Idlers/front_idler_left_lower.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Front_Idlers/front_idler_right_upper.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Gantry/Front_Idlers/front_idler_right_lower.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Front_Idlers/[a]_tensioner_left.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Front_Idlers/[a]_tensioner_right.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/X_Axis/XY_Joints/xy_joint_left_upper_MGN12.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/X_Axis/XY_Joints/xy_joint_left_lower_MGN12.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/X_Axis/XY_Joints/xy_joint_right_upper_MGN12.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/X_Axis/XY_Joints/xy_joint_right_lower_MGN12.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Z_Idlers/z_tensioner_bracket_a_x2.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Z_Idlers/z_tensioner_bracket_b_x2.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Z_Idlers/[a]_z_tensioner_9mm_x4.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Z_Joints/z_joint_upper_x4.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Z_Joints/z_joint_lower_x4.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/Z_Joints/z_joint_upper_hall_effect.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/[a]_z_belt_clip_lower_x4.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Purchased CNC replacement from outset.
STLs/Gantry/[a]_z_belt_clip_upper_x4.stl	0	replaced_omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Metal replacement from the outset; upper stock clip omitted only after M8 confirms four CNC upper interfaces.
Live Shaft XY/Endstop_PCB_Mount_h eat_inserts.step	Unverified / unset	unresolved	Live_Shift_XY.zip query v=1779734631; hash in sources	Select/export after actual PCB and bump fit; no manufacturer print settings, insert/screw spec or reuse permission established.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
Live Shaft XY/2x_PCB_Spacer.step	Unverified / unset	unresolved	Live_Shift_XY.zip query v=1779734631; hash in sources	Select/export after actual PCB and bump fit; no manufacturer print settings, insert/screw spec or reuse permission established.
Live Shaft XY/Endstop_Bump.step	Unverified / unset	unresolved	Live_Shift_XY.zip query v=1779734631; hash in sources	Select/export after actual PCB and bump fit; no manufacturer print settings, insert/screw spec or reuse permission established.
Live Shaft XY/Endstop_Bump_Deep.s tep	Unverified / unset	unresolved	Live_Shift_XY.zip query v=1779734631; hash in sources	Select/export after actual PCB and bump fit; no manufacturer print settings, insert/screw spec or reuse permission established.
extrusion_backers/STL s/XY_cable_chain_bridg e-3hole- 3mm_backer.stl	Unverified / unset	unresolved	62268ed817878e54d6a11 86882060aa8368d4f0f	Candidate only if actual cable-chain routing requires it; 3mm name does not verify 3.2mm backer fit.
extrusion_backers/STL s/XY_cable_chain_bridg e-Igus-3mm_backer.stl	Unverified / unset	unresolved	62268ed817878e54d6a11 86882060aa8368d4f0f	Candidate only if actual cable-chain routing requires it; 3mm name does not verify 3.2mm backer fit.
STLs/Leviathan_bracke t_set.stl	Unverified / unset	unresolved board- dependent	878c3c4893ea353552c95 7f574509f9890e1e37a	One bracket set if actual Leviathan matches; RevD BOM claims left/right included; confirm E1
STLs/usb_adapter_moun t_partial_cover.stl	1	required after E1/T1	42ae497cfd627acde0f6ad 81c8dd26f51efcbb48	V2 adapter partial cover if exact docs/revision confirmed E1; kit inclusion unverified
STLs/cw2_captive_pcb_ cover.stl	1	required after E1/T1	be93526b948d3339c755c b32116baef221ce1a8d	Reused original source part linked by V2 docs; verify actual board/clearance E1/T1
STLs/cw2_chain_anchor _tilted.stl	Unverified / unset	unresolved board- dependent	be93526b948d3339c755c b32116baef221ce1a8d	One if confirmed cable- chain route; USB umbilical/backer/probe cable rating E4/M3 may require different relief
STLs/usb_adapter_moun t.stl	Unverified / unset	unresolved board- dependent	be93526b948d3339c755c b32116baef221ce1a8d	One only for original SB adapter after E1; do not install original/V2 alternate mounts together
STLs/bed_wago_mount.s tl	1	included by LDO	5b0496a5ae8d0d6a3f782 dfa7e06eb9e1c102ba5	RevD includes one; verify actual connector, MRW mounting/strain relief and ratings E3

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/BTT Pi TFT4.3 Mount/[a]_faceplate.stl	Unverified / unset	unresolved	5b0496a5ae8d0d6a3f782dfa7e06eb9e1c102ba5	One for matching actual LDO/BTT4.3 display E1/P2
STLs/BTT Pi TFT4.3 Mount/mount.stl	Unverified / unset	unresolved	5b0496a5ae8d0d6a3f782dfa7e06eb9e1c102ba5	One standard mount OR thick mount, not both; verify actual panel/display E1/P2
STLs/BTT Pi TFT4.3 Mount/mount_thick.stl	Unverified / unset	unresolved	5b0496a5ae8d0d6a3f782dfa7e06eb9e1c102ba5	One thick-panel variant if matching actual panel; do not install both alternatives
STLs/Electronics_Bay/Controller_Mounts/BIGDIPPER_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/BTT_MOT_EXP_bracket.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/Duet2_Duet3Mini5_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/GTR_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/Octopus_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/S6_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/SKR_Pro_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/SKR_bracket_inline_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Controller_Mounts/Spider_bracket_set.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/Other_PS_Mounts/UHP_200_Mount_x2.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Match actual PSU/host board revision before selecting E1.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Electronics_Bay/ Other_PS_Mounts/UHP_35 0_Mount_x2.stl	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Match actual PSU/host board revision before selecting E1.
STLs/Electronics_Bay/ PSU_stabilizer_50mm.st l	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Match actual PSU/host board revision before selecting E1.
STLs/Electronics_Bay/ lrs_200_psu_bracket_x2 .stl	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Two brackets if actual LRS-200-24 confirmed; no PSU substitution assumed.
STLs/Electronics_Bay/ pcb_din_clip_x3.stl	4	included by LDO	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Rev D BOM claims four clips; verify delivered count and actual board brackets E1. File name x3 is stock count.
STLs/Electronics_Bay/ raspberrypi_bracket.st l	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Match actual PSU/host board revision before selecting E1.
STLs/Electronics_Bay/ rs25_psu_bracket.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Provisional Leviathan replaces alternate controller and separate 5V PSU mounts.
STLs/Electronics_Bay/ wago_221- 415_mount_3by5.stl	1	required	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Exhaust_Filter/[a]_exhaust_fan_grill.s tl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Stock exhaust-fan assembly omitted; its fan ownership is not confirmed. LDO cover closes opening.
STLs/Exhaust_Filter/[a]_exhaust_filter_moun t_x2.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Stock exhaust-fan assembly omitted; its fan ownership is not confirmed. LDO cover closes opening.
STLs/Exhaust_Filter/[a]_filter_access_cover .stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Stock exhaust-fan assembly omitted; its fan ownership is not confirmed. LDO cover closes opening.
STLs/Exhaust_Filter/e xhaust_filter_grill.st l	1	required	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Exhaust_Filter/e xhaust_filter_housing. stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Stock exhaust-fan assembly omitted; its fan ownership is not confirmed. LDO cover closes opening.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Gantry/AB_Drive_Units/[a]_cable_cover.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Confirm retained chain/USB umbilical/backer clearance at M3/M6/E4 before selecting.
STLs/Gantry/AB_Drive_Units/[a]_z_chain_retainer_bracket_x2.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Confirm retained chain/USB umbilical/backer clearance at M3/M6/E4 before selecting.
STLs/Gantry/X_Axis/XY_Joints/[a]_endstop_pod_D2F_switch.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/XY_Joints/[a]_endstop_pod_hall_effect.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/XY_Joints/[a]_xy_joint_cable_bridge_2hole.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/XY_Joints/[a]_xy_joint_cable_bridge_3hole.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/X_Carriage/pinda_adapter.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/X_Carriage/probe_retainer_bracket.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/X_Carriage/probe_retainer_bracket_9mm.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/X_Carriage/x_frame_V2TR_MGN12_left.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/X_Axis/X_Carriage/x_frame_V2TR_MGN12_right.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC carriage/live joints replace stock geometry, endstop and probe supports.
STLs/Gantry/z_chain_bottom_anchor.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Confirm retained chain/USB umbilical/backer clearance at M3/M6/E4 before selecting.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Gantry/z_chain_guide.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Confirm retained chain/USB umbilical/backer clearance at M3/M6/E4 before selecting.
STLs/Panel_Mounting/Front_Doors/door_hinge_x6.stl	6	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/Front_Doors/handle_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/Front_Doors/handle_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/Front_Doors/latch_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/bottom_panel_clip_x4.stl	4	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/bottom_panel_hinge_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/corner_panel_clip_4mm_x8.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.
STLs/Panel_Mounting/corner_panel_clip_6mm_x8.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.
STLs/Panel_Mounting/deck_support_3mm_x8.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.
STLs/Panel_Mounting/deck_support_4mm_x8.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Panel_Mounting/m idspan_panel_clip_4mm_ x7.stl	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.
STLs/Panel_Mounting/m idspan_panel_clip_6mm_ x8.stl	Unverified / unset	unresolved	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Choose one clip-thickness family after actual panel measurement F2/P2; filename gives count for chosen family.
STLs/Panel_Mounting/z _belt_cover_a_x2.stl	2	required	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Panel_Mounting/z _belt_cover_b_x2.stl	2	required	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/250/front _skirt_a_250.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/250/front _skirt_b_250.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/250/rear_ center_skirt_250.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/250/side_ skirt_a_250_x2.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/250/side_ skirt_b_250_x2.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/300/front _skirt_a_300.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/300/front _skirt_b_300.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/300/rear_ center_skirt_300.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/300/side_ skirt_a_300_x2.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.
STLs/Skirts/300/side_ skirt_b_300_x2.stl	0	replaced/omit	a192410e27ea345644ae5 c4b29b4c9c40cbe1a73	Wrong size for provisional 350 target; revisit only if I1 changes it.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Skirts/350/front_skirt_a_350.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/350/front_skirt_b_350.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/350/rear_center_skirt_350.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/350/side_skirt_a_350_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/350/side_skirt_b_350_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_belt_guard_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_belt_guard_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_btt_knob_light_shield.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/[a]_fan_g_rill_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_fan_g_rill_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_fan_g_rill_open_optional_x2.stl	2	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Alternative open grille; not an additional installed pair.
STLs/Skirts/[a]_fan_g_rill_retainer_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/[a]_keystone_blank_insert.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Skirts/[a]_mini12864_case_front_insert.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/[a]_mini12864_case_hinge.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/keystone_panel.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Skirts/mini12864_case_front.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/mini12864_case_rear.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/power_inlet_IECGS_1.2mm.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/power_inlet_IECGS_1mm.stl	Unverified / unset	unresolved	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Required if actual inlet is LDO 1.0mm; verify E1/P2 before print.
STLs/Skirts/power_inlet_filtered.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Not the provisional LDO D-family display/inlet path; verify actual options E1/P2.
STLs/Skirts/side_fan_support_x2.STL	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Spool_Management/bowden_retainer.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Spool_Management/spool_holder.stl	1	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Superceded_Parts/Frame_Brackets/dual_ramps_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/duet_duex_bracket_frame_x2.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Superceded_Parts/Frame_Brackets/lrs_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/mks_gen_1.4_bracket_frame_x2.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/mks_gen_L_bracket_frame_x2.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/raspberry_pi_bracket.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/rs_25_psu_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/rs_35_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/skr_1.3_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/Frame_Brackets/skr_mini_e3_bracket_frame.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/[a]_belt_clamp_MGN9_x2.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/x_carriage_frame_left_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/x_carriage_frame_right_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/x_carriage_pivot_block_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/xy_joint_left_lower_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/xy_joint_left_upper_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/MGN9_X/xy_joint_right_lower_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Superceded_Parts/MGN9_X/xy_joint_right_upper_MGN9.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/XY/zipchain2_xy_end.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/XY/zipchain2_xy_link_a.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/XY/zipchain2_xy_link_b.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/Z/zipchain2_z_end.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/Z/zipchain2_z_link_a.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/Z/zipchain2_z_link_b.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/ZipChain/Z/zipchain2_z_link_b_locking.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/[a]_stopgap_80T_hubbed_gear.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Superceded_Parts/[a]_z_tensioner_x4_6mm.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Superseded upstream alternative; not the R2 main path.
STLs/Test_Prints/Filament_Card.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Test_Prints/Filament_Card_Caddy_25.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Test_Prints/Heatset_Practice.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Test_Prints/Thread_Test_1_x1_Rev1.STL	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Test_Prints/Thread_Test_2_x1_Rev1.STL	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Test_Prints/Thread_Test_3_x1_Rev1.STL	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Test_Prints/Voron_Design_Cube_v7.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Tools/MGN12_rail_guide_x2.stl	2	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Tools/MGN9_rail_guide_x2.stl	2	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Tools/bed_hole_marking_template_x1_Rev2.STL	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Tools/bottom_panel_template.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Tools/pulley_jig.stl	1	optional	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	Calibration or assembly aid; not an installed machine part.
STLs/Z_Drive/[a]_belt_tensioner_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/[a]_belt_tensioner_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/[a]_z_drive_baseplate_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/[a]_z_drive_baseplate_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/z_drive_main_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/z_drive_main_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/z_drive_retainer_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/z_drive_retainer_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.

Exact filename	Planned qty	Disposition	Pinned source revision	Guidance / gate
STLs/Z_Drive/z_motor_mount_a_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Drive/z_motor_mount_b_x2.stl	2	required	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	KEEP; upstream filename count and retained manual assembly. ABS guidance p4.
STLs/Z_Endstop/nozzle_probe.stl	0	replaced/omit	a192410e27ea345644ae5c4b29b4c9c40cbe1a73	CNC tensioners or Cartographer replace this function.
STLs/nozzle_probe_ldo.stl	0	replaced/omit	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Cartographer Z homing omits kit nozzle Z endstop; keep supplied piece as surplus.
STLs/led_fan_pcb_spacer_x2.stl	2	included by LDO	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Two in Rev D BOM; verify carton and PCB fit.
STLs/cw2_offset_chain_anchor.stl	Unverified / unset	unresolved	8270e8cf6c7ba29a7fd11d143c287bd3af576934	LDO original two-hole chain anchor; verify actual Nitehawk/routing/backer fit.
STLs/exhaust_cover.stl	1	required	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Seal unused exhaust with stock grill; no extra filter modification added.
STLs/handlebar_spacer_x4.stl	4	optional	8270e8cf6c7ba29a7fd11d143c287bd3af576934	If actual included handles installed; M5x14 + hammerhead per LDO supplement.
STLs/z_rail_stop_x4.stl	4	optional	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Captive-rail safety aid; temporary stops needed while handling.
STLs/ldo_bestagon_insert.stl	1	optional	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Decorative kit insert only.
STLs/z_belt_cover_alled.stl	2	optional	8270e8cf6c7ba29a7fd11d143c287bd3af576934	Alternative A cover if LED wires route through Z motor opening; not added to stock covers.
STLs/Purge Bucket/brush_holder_sheet_stop.stl	0	replaced/omit	8270e8cf6c7ba29a7fd11d143c287bd3af576934	No automatic brass brush route across Cartographer coil; manual initial cleaning.
STLs/Purge Bucket/individual_sheet_stop.stl	Unverified / unset	unresolved	8270e8cf6c7ba29a7fd11d143c287bd3af576934	MRW surface/Wings/probe envelope must determine sheet retention, B1/B5/T4.

Unresolved questions and verification gates

All gates are open at publication because no delivered-hardware inspection or physical measurement was supplied. Close only with the precise evidence, then update the manifest, affected chapter and validation record. Continue independent work; stop before the affected operation.

Gate	Evidence needed	Affected operation / block	Procedure
B1	Identify MRW350 exact SKU/variant, magnet arrangement, actual dimensions, heater option and top/underside hole photos	Bed adhesive/surface assembly; Cartographer scanning; assembly and calibration	Before step
B2	Count technical package contents and revision-matched paper instructions; resolve M3 lengths/M2 head type; resolve Wings 2x unit and pieces per set	Kinematic seats and installed/surplus counts; assembly	Before step
B3	Manufacturer Wings size/revision map of three contact positions, pin blocks, third-seat support and mounting screws; physical plate/frame/deck/DIN hole and dry-fit measurements. Finished gantry/nozzle/belt envelope remains B5/M10.	Early support/frame/deck/DIN mapping, then Wings/frame/balls final assembly; assembly	Before step
B4	Exact heater label/install sheet, voltage/power/sensor, fuse selection, M5 attachment hardware and PE method, compatible surface ownership	Adhesion, thermal safety and heater route; assembly and electrical commissioning	Before step
B5	Measured bed top height, complete UHF nozzle/coil position, manual full-envelope clearance record with intended removable surface	Ztravel/XYreach limits and powered motion; electrical commissioning and calibration	Before step
T1	Motor/gears/extruder identification; carriage width/revision and Xrail type; belt capture/fasteners/endstop hardware and mixed-gantry alignment	CW2/carriage assembly and belt/endstop fastening; assembly	Before step
T2	Exact manufacturer-supported UHF cartridge and SB/CW2 shell fit; candidate CAD comparison plus cold physical fit and duct alignment	Printing/assembly of UHF cartridge; assembly	Before step
T3	Delivered exact 2F revision,label,instructions,nozzle, PT1000 and heater specs; revision-matched support/mount screws	Hotend assembly and load handoff; assembly and electrical commissioning	Before step

Gate	Evidence needed	Affected operation / block	Procedure
T4	V4Standard and CNCextender variant identification; measured coil2.6-3mm above tip; actual offsets, keepout and full envelope clearance	Probe installation and probing; assembly and calibration	Before step
E1	Photos of mainboard/toolboard/fan adapter silkscreen and MCU; PSU labels/selectors; actual fuse/harness labels; kit packing list, separate from current store listing	assembly wiring; firmware selection; assembly wiring; firmware selection	Before step
E2	Exact hotend revision/voltage/rated current/cold resistance/startup/peak documentation; PCB continuous/pulsed rating; MOSFET, terminal, umbilical/contact, wire, fuse and PSU shared-load ratings	hotend heater wiring; heater commissioning; hotend heater wiring; heater commissioning	Before step
E3	Exact bed variant and heater/sensor options; voltage/power/physical bond instructions; thermal fuse data/mounting; SSR/heatsink and fuse ratings; PE lug procedure; qualified inspection	bed wiring; power-on approval; bed heater commissioning; bed wiring; power-on approval; bed heater commissioning	Before step
E4	V4 Standard marking/firmware interface; cable/contact identities; oriented probe/host or downstream P8 pin-to-pin mapping; supply voltage and usable current; chain/umbilical bend/strain requirements	probe wiring; probe energization; probe wiring; probe energization	Before step
E5	Actual board bootloader/firmware and host Klipper versions; compatible released new-plugin/Survey pair; exact binary/hash and installed plugin version/commit; no legacy sections/macros	firmware configuration; Z homing; probing/calibration; firmware configuration; Z homing; probing/calibration	Before step
E6	Manual gantry square and smooth full-Z travel; individual motor/corner map and actual motor-current ratings; physical X/Y endstop mount/state; conservative measured travel envelope	powered motion; Z homing; QGL; powered motion; Z homing; QGL	Before step

Gate	Evidence needed	Affected operation / block	Procedure
E7	MRW magnet arrangement (embedded magnets fail current scan path), installed spring sheet, measured probe/nozzle offsets and 2.6-3mm coil/nozzle relationship, safe central reference, QGL points, nozzle/coil reachable mesh bounds	scan calibration; Z homing; QGL; mesh; first print; scan calibration; Z homing; QGL; mesh; first print	Before step
M1	Identify revisions, actual bag counts and independent 6mm XY / 9mm Z widths.	Assembly of affected CNC subassembly; Assembly of affected CNC subassembly	Before step
M2	Obtain frame mounting drawing/fasteners and verify Z-idler alignment and top driver access.	Z tensioner mounting; Z tensioner mounting	Before step
M3	Check backer build-size variant, measured lengths/holes/thickness, T-nuts and routing clearance.	Backer mounting and closing gantry ends; Backer mounting and closing gantry ends	Before step
M4	Measure actual A/B shaft; prove support-race engagement, pulley plane and approved motor screws.	Motor mounting and belt loading; Motor mounting and belt loading	Before step
M5	Confirm CNC/extrusion and rail mount screws, orientation and combined belt planes with backers.	Final CNC gantry attachment; Final CNC gantry attachment	Before step
M6	Identify actual X/Y PCB; obtain matching printable export, inserts/screws and bump trigger fit.	Endstop mounting and XY homing; Endstop mounting and XY homing	Before step
M7	Obtain rigid-joint installation drawing, orientation, screw-to-hole assignment and approved rail fit.	Rigid-joint fastening and cutting Z belts; Rigid-joint fastening and cutting Z belts	Before step
M8	Count clamps and prove proposed front/rear top allocation by hole/tooth width and screw fit.	Z belt capture; Z belt capture	Before step
M9	Supported hand inspection through safe Z travel for binding, carriage lift and belt tracking.	Powered Z and quad gantry leveling; Powered Z and quad gantry leveling	Before step
M10	Full hand XYZ travel check with final carriage/toolhead/bed/cables and measured safe envelope.	Powered XY and software travel limits; Powered XY and software travel limits	Before step
I1	Actual kit label/packing list; measure all extrusion lengths and identify kit size, LDO revision/batch, options. Do not commit private receipt.	Size-dependent printing, belts and assembly; Size-dependent printing, belts and assembly	Before step

Gate	Evidence needed	Affected operation / block	Procedure
P1	Resolve filenames/interface before printing any UHF hotend, endstop or PCB mount; count LDO supplied parts.	Affected printed part preparation; Affected printed part preparation	Before step
F1	Verified Wings/kinematic three-point layout, frame attachments and cable route; do not fix bed supports at stock spacing. Verify fixed deck/DIN holes align with resolved support positions before closing deck; obtain documented replacement layout if not.	Bed support fastening, final deck/DIN installation and passthrough location; Bed support fastening, final deck/DIN installation and passthrough location	Before step
F2	Measure shipped deck thickness; resolve FAQ 4mm vs RevD BOM 3mm discrepancy and select clips.	Deck clip selection and mounting; Deck clip selection and mounting	Before step
P2	Panel thickness, skirt size, actual inlet/display/fan voltage and supplied mounts.	Electronics enclosure closure and display wiring; Electronics enclosure closure and display wiring	Before step

Assembly gaps block only affected subassemblies. Missing board/load/cable/fuse/PE evidence blocks electrical commissioning; missing offsets and travel checks blocks calibration. A software limit is not evidence of safe electrical hardware.

Source lock and illustration provenance

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mrw350	Unversioned; snapshot hash	2026-10-09T06:22:35+00:00	9e8f262530183978c7ac60105acf771c298ce0d12a79dcb513608070fe8ab76d	Unverified reuse terms; link only
mrw-kinematic-product	Unversioned; snapshot hash	2026-10-09T06:22:35+00:00	56eea44d0c76325687e3b44b89ca4d8c4874e720095e096b88bb7f775026108b	Unverified reuse terms; link only
mrw-kinematic-pdf	PDF metadata: created/modified 2022-04-08 20:40:58 UTC	2026-10-09T06:22:35+00:00	ff56419b66ec20d32bb5f6968667ccb22778ca8261e004428ba04d58198dcd8e	Unverified reuse terms; link only
mrw-wings	Unversioned; snapshot hash	2026-10-09T06:22:35+00:00	f9699fda5c1fa4230c00c56b5733e018a3c2b54d209e3ac88beaf2e501c1187d	Unverified reuse terms; link only
mrw-wings-variant-data	Unversioned; snapshot hash	2026-10-09T06:22:35+00:00	a7b010b342a64a5b833d7378b925c9ded76a88618822b52ba372e083c4fd7291	Unverified reuse terms; link only
mrw-beta-observations	Pre-production tester observations; four unnumbered pages	2026-10-09T06:22:35+00:00	89c2b73cb03ebac432726586a0a5b857be93d1190f371a24e5f4267d8bad08e2	Unverified reuse terms; link only
west3d-rapido2f	Unversioned; snapshot hash	2026-10-09T06:22:35+00:00	5bb658f55d978e852ca9259156483decdbc258cc06cdb9ad1dc468228924cf94	Unverified reuse terms; link only
stealthburner-repo	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	2026-10-09T06:22:35+00:00	1342dd0d72987781ee60bbded4607019c290d6628473715916806f10318366dbb	GPL-3.0 repository; preserve attribution
stealthburner-manual	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	2026-10-09T06:22:35+00:00	9cea9b533c114658a22c59d51c0766e6b2e51d5b4584277c873858b392e87cb4	GPL-3.0 repository; preserve attribution
stealthburner-rapido-v2	8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b	2026-10-09T06:22:35+00:00	aafdc6020b3b3a3d94c42d78b70770e85b538f9baa25fdc3d3316b026bcedb23	GPL-3.0 repository; preserve attribution
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phaetus-rapido2f-cad	7401934e28d56844ba4f311e9d588b1e4d0510dd	2026-10-09T06:22:35+00:00	acf694696a8a783eade95ad3df0229f18a88b23efecfbad0aa4400dd305e19	Models personal use only; commercial use prohibited; link only

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cartographer-v4-standard-cad	bf01749f16f239b5310aaf03e96b18e9dfb9b832	2026-10-09T06:22:35+00:00	7496da4af02fd3bfe8eec6fcdee32f5b6c41682f26b159305d24595222e6111	Unverified reuse terms; link only
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cartographer-faq	Page labels last update 2024-12-17; retrieved current	2026-10-09T06:22:35+00:00	bbe340924d1dfaa6f748ea77f4db6eb7d5888c1136176a49a145b6617d8ab38b	Unverified reuse terms; link only
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E-NHV2-FW	42ae497cfd627acde0f6ad81c8dd26f51efcbb48	2026-10-09	a4ba1b675520a4f7907c521ab9e421ac09b6c01457ed62c2cc775b8cc3b939c1	unknown; no LICENSE/COPYING in pinned recursive tree

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E-NH-PINOUT	photographed V1.4; not proof of actual board	2026-10-09	8dc20dde15498621161751e8b4188bda7c11e0bd887d64c49a3b25620c94a4e6	all rights reserved (LDO page metadata alr)
E-LEV-MANUAL	878c3c4893ea353552c957f574509f9890e1e37a	2026-10-09	e64a191ec1c6f2f9d7dd744446acd592b5734a5cb23135e5b2e13d62768e1872	GPL-3.0
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Build, check, browse and export

Read [the guide entry point](#). The output is a portable static site with local illustrations, a responsive navigation menu and a complete print edition. It needs no server, CDN, account or network to browse after building. Manufacturer manuals linked for uncertain reuse permission still require access to their source.

Reproducible commands

From repository root with Python 3.10 or newer (validated with 3.10.12):

```
python3 -m venv docs/custom-build/.venv
docs/custom-build/.venv/bin/pip install -r docs/custom-build/requirements.txt
docs/custom-build/.venv/bin/python docs/custom-build/tools/build.py
docs/custom-build/.venv/bin/python docs/custom-build/tools/check.py
```

Open `docs/custom-build/site/index.html` in a browser, or serve it locally:

```
python3 -m http.server 8000 --directory docs/custom-build/site
```

The data/source Markdown are authoritative. To edit a purchase allocation, change `data/build-manifest.json` and the corresponding chapter; update the reference tables with `python3 docs/custom-build/tools/reference.py`, rebuild and check. Research records retain the evidence behind the lockfile and gates. Do not refresh a dependency or source commit without reviewing the resulting changed geometry, instructions and licenses.

PDF export and visual checks

The base dependency set builds/checks HTML. PDF export and browser QA additionally use Playwright 1.57.0 and an installed Chrome; no browser download is implicit.

```
docs/custom-build/.venv/bin/pip install playwright==1.57.0
docs/custom-build/.venv/bin/python docs/custom-build/tools/export_pdf.py
docs/custom-build/.venv/bin/python docs/custom-build/tools/browser_check.py
```

The PDF is `site/custom-build-guide.pdf`. The export uses print CSS, local fonts and local images. If Chrome cannot run in the environment, the command fails explicitly; report export unavailable instead of claiming success. Regenerating the upstream reference PNGs also needs PyMuPDF 1.27.2.2:

```
docs/custom-build/.venv/bin/pip install PyMuPDF==1.27.2.2
docs/custom-build/.venv/bin/python docs/custom-build/tools/render_stock.py
```

To regenerate the separately licensed StealthBurner figures too, supply the pinned manual downloaded from the source lockfile: `python3 docs/custom-build/tools/render_stock.py --stealthburner-manual PATH`. Without that option the committed StealthBurner figures are retained.

See [VALIDATION](#) for the actual versions, command outcomes and visual review performed for this revision. Rebuilding unchanged content does not mean a physical printer was assembled or tested.

GitHub publication

The actual parent fork is [obnauticus/Voron-2](#), branch `Voron2.4`, preserving upstream history. The custom guide and generated HTML/PDF are committed in that branch. No upstream pull request is part of this project. GitHub Pages publishes the committed output from `Voron2.4` and `/docs`; `docs/index.html` points to the complete site and `docs/.nojekyll` preserves it as static output. Run the documented build, PDF export and checks **before each push**; this publication mode does not run the custom checker remotely. Its actual deployment state and URL are recorded in the validation report only after verified. Do not change repository visibility or account domain settings.

The current GitHub authorization rejected an active Actions workflow because it lacks `workflow` scope. An optional pinned workflow is retained as `tools/custom-guide.yml.example`, outside the active workflow directory. If that grant is available later, install it in `.github/workflows/` and select GitHub Actions publishing in Pages settings. It installs the pinned dependencies, builds/checks the committed PDF and HTML, then deploys. For source-branch Pages configuration, see [GitHub's Pages API documentation](#).

For a local derivative without an authenticated fork, the remaining publication procedure would be: fork `VoronDesign/Voron-2` in the intended account, set that fork as `origin`, verify `upstream` points to `VoronDesign`, push `Voron2.4`, then enable Pages if desired. This session's actual state supersedes that fallback.

Attribution and asset licenses

This is an **unofficial custom guide**, modified 2026-10-09 for the reported purchases. Voron Design, LDO, Phaetus, Cartographer, Mandala Rose Works and Vitalii3D have not reviewed or endorsed the combined assembly.

The upstream repository remains intact at base commit `a192410e27ea345644ae5c4b29b4c9c40cbe1a73` , branch `Voron2.4` . Its original CAD, STLs, PDF, README attribution and GPL-3.0 LICENSE are preserved. There was no editable Markdown/InDesign manual source in the inspected tree; the custom source is maintained Markdown rather than a modified extracted PDF.

Local asset family	Author/source	Reuse basis and limits
stock-pNNN.png	Voron Design project, retained Manual/Assembly_Manual_2.4r2.pdf	Rendered unmodified reference pages from the GPL-3.0 repository, cited by printed page and zero-based PDF index; original attribution appears in each page. They show STOCK hardware, not the custom CNC assembly.
sb-stock-pNNN.png	Voron Design project, separate StealthBurner commit 8bcb9c246fac19d8ac03931ef97fa07c5e5f0f2b , manual version 2023-07-07	Separately checked GPL-3.0 repository source, rendered stock CW2/SB reference pages. These figures do not establish UHF compatibility.
motion-*.svg , bed-*.svg , toolhead-*.svg , electronics-*.svg	Original custom documentation schematics, 2026	GPL-3.0; labeled non-dimensional relationships or conditional plans. They are not manufacturing drawings or proof of fit.
Custom Markdown, Python tooling, stylesheet and generated HTML/PDF	Custom guide derivative, 2026	GPL-3.0 with upstream attribution; no warranty. Maintained source included in this repository.
Store photos, Vitalii/MRW/Phaetus PDF diagrams, LDO web images	Their respective manufacturers	No blanket permission inferred. Source PDFs/images remain linked, not redistributed, where reuse terms are unknown or reserved. Private research scratch copies are not committed.

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The installed upstream LICENSE is at repository root. The built site includes an identical `LICENSE.txt` . The complete corresponding custom source is this directory. Static checks do not certify physical safety, hardware compatibility or fitness for a particular purpose.

Validation results

Reviewed on October 9, 2026. This is a completed, illustrated documentation project with hardware verification gates. It is **not a physically validated or heater-ready printer configuration**. No firmware, motion, heating or physical assembly was performed.

Actual local results

- `python3 docs/custom-build/tools/reference.py` : passed; generated records for all **14 purchases, 79 sources, 31 open gates, 23 compatibility interactions, 200 printed-parts rows**, and **84 licensed local assets**.
- `python3 docs/custom-build/tools/build.py` : passed; **22 navigable pages** and a complete print edition, with local assets, responsive layout and print CSS.
- `python3 docs/custom-build/tools/check.py` : passed with **zero errors**; **1,259 local links, 93 image references**, all **263 upstream PDF indices** covered once, purchase dispositions and omitted stock parts checked, exact source filenames checked against pinned repository/archive inventories, asset hashes/licenses checked, and all four configuration examples confirmed entirely commented. Four deliberately invalid inventory/allocation/readiness/ configuration inputs were rejected. Export metadata checks the PDF hash and its HTML, stylesheet and local-illustration input hash.
- `python3 docs/custom-build/tools/browser_check.py` : passed; eight representative pages at **1440 × 1000** and **390 × 844, 16 viewport checks, zero failures**. Images loaded and the document body had no horizontal overflow. Desktop and mobile screenshots were inspected, including the motion and toolhead chapters.
- `python3 docs/custom-build/tools/export_pdf.py` : passed using installed Chrome; exported an A4 PDF from the complete local print edition. Representative pages were visually inspected: contents, bed, toolhead, wiring and printed-parts tables. The PDF opens, every page has text, and images are required to decode before export. The current file and input hashes are in [the export record](#).
- Three independent documentation reviews examined motion, bed/toolhead and electrical/software safety interactions. Remaining physical dependencies are explicit gates before their affected steps. Final review corrected bed dry-fit and toolhead-return ordering, and requires X/Y homing before commanded axis moves.
- Original upstream manual, CAD, STLs and license were retained. Private order documents, payment information, addresses and account credentials are excluded.

Versions and corrected failures

Python **3.10.12**, Python-Markdown **3.3.7**, Beautiful Soup **4.12.2**, Soup Sieve **2.3.1**, Playwright **1.57.0**, PyMuPDF **1.27.2.2**, and Chrome **145.0.7632.45** were used. The base dependency versions are pinned in `requirements.txt`; browser/export dependencies are documented separately.

Initial static validation found two missing original-SVG provenance fields; the source normalizer was corrected. Initial browser review found mobile overflow in the toolhead page; wrapping was corrected and all 16 checks passed on rerun. Adding the report exposed an output-artifact link mapping error; the builder now maps source Markdown's

`site/` links correctly into HTML. The restricted sandbox prevented Chrome startup; approved browser execution outside that sandbox completed both browser review and PDF export. These failures are recorded rather than treated as passing runs.

Publication

The actual new fork is [obnauticus/Voron-2](#), branch `Voron2.4`, derived from upstream commit `a192410e27ea345644ae5c4b29b4c9c40cbe1a73`. Remote guide push and Pages deployment are pending at this local-report stage; the publication record will be updated only after verification.

The initial guide push was rejected because the existing OAuth grant lacks permission to create an active Actions workflow. The workflow is retained as an inactive example; publication uses the checked committed `/docs` output. No broader authorization grant or account-domain change was requested.

Limits of these results

Static links, screenshots, PDF layout and consistency checks do not prove physical assembly compatibility, electrical ratings, protective-earth safety, probe performance or print quality. The delivered kit size/batch/PCB revisions, Wings package and support map, double-shear shaft engagement, UHF duct/cartridge geometry, heater startup load and complete electrical path remain unverified. An embedded-magnet MRW bed is manufacturer-incompatible with the selected Cartographer scan path; identify the delivered bed before that path proceeds. All **31 hardware/evidence gates remain open**. Complete independent assembly work, but stop before each blocked operation. All configuration examples remain non-deployable until their hardware and measured values are resolved.